TEL AVIV UNIVERSITY
The Iby and Aladar Fleischman Faculty of Engineering
Department of Electrical Engineering

Deep Learning Architectures for Two-Hop Relay Communication: A Comparative Study of Classical and Neural Network Relay Strategies

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
In Tel Aviv University

by
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This research work was carried out at Tel-Aviv University
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Abstract

This thesis presents a systematic comparison of classical and learned relay strategies for two-hop communication over SISO Rayleigh fading channels. Nine relay methods are considered: amplify-and-forward (AF), decode-and-forward (DF), multilayer perceptrons (MLPs), generative models based on a variational autoencoder (VAE) and a conditional generative adversarial network (cGAN), and sequence models based on the Transformer, Mamba-S6, and Mamba-2-SSD architectures. All methods are assessed through Monte Carlo simulation using bit-error rate (BER), confidence intervals, and statistical significance testing.

Under a canonical matched-channel benchmark, the best learned relays outperform AF at low signal-to-noise ratio (SNR) and attain BER comparable to DF at medium and high SNR, while DF achieves the lowest BER among the conventional schemes without any trainable parameters. Although the literature establishes strong information-theoretic results for DF in asymptotic capacity regimes, these results do not directly imply optimality under the finite-length, uncoded conditions studied here. A complexity analysis indicates that performance is governed primarily by model capacity rather than architectural sophistication: a compact MLP with only 169 trainable parameters attains the BER of networks more than two orders of magnitude larger, and Mamba-2 SSD matches the BER of Mamba S6 at substantially lower training cost.

The robustness of the evaluated methods is then examined in regimes where the modeling assumptions underlying conventional receivers no longer hold. For QPSK, the conclusions of the canonical benchmark are unchanged; for 16-QAM, a joint two-dimensional classifier eliminates the BER floor incurred when the in-phase and quadrature components are processed independently. The principal contribution is a systematic evaluation under unknown and mismatched channels, in which conventional model-based relays degrade substantially on a channel with an unknown intersymbol-interference (ISI) filter, whereas a learned relay with only 169 trainable parameters approaches, without explicit channel knowledge, the performance of genie-aided Viterbi maximum-likelihood sequence estimation (MLSE) matched to that filter.

These results characterize the operating regimes of the evaluated approaches. When the assumed channel model accurately represents the true channel, DF attains the lowest BER among the evaluated per-symbol relays without training **AK:** Why? DF cuts the code blocklength by half. Can't this be beaten?[1]; under model uncertainty or mismatch, learned relays are more robust to modeling errors. Their principal advantage therefore lies not in surpassing optimal model-aware receivers under accurate modeling, but in preserving performance when the channel model is unavailable or fails to capture channel memory or distortion — for example, when a channel filter introduces intersymbol interference (unmodeled memory), the regime studied in the unknown/mismatched-channel study (Chapter 7). **AK:** MLSE w.r.t. what? Did you assume colored noise or a channel filter? Did you use convolutional codes? The System MOdel in Chapter 1.1 appears memoryless with white noise[2]

[1] In the uncoded, per-symbol setting of this thesis, the relay makes a one-shot symbol decision and $\mathrm{sign}(y)$ is the MAP decision, hence optimal among per-symbol relay functions. The blocklength/coding argument does not apply here because no channel code is used; coded block-DF and soft-information (LLR) forwarding, which could indeed do better, are identified as future work (Remark 1.2) **AK:** AF outperforms DF for uncoded symbols for low SNR values. DF is not universally better across all SNR values.**GZ:** Checked directly against this thesis's own data (Table 5.4, canonical Rayleigh channel, and its AWGN counterpart): DF's per-symbol slicing has a strictly lower BER than AF at every SNR point evaluated, 0–20 dB, on both channels, so there is no low-SNR crossover within the range studied here. The general point remains valid in the wider literature: *naive* (always-forward) DF can underperform AF at sufficiently low SNR because the relay regenerates and confidently retransmits its own decoding errors, which is why *selective* DF (relaying only above a reliability threshold) is the standard mitigation. That effect simply does not appear within the evaluated SNR range and configuration, so the abstract's claim is accurate as an empirical statement about the results reported here rather than a universal one.

[2] MLSE is with respect to the FIR channel filter (3-tap ISI) introduced only in the unknown/mismatched-channel study of Chapter 7; the noise remains white and no convolutional code is used. The canonical model of Chapter 1 is indeed memoryless — "channel memory and distortion" in the abstract refers specifically to that deliberate extension, not the canonical setup. The abstract sentence now names the FIR/ISI channel filter explicitly so that MLSE has a well-defined referent **AK:** This si unclear from the abstract. You cannot refer to a model that you've never mentioned.**GZ:** Fixed at the source rather than by cross-reference: the earlier sentence introducing MLSE now names the impairment inline ("a channel with an unknown intersymbol-interference (ISI) filter"), so the abstract is self-contained and MLSE has a well-defined referent without requiring the reader to have already seen Chapter 7.

Acknowledgments

I would like to thank my supervisors, Dr. Anatoly Khina and Prof. Raja Giryes at Tel Aviv University, for their guidance and support throughout this research. This work was conducted in the School of Electrical and Electronic Engineering, Tel Aviv University.

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AK: You need to use square-brackets to display more concise table and figure titles in the corresponding lists.

Chapter 1

Introduction

Relay communication is a key technique in contemporary wireless systems that uses intermediate nodes between a source and a destination to extend coverage, enhance reliability, and increase throughput. Classical relay strategies such as amplify-and-forward (`AF`) and decode-and-forward (`DF`) have been extensively analyzed and are widely used as baseline protocols; in several canonical relay-channel models, block-based `DF` is capacity-achieving or capacity-approaching under asymptotically long codewords [2, 3, 4, 5, 6]. As clarified in Remark 1.2, these results concern coded block operation and do not directly characterize the uncoded, symbol-wise setting studied in this thesis.

Recent advances in machine learning have motivated the application of neural networks to physical-layer communication problems, including detection, decoding, and end-to-end system optimization. Neural models can approximate nonlinear mappings that are analytically intractable and have been reported to offer advantages in low-SNR or non-ideal regimes.

This motivates the central question of this thesis: *under what conditions can neural network–based relays improve upon or complement classical relay strategies in two-hop relay communication?*

This thesis presents a systematic comparative study of classical and neural relay strategies within a unified simulation framework. Nine relay methods are evaluated: two classical relay strategies (`AF` and `DF`) and seven neural relays spanning supervised learning (`MLP`, SNR-adaptive `Hybrid` relay), generative modeling (variational autoencoder `VAE` and conditional GAN, `cGAN`, with `WGAN-GP`), and sequence-based architectures (`Transformer`, `Mamba-S6`, and `Mamba-2 SSD` with structured state-space duality).

The main objective of this thesis is not to propose a new relay design, but to quantify *performance vs. complexity* tradeoffs and identify regimes where learning-based relays are effective. Existing studies typically focus on a limited set of architectures or channel models, making it difficult to disentangle the effects of model structure, parameter count, and channel conditions.

Scope of this thesis. This thesis studies a single, fixed communication problem, the classical source–relay–destination (two-hop) link, under *one canonical setup*: SISO on both hops, i.i.d. Rayleigh fading, complex baseband, uncoded BER, on the two constellations that baseband natively carries, BPSK and QPSK. The only axis varied is the *relay function*. A scoped extension to 16-QAM (Chapter 6) and a dedicated unknown-channel contribution (Chapter 7) are included; all other departures from the canonical setup are identified as future work in Chapter 8:

Table 1.1: Scope of the thesis: the canonical setup (BPSK and QPSK on Rayleigh), the 16-QAM extension, and the unknown-channel contribution. No other topology, relay protocol, or performance metric is evaluated in the main line of this work; all such variations are deferred to future work.

Axis	**Canonical value (future work in parentheses)**
Topology	SISO on both hops *(future work: MIMO second hop)*
Channel	i.i.d. Rayleigh fast fading (canonical), applied identically on both hops; AWGN solely as the analytical baseline the simulator is calibrated against (Section 5.2); unknown, mismatched, and time-varying channels are studied in the dedicated contribution (Chapter 7) *(future work: Rician and other channel families)*
Relay Function *(the studied axis)*	*Classical:* AF, DF. *Learned:* MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD
Modulation	Each constellation is paired with one channel, giving three configurations in total (Table 4.1): BPSK on AWGN (calibration), QPSK on Rayleigh (canonical, Chapter 5), and 16-QAM on Rayleigh (scoped extension, Chapter 6); QPSK is additionally re-checked under the unknown-ISI channel of Chapter 7 *(future work: 16-PSK, 64-QAM and beyond)*
Performance Metric	Uncoded BER vs. SNR, Monte-Carlo averaged

Everything else discussed in the literature review (the general relay channel with a direct link, its capacity bounds, and the other classical relay protocols such as compress-, compute-, estimate-, and filter-and-forward) is provided as *background context only* and is not part of the studied system.

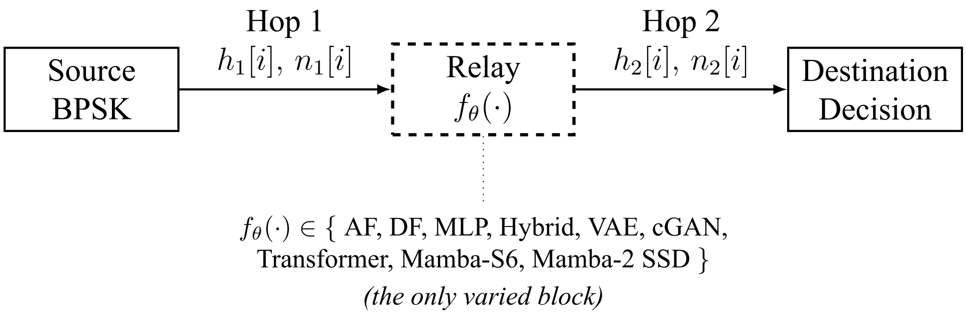

Figure 1.1: Canonical two-hop relay benchmark used throughout the thesis.The source, both channels, and the destination are fixed; only the relay processing function $f_\theta(\cdot)$ (dashed) is varied among the nine candidate strategies. Observed BER differences are therefore attributable to the relay strategy alone.

1.1 System Model

System model studied in this thesis. Throughout this work, a single canonical setup underlies the main line of argument: a half-duplex two-hop link with a single relay and *no* direct source-to-destination path, in which the source S transmits during a listening phase to the relay R, and the relay then transmits during a transmission phase to the destination D. Both hops are single-input single-output (SISO) complex-baseband channels with i.i.d. Rayleigh *fast* fading, a fresh realization $h \sim \mathcal{CN}(0, 1)$ per channel use, known perfectly at the corresponding receiver. The modulation is BPSK, and the performance metric is the *uncoded* bit-error rate (BER) as a function of SNR, averaged by Monte Carlo over the fading ensemble. Since BPSK carries information on the real axis only, each receiver applies coherent phase compensation ($y \cdot h^*/|h|$) and takes $\Re\{\cdot\}$ of the result; the imaginary component then contains noise only, so this projection is the sufficient statistic and loses nothing. The learned component of the system is confined to the *relay processing function* $f_\theta(\cdot)$, the block for which AF and DF are the classical alternatives; the source, channels, and destination demodulator are fixed, so observed BER differences are attributable to the relay strategy alone. The AWGN channel is used primarily as the analytical calibration limit that validates the simulator, and a companion nine-relay comparison on it is additionally reported in Section 5.3 alongside the canonical Rayleigh result. Beyond the canonical core, the thesis makes two further contributions of unequal weight. The canonical setup covers both constellations the complex baseband natively carries, BPSK and QPSK. A scoped extension (Chapter 6) takes the further step to 16-QAM, where per-axis I/Q processing breaks down, including a joint 2D N-class classification formulation of the relay. The principal experimental contribution (Chapter 7) is a systematic study of learned relaying under *unknown and mismatched* channels that deliberately violate the assumptions of the canonical memoryless model, mapping, along every axis relevant to practical deployment, precisely when a minimal learned relay adds value over the strongest classical baselines and when it does not (H6). That chapter also revisits, under an unknown-ISI channel, whether the QPSK generalization established for the canonical benchmark (Chapter 6) still holds: the memoryless-relay failure mode does, but the ordering between the learned relay and genie-CSI Viterbi MLSE does not (Section 7.1.2). All other departures (other channel families, a MIMO second hop, learned constellations) are identified as future work in Chapter 8. Capacity-theoretic results and the cooperative-diversity order are quoted in Chapter 2 as classical *background only*; they are not the operating point evaluated in the experiments.

Figure 1.1 summarizes this canonical benchmark: the source, channels, and destination are fixed, while only the relay function $f_\theta(\cdot)$ is varied.

Scope of the canonical model. The canonical physical channel considered in Chapters 5–6 is memoryless, with white noise, perfect receiver-side CSI, and uncoded transmission. Therefore, the current observation $y_R[i]$ is a sufficient statistic for detecting $x[i]$, and the classical AF and DF baselines operate per symbol. Some neural architectures nevertheless receive a finite window of neighboring observations, as specified in Chapter 4. This is an architectural input convention used for the evaluated feedforward and sequence models; it neither introduces channel memory nor provides additional information about $x[i]$ under the canonical i.i.d. assumptions. Actual channel memory, together with channel estimation, blind equalization, and sequence detection via Viterbi MLSE, is introduced only in Chapter 7, where a 3-tap FIR/ISI channel deliberately violates the canonical memoryless assumption.

1.1.1 Two-Hop Relay Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source $\rightarrow$ Relay), listening phase:

$$y_R[i] = h_1[i]\, x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0,1), \quad n_1[i] \sim \mathcal{CN}(0,\sigma^2) \tag{1.1}$$

Slot 2 (Relay $\rightarrow$ Destination), transmission phase:

$$y_D[i] = h_2[i]\, x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0,1), \quad n_2[i] \sim \mathcal{CN}(0,\sigma^2) \tag{1.2}$$

where $x[i] \in \{-1, +1\}$ is the BPSK symbol transmitted by the source at time i, $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i}) \,,$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function. For AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$. Some evaluated neural architectures instead receive a finite window of neighboring observations, with the exact input dimensions specified in Chapter 4. This finite context is an architectural input convention and does not introduce memory into the canonical physical channel. Under the canonical i.i.d. assumptions, $y_R[i]$ is already a sufficient statistic for detecting $x[i]$, so neighboring observations contain no additional information about the current symbol. A different case is introduced in Chapter 7, where a 3-tap FIR/ISI channel creates genuine temporal memory and makes neighboring observations informative. [3] Because the fading coefficient is known at each receiver, coherent compensation ($y \cdot h^*/|h|$ followed by $\Re\{\cdot\}$) reduces each hop to an equivalent real channel with a random per-symbol effective gain $|h[i]|$,

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \qquad \tilde{y}_D[i] = |h_2[i]|\, x_R[i] + \tilde{n}_2[i], \tag{1.3}$$

with $\tilde{n}[i] \sim \mathcal{N}(0, \sigma^2)$ real. **AK:** Since the channel is complex, you could've sent QPSK instead of BPSK.[4]**GZ:** The distinction between the canonical benchmark and the modulation extension is now made explicit. BPSK remains the canonical modulation used to isolate relay processing, while QPSK and 16-QAM are defined and evaluated separately in Chapter 6. The earlier claim that BPSK "loses no generality" was removed because it was too broad: QPSK reproduces the BPSK behavior under independent I/Q processing, but 16-QAM requires a joint 2D formulation. The *AWGN calibration limit* used to validate the simulator (Chapter 2) is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

Remark 1.1 (Realizability of the symmetric window)**.** The window $y_R[i - w : i + w]$ uses look-ahead samples and is therefore non-causal in the *streaming* sense. It is nonethe-

[3] paragraph revised

[4] QPSK could place an independent bit on the quadrature axis, but for rectangular constellations the I and Q axes are independent, so QPSK is exactly two parallel BPSK channels and loses no generality. This is confirmed experimentally in (Chapter 6. BPSK is thus the minimal modulation that isolates the relay-denoising problem, the only varied axis in this thesis **AK:** I am aware of that. In the abstract you talked about QPSK and larger QAM constellations, but you do not mention them here. What is the source of this mistmatch? If you use compelx channels, stick to 2D constellations or avoid talking about them in the sequel as well.

less fully realizable in the half-duplex protocol studied here: the relay buffers the *entire* listening-phase block before the transmission phase begins, so by the time $x_R[i]$ is computed, all samples $y_R[i-w:i+w]$ have already been received **AK:** Since you assume uncoded transmission, what reason is there to use $\omega > 0$?[5]. The symmetric window is thus a block-level (store-and-forward) operation with a fixed processing latency of w symbols. The classical baselines AF and DF are memoryless in their per-symbol operation ($w = 0$) and are strictly causal.

The noise terms are independent and identically distributed with variance

$$\sigma^2 = \frac{P_s}{\gamma} \tag{1.4}$$

where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for BPSK) and γ denotes the per-hop linear signal-to-noise ratio (SNR), equal on both hops.

The choice of a relay processing function $f(\cdot)$ fundamentally determines system performance and is the central object of study in this thesis. Classical approaches include amplify-and-forward (AF), which simply scales the received signal, and decode-and-forward (DF), which regenerates the signal through demodulation and re-modulation. Each has well-understood performance characteristics: AF is simple but propagates noise, while DF eliminates first-hop noise but introduces error propagation when decoding fails [2].

Remark 1.2 (Terminology: symbol-wise DF vs. information-theoretic DF)**.** The term "decode-and-forward" is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [2, 3], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless

[5] Agreed. For the canonical memoryless channel, $w = 0$ is sufficient because $y_R[i]$ is already a sufficient statistic for $x[i]$. The window is retained primarily for the unknown-channel study of Chapter 7, where the relay must span channel memory (3-tap ISI). Using a common architecture throughout the thesis keeps the relay formulation fixed across experiments and isolates the effect of the channel assumptions rather than the network structure. **AK:** The goal of the "system model" is to provide the model. Either drop this part from this section and define explicitly in Chapter 7 or define in detail what is the competing model you compare yourself against.**GZ:** Kept, but scoped explicitly rather than dropped: this remark defines only the *architectural* convention, that a symmetric window is realizable because the half-duplex protocol buffers the entire listening-phase block, not a competing classical model. That convention applies uniformly to every windowed relay in the thesis, including the canonical setup where $w = 0$, so it belongs here rather than being duplicated per chapter. The actual competing classical model against which the window is exercised, a 3-tap FIR/ISI channel with genie-CSI and pilot-LS Viterbi MLSE, is defined precisely, and only, in Chapter 7 (Section 7.1), which now also reports the same comparison for QPSK (Section 7.1.2). No channel-memory model is introduced in this chapter.

and, over a single link, can operate reliably at any rate below the first-hop capacity.

The system studied in this thesis is *uncoded*: one BPSK symbol carries one information bit and no outer error-correction code is applied. Consequently, the “DF” baseline throughout this work refers to *symbol-wise slicing*: per-symbol demodulation followed by re-modulation, i.e., $f_{\mathrm{DF}}(y_R[i]) = \mathrm{sign}(y_R[i])$ with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used here, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance. Extending the comparison to the coded setting, where the relay (classical or learned) aggregates information across a full code block, is an important direction for future work and is discussed in Chapter 8.

Chapter 2

Background and Related Work

Scope relative to the canonical model. Unless stated otherwise, the relay-processing methods reviewed in this chapter are discussed in the context of the canonical memoryless channel model adopted throughout Chapters 5– 6. Channel-memory effects, sequence detection, and channel models with intersymbol interference (ISI) are intentionally deferred to the unknown-channel study of Chapter 7.

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ outperform the classical relay functions (AF scaling, DF regeneration) by exploiting patterns in the received signal that analytical methods do not capture? This question is particularly relevant at low SNR, where both AF (noise amplification) and DF (decoding errors) have well-known limitations, and where the non-linear decision boundary learned by a neural network may provide a potential advantage.[6]

2.1.1 Amplify-and-Forward (AF)

[7] The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \tag{2.1}$$

[6] “substantive advantage” $\rightarrow$ “potential advantage”: the section frames an open question, so the wording no longer pre-judges its own answer (consistent with the “may offer”$\rightarrow$“have been reported” fix in the Introduction).

[7] Removed stray period in the subsection title.

The end-to-end SNR for AF relay is the standard harmonic-mean expression [4, 7]: (Equation 2.2)

$$\mathrm{SNR}_{\mathrm{eff}}^{\mathrm{AF}} = \frac{\mathrm{SNR}_1 \cdot \mathrm{SNR}_2}{\mathrm{SNR}_1 + \mathrm{SNR}_2 + 1} \tag{2.2}$$

This form assumes *CSI-assisted variable-gain* AF, in which the relay knows its instantaneous first-hop gain and normalizes its output power per symbol (Equation 2.1); this is the variant implemented throughout this thesis. The additive $+1$ in the denominator originates from the relay's own noise being re-amplified along with the signal, and it is the term that distinguishes this convention from the fixed-gain (blind, long-term-power-normalized) variant, whose end-to-end SNR takes a slightly different form.

This expression reveals a fundamental limitation: even when one hop has high SNR, the effective SNR is bottlenecked by the weaker hop. Moreover, the relay amplifies both signal and noise from the first hop, leading to noise accumulation at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \mathrm{demod}(y_R), \quad x_R = \mathrm{mod}(\hat{b}_R) \tag{2.3}$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\mathrm{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \tag{2.4}$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.40). For BPSK modulation over AWGN (real-valued noise with variance $1/\mathrm{SNR}$), each hop's BER is $P_e = Q\left(\sqrt{\mathrm{SNR}}\right)$. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 1.2, the "DF" baseline used here is the uncoded symbol-wise slicer, not information-theoretic block-DF; its optimality claims are therefore confined to the uncoded setting.[8]

[8] Added cross-reference to the symbol-wise/block-DF distinction, mirroring the abstract fix that avoids

2.1.3 Other Classical Relay Techniques

AF and DF represent two canonical extremes of the relay processing spectrum: AF performs minimal processing (linear scaling) while DF performs maximal classical processing (full regeneration). Additional techniques (e.g. CF, EF, CoF) occupy intermediate points along this spectrum. In this thesis, the AI-based relay strategies are designed to *learn* the relay processing function from data, potentially discovering strategies that subsume or outperform these fixed classical schemes. The focus on AF and DF as classical baselines is therefore deliberate: on the canonical matched-channel benchmark they bracket the classical design space and serve as representative reference baselines against which the learned strategies are evaluated; they should not be read as strict lower and upper performance bounds, since under channel mismatch a learned relay can outperform DF (Chapter 7).[9] These additional relay strategies are reviewed for completeness; they are not simulated in this thesis because the studied system assumes a two-hop link without a direct source-destination path.

2.2 Machine Learning in Wireless Communication

The application of machine learning to physical-layer wireless communication has gained significant momentum in recent years, driven by the ability of neural networks to learn complex non-linear mappings directly from data. Deep learning has been applied to channel estimation [8], signal detection [9], autoencoder-based end-to-end communication [10], and resource allocation [11]. These approaches learn complex mappings from data, potentially outperforming hand-crafted algorithms in scenarios where analytical solutions are intractable or suboptimal.

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, the **universal approximation theorem** [12] guarantees that a feedforward network with a single hidden layer and a non-linear activation function can approximate any continuous function on a compact domain to arbitrary accuracy, given sufficient width. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \to [-1, 1], \quad f^*(y_{i-w}, \ldots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \ldots, y_{i+w}] \tag{2.5}$$

presenting DF as a universal optimum.

[9] Rephrased "provide clear lower and upper bounds" $\to$ representative baselines confined to the matched benchmark, because a learned relay approaching MLSE under mismatch shows DF is not a universal upper bound (same non-optimality concern raised by AK on the abstract).

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \tag{2.6}$$

which is a smooth sigmoid-like function that approaches the hard-decision signum function as $\sigma^2 \to 0$ (high SNR). A neural network with a single hidden layer and tanh output can represent this posterior *exactly for a fixed* σ^2, a single tanh unit with the appropriate input scaling suffices. The deployed relay, however, is trained at SNR $\in \{5, 10, 15\}$ dB and evaluated over 0–20 dB, so one set of weights must approximate an entire *family* of posteriors indexed by σ^2 (and, on the fading channel, a mixture over random effective SNRs). That a 169-parameter network approximates this family closely enough to track the classical baselines is an empirical finding of this thesis, not a consequence of the exact single-σ^2 representation.

Second, the **bias-variance decomposition** provides a framework for understanding model complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma^2_{\text{irreducible}} \tag{2.7}$$

The irreducible noise $\sigma^2_{\text{irreducible}}$ represents the minimum achievable MSE, determined by the channel noise. Increasing model complexity (more parameters) reduces bias but increases variance. For the relay denoising task, the target function f^* is simple (essentially a soft threshold), so the bias term is already small for modest networks. Adding parameters beyond this point primarily increases variance (overfitting), explaining the U-shaped relationship between model size and BER observed in this thesis.[10]

2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

Ye et al. [8] demonstrated that deep neural networks can jointly perform channel estimation and signal detection in OFDM systems, achieving near-optimal performance with lower complexity than separate estimation and detection stages. Their key insight was that the DNN implicitly learns the channel's statistical structure, eliminating the need for explicit

[10] "inverted-U" $\to$ "U-shaped": by the bias–variance decomposition, BER *decreases* then *increases* with model size, which is a U-shaped (not inverted-U) curve; this also matches the "plateau or degrade beyond a modest model size" statement below.

pilot-based estimation.

Samuel et al. [9] proposed DetNet, an unfolded projected gradient descent network for MIMO detection. By unrolling iterative optimization into a fixed number of neural network layers, DetNet achieves near-maximum-likelihood detection performance with $O(N_t^2)$ complexity instead of the exponential complexity of exhaustive ML search. This demonstrates the power of incorporating domain knowledge (the iterative structure of the detection problem) into the network architecture.

Dorner et al. [10] proposed treating the entire communication system: modulator, channel, and demodulator: as an autoencoder, trained end-to-end to minimize bit error rate. This approach jointly optimizes all components, potentially discovering novel modulation and coding schemes that outperform separately designed modules. The autoencoder paradigm is conceptually related to the relay processing task: both involve learning an encoder-decoder pair that maps through a noisy channel.

Sun et al. [11] applied deep learning to the NP-hard problem of resource allocation in interference networks, demonstrating that a DNN can learn near-optimal power control policies orders of magnitude faster than conventional iterative algorithms.

In the context of relay-assisted communication systems, most existing works employ neural networks primarily for *relay-selection* and *resource-allocation*, particularly in decode-and-forward (DF) relaying scenarios, where learning-based classifiers or predictors are used to identify the optimal relay based on channel state information, signal-to-noise ratios, or energy constraints, while the underlying relay operation remains conventional [13].

A broader perspective on the role of machine learning in wireless communications is provided in [14], which highlights the potential of learning-based methods across multiple protocol layers, including cooperative communications.

Another related line of work exploits the structural similarities between amplify-and-forward (AF) relay networks and neural networks, treating relay nodes as neurons with nonlinear transfer functions and using deep-learning optimization tools to tune relay gains and exploit hardware-induced nonlinearities [15, 16]. While these approaches demonstrate significant gains over classical linear AF schemes, they focus on analog gain optimization rather than symbol-level detection.

Classical cooperative communication frameworks based on AF and DF relaying remain well understood and widely adopted [6], but they rely on fixed linear forwarding or hard-decision rules. Consequently, despite substantial progress, relatively little attention has been devoted to learning the relay's symbol-level input-output mapping itself, motivating neural network-based detection at the relay as a generalized, data-driven alternative to conventional AF and DF processing.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \tag{2.8}$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w+1$ received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships.[11] The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1\mathbf{w} + \mathbf{b}_1), \quad \hat{x} = \tanh(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2) \tag{2.9}$$

The sliding window formulation ($i \in [i-w : i+w]$) is motivated by the observation that adjacent received symbols share common noise statistics and channel conditions. Under the canonical model (i.i.d. fast fading, white noise), adjacent symbols are statistically independent, so a single-symbol estimator ($w=0$) is already a sufficient statistic and a window confers no theoretical benefit. The window is therefore not motivated by temporal correlation in the canonical setting; it is retained because it is required by the unknown-channel study of Chapter 7, where a 3-tap ISI channel couples adjacent symbols and the relay must span that memory. Keeping a fixed windowed architecture across all experiments isolates the effect of the channel assumptions from the network structure. [12]

Multi-SNR training is a key design choice: by training on data generated at multiple SNR levels (5, 10, 15 dB in this thesis), the network learns a denoising function that generalizes across operating conditions rather than specializing to a single noise level. This is analogous to training a robust estimator that operates well across a range of noise

[11] “An well known example for” → “A well-known example of”; “Multi-layer-perceptron” → “multilayer perceptron” to match the abstract; double spaces removed.

[12] Corrected: the earlier text invoked “correlated fading” to justify $w > 0$, contradicting the canonical i.i.d. fast-fading, white-noise assumption (AK). The window is now correctly justified only by the Chapter 7 ISI extension, consistent with Remark 1.1.

variances, a concept related to minimax estimation in statistical decision theory.

A critical question in applying neural networks to relay processing is the relationship between model complexity and performance. Prior work has generally assumed that larger models yield better performance, but this assumption has not been rigorously tested in the relay communication context. The bias–variance framework predicts that for a low-complexity target function like relay denoising, performance should plateau or degrade beyond a modest model size: a prediction that this thesis confirms empirically. Understanding this relationship is essential for practical deployment, where relay nodes may have limited computational resources.

2.3 Generative Models for Signal Processing

Generative models offer an alternative paradigm for relay signal processing. Rather than directly learning a denoising function (discriminative approach), generative models learn the distribution of clean signals $p(\mathbf{x})$ and use this knowledge to reconstruct the transmitted signal from noisy observations via Bayes' rule:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \tag{2.10}$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [17] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\mathrm{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \tag{2.11}$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{Latent space regularity}} \tag{2.12}$$

The reconstruction term encourages the decoder to accurately reproduce the input from the latent code, while the KL term regularizes the latent space to be close to the standard Gaussian prior $p(\mathbf{z})$. The β parameter (β-VAE) [18] in Equation 2.12 controls the trade-off between reconstruction quality and latent space smoothness: $\beta < 1$ prioritizes reconstruction (beneficial for the relay task where accurate signal recovery is paramount), while $\beta > 1$ encourages disentangled latent representations.[13]

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \tag{2.13}$$

Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{2.14}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allowing backpropagation through the encoder.

For relay signal processing, the VAE learns a compressed latent representation of the clean signal manifold. During inference, the noisy received signal is encoded into the latent space, and the decoder maps it back to a denoised estimate. The regularized latent space provides implicit denoising: noisy inputs that map to regions far from the learned signal manifold are pulled back toward high-probability regions during decoding. However, the stochastic sampling introduces additional variance into the estimate, which can be harmful for the deterministic BPSK denoising task: a trade-off that manifests as the consistent VAE under-performance observed in this thesis.

2.3.2 Conditional Generative Adversarial Networks

Conditional GANs (cGANs) [19] learn signal denoising through adversarial training, building on the foundational GAN framework [20].[14] The original GAN formulates generative modeling as a two-player minimax game between a generator G and a discriminator D: (Equation 2.15)

[13] HTML artifacts `</>` restored to </> for compilation.

[14] Unified naming “CGANs”→“cGANs” (and all later “CGAN”→“cGAN”) to match the abstract’s `cGAN`.

$$\min_G \max_D \; \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))] \tag{2.15}$$

At the Nash equilibrium, G generates samples indistinguishable from real data and D outputs 1/2 everywhere. The conditional variant conditions both G and D on auxiliary information (the noisy signal $\mathbf{y}$), enabling the generator to learn a noise-conditioned mapping.

A fundamental challenge with the original GAN objective is *training instability*: the Jensen-Shannon divergence that the discriminator implicitly estimates can produce vanishing gradients when the generator distribution and data distribution have disjoint supports (which is common early in training). The *Wasserstein GAN (WGAN)* addresses this by replacing the JS divergence with the Earth Mover (Wasserstein-1) distance: (Equation 2.16)

$$W(p_{\text{data}}, p_G) = \inf_{\gamma\in\Pi(p_{\text{data}}, p_G)} \mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\gamma}[||\mathbf{x} - \mathbf{y}||] \tag{2.16}$$

The Wasserstein distance provides meaningful gradients even when distributions do not overlap, enabling stable training. By the Kantorovich-Rubinstein duality, the Wasserstein distance can be computed via a supremum over 1-Lipschitz functions, which the critic network approximates. The *gradient penalty (GP)* formulation [21] enforces the Lipschitz constraint softly by penalizing the gradient norm of the critic at interpolated points: (Equation 2.17)

$$\text{GP} = \mathbb{E}_{\hat{\mathbf{x}}\sim p_{\hat{\mathbf{x}}}}\left[(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2\right] \tag{2.17}$$

where

$$\hat{\mathbf{x}} = \epsilon\mathbf{x}_{\text{real}} + (1-\epsilon)\mathbf{x}_{\text{fake}}, \epsilon \sim \text{Uniform}(0,1). \tag{2.18}$$

The full WGAN-GP training objectives used in this thesis are:

$$\mathcal{L}_G = -\mathbb{E}[D(G(\mathbf{y},\mathbf{z}),\mathbf{y})] + \lambda_{L_1}\|\hat{\mathbf{x}} - \mathbf{x}\|_1 \tag{2.19}$$

$$\mathcal{L}_D = \mathbb{E}[D(G(\mathbf{y},\mathbf{z}),\mathbf{y})] - \mathbb{E}[D(\mathbf{x},\mathbf{y})] + \lambda_{\text{GP}} \cdot \text{GP} \tag{2.20}$$

The L_1 reconstruction loss in the generator objective serves a dual purpose: it provides a strong sample-level reconstruction signal (complementing the adversarial signal which provides a distributional match), and it prevents *mode-collapse* (the generator converging to a single output regardless of input).[15] For relay denoising, the L_1 term dominates at $\lambda_{L_1} = 100$, making the cGAN behave primarily as a supervised denoiser with an adversarial regularizer that encourages outputs to lie on the manifold of clean signals.

2.3.3 Generative vs. Discriminative Paradigms for Relay Processing

The application of generative models to relay processing has, to the best of our knowledge, not been extensively studied.[16] The key question is whether the generative inductive bias (learning the data distribution rather than just the input-output mapping) provides any advantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1, +1\}$), suggesting that the generative overhead may not be justified. This thesis provides, to the best of our knowledge, the first systematic comparison of VAE and cGAN-based relay processing against classical and supervised learning methods, and the results show that the generative paradigm provides no significant advantage for this particular task. The VAE is a consistent underperformer, but that specific result should be read with a caveat: the VAE relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic-decoding variant (substituting the posterior mean $\boldsymbol{\mu}$ at test time, which is standard practice) was not evaluated here. The VAE's gap may therefore reflect this single inference-time choice rather than the generative paradigm as such; disentangling the two is left to future work.

2.4 Sequence Models: Transformers, State Space Models

Recent advances in sequence modeling have produced two competing paradigms with distinct computational properties and inductive biases. Both have achieved state-of-the-art results in natural language processing, but their relative merits for physical-layer signal processing (where sequences are real-valued temporal signals rather than discrete tokens) have, to the best of our knowledge, not been previously investigated.[17]

[15] "pixel-level" $\rightarrow$ "sample-level": the relay operates on modulation symbols, not image pixels (register/precision).

[16] Softened the novelty claim with "to the best of our knowledge" (same caution applied to the "first"/"unexplored" claims elsewhere in this chapter).

[17] Softened the novelty claim with "to the best of our knowledge".

2.4.1 Transformers and the Attention Mechanism

Transformers [22] use multi-head self-attention to capture global dependencies in sequences. The core attention mechanism computes a weighted combination of value vectors, where the weights are derived from the compatibility of query and key vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2.21}$$

where $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d_k}$ and $\mathbf{V} \in \mathbb{R}^{n \times d_v}$ are obtained from the input via learned linear projections W^Q, W^K, W^V. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in magnitude, which would push the softmax into saturation regions with vanishing gradients.

Multi-head attention extends this by running h parallel attention heads, each with independent projections, and concatenating their outputs: (Equation 2.22)

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \text{head}_i = \text{Attention}(\mathbf{X}W_i^Q, \mathbf{X}W_i^K, \mathbf{X}W_i^V) \tag{2.22}$$

Each head can attend to different aspects of the input: for signal processing, one head might focus on the immediate neighbors (local denoising) while another captures longer-range patterns. The total parameter count for multi-head attention is:

$$3d_{\text{model}}^2 + d_{\text{model}}^2 = 4d_{\text{model}}^2 \tag{2.23}$$

for Q, K, V projections and the output projection. The attention matrix:

$$\mathbf{A} = softmax(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}) \in \mathbb{R}^{n \times n} \tag{2.24}$$

computes pairwise interactions between all n positions, yielding a time and memory complexity of: $O(n^2)$. For the 11-symbol relay window used in this thesis, this is negligible ($11^2 = 121$ entries). However, the quadratic cost becomes prohibitive for longer sequences (e.g., $n = 1000$ symbols), motivating linear-time alternatives.

Positional encoding is necessary because the attention mechanism is permutation-equivariant (it treats input positions symmetrically). This thesis uses sinusoidal positional encoding: (Equation 2.25)

$$PE_{(pos,2k)} = \sin(\frac{pos}{10000^{2k/d}}), \quad PE_{(pos,2k+1)} = \cos(\frac{pos}{10000^{2k/d}}) \tag{2.25}$$

which injects position information into the embeddings, enabling the model to distinguish between symbols at different positions in the window.

2.4.2 Structured State Space Models

Structured state space models (SSMs) [23] are a class of sequence models derived from continuous-time linear dynamical systems. The continuous-time SSM maps an input signal $u(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a latent state $\mathbf{x}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \quad y(t) = \mathbf{C}\mathbf{x}(t) + Du(t) \tag{2.26}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ governs the state dynamics, $\mathbf{B} \in \mathbb{R}^{N\times 1}$ controls input injection, $\mathbf{C} \in \mathbb{R}^{1\times N}$ is the output projection, and $D \in \mathbb{R}$ is the feedthrough. The connection to classical signal processing is immediate: this is a linear time-invariant (LTI) filter, and the state space dimension N determines the order of the filter (number of poles/zeros in the transfer function).

To process discrete-time sequences, the continuous SSM is **discretized** using a step size $\Delta > 0$. Applying the zero-order hold (ZOH) discretization: (Equation 2.27)

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \approx \Delta\mathbf{B} \tag{2.27}$$

yields the discrete recurrence: (Equation 2.28)

$$\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k, \quad y_k = \mathbf{C}\mathbf{x}_k + Du_k \tag{2.28}$$

The **HiPPO (High-order Polynomial Projection Operators)** framework provides a principled initialization for $\mathbf{A}$: the HiPPO-LegS matrix is designed so that the state $\mathbf{x}_k$ stores a compressed representation of the input history as coefficients of a Legendre polynomial expansion. This initialization enables long-range memory without the vanishing gradient problems of standard RNNs.

The S4 model [23] introduced the key insight that structured (diagonal or low-rank) parameterizations of $\mathbf{A}$ enable efficient computation. With a diagonal $\mathbf{A} = \mathrm{diag}(a_1, \ldots, a_N)$, the recurrence decouples into N independent scalar recurrences, each computable in $O(n)$ time. The total cost is $O(nN)$ for a length-n sequence with state dimension N: linear in sequence length.

2.4.3 Mamba: Selective State Spaces

Mamba [24] extends the SSM framework by making the parameters **input-dependent** (selective), transforming the LTI system into a linear time-varying (LTV) system:

$$\Delta_k = \mathrm{softplus}(W_\Delta u_k + b_\Delta), \quad \mathbf{B}_k = W_B u_k, \quad \mathbf{C}_k = W_C u_k \tag{2.29}$$

where W_Δ, W_B, W_C are learned projection matrices. Crucially, the state matrix is parameterized as $\mathbf{A} = -\exp(A_{\log})$, so that $\mathbf{A}$ is diagonal with strictly negative entries (Hurwitz, $\mathbf{A} \prec 0$) and the discretized transition $\exp(\Delta_k \mathbf{A})$ is a contraction for every $\Delta_k > 0$. This sign convention is what makes the selectivity mechanism well behaved: it lets the model dynamically control which inputs are stored in the state and which are forgotten.

A **large** Δ_k (triggered by a relevant input) drives the transition toward zero,

$$\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A}) \approx \mathbf{0}, \tag{2.30}$$

which resets the state and lets the new input dominate via

$$\bar{\mathbf{B}}_k \approx \Delta_k \mathbf{B}_k. \tag{2.31}$$

Conversely, a **small** Δ_k (triggered by an irrelevant or noisy input) makes the transition approximately the identity,

$$\bar{\mathbf{A}}_k \approx \mathbf{I}, \tag{2.32}$$

thereby preserving the current state and ignoring the input.

For relay signal processing, this selective mechanism is particularly well-suited: the model can learn to attend to the actual signal component of each received sample while suppressing the noise component, effectively implementing an *adaptive filter* whose coefficients are conditioned on the input. This generalizes the Wiener filter, the optimal linear MMSE denoiser under wide-sense-stationary assumptions, which cannot adapt its coefficients on a per-sample basis.[18]

Remark 2.1. These aforementioned adaptive-filtering connections become directly relevant only in the unknown-channel study of Chapter 7; on the canonical memoryless channel the relay reduces to a per-symbol denoiser.

The Mamba architecture wraps the selective S_6 layer in a gated architecture: the input is projected to $2 \times d_{\text{inner}}$ channels (expand factor 2), and split into two branches. The main branch passes through a causal 1D convolution (*Conv1D*) to mix local temporal context, followed by a *SiLU* activation and the S_6 selective scan. The second branch acts as a *SiLU* gate. The two branches are then multiplicatively merged and contracted back to d_{model}. This gating mechanism provides a multiplicative interaction that helps the model learn sharp non-linear decision boundaries.

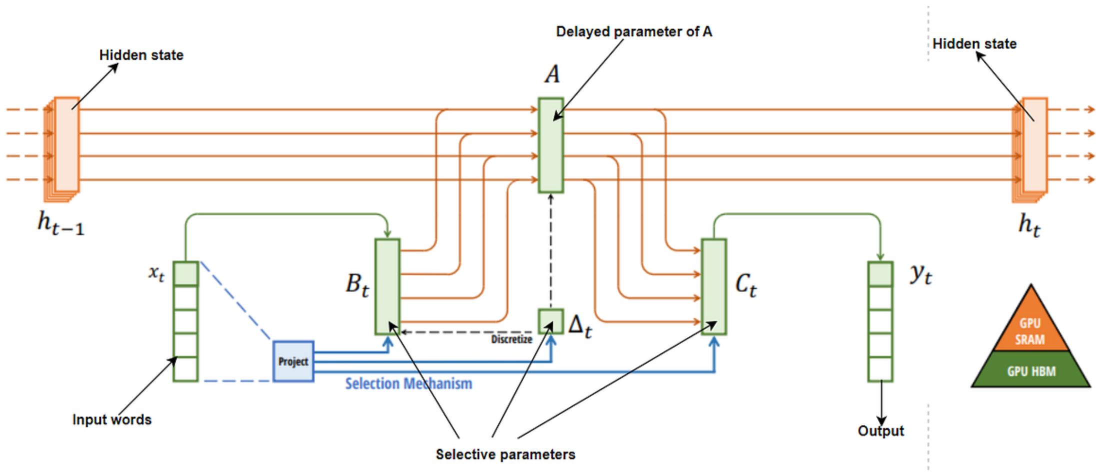

Figure 2.1: Mamba: Linear-Time Sequence Modeling with Selective State Spaces. Structured SSMs independently map each channel (e.g. $D = 5$) of an input x to output y through a higher dimensional latent state h (e.g. state dimension $N = 4$). Prior SSMs avoid materializing this large effective state (DN times batch size B and sequence length L) through clever alternate computation paths requiring time-invariance: the (Δ, A, B, C) parameters are constant across time. The selection mechanism adds back input-dependent dynamics, which also requires a careful hardware-aware algorithm to only materialize the expanded states in more efficient levels of the GPU memory hierarchy.

[19]

2.4.4 Mamba-2: Structured State Space Duality

Mamba-2 (SSD) [25] reformulates the selective state space model through an algebraic duality between linear recurrences and structured matrix multiplications. The key theoret-

[18] Fixed the double "which" and the imprecise "optimal LTI denoiser"; the Wiener filter is the optimal *linear* (MMSE) estimator under WSS assumptions.

[19] Caption: the latent-state dimension is the SSM state size N, not D (the number of channels); corrected "$D = 4$"→"$N = 4$" to match the surrounding notation.

ical insight is that the SSM output can be expressed as:

$$y_i = \sum_{j \le i} \mathbf{C}_i^\top \left(\prod_{k=j+1}^{i} \bar{\mathbf{A}}_k \right) \mathbf{B}_j u_j \tag{2.33}$$

Defining the **SSM matrix** $M \in \mathbb{R}^{n \times n}$ with entries:

$$M_{ij} = \begin{cases} \mathbf{C}_i^\top \bar{\mathbf{A}}_{j+1:i} \mathbf{B}_j & i \ge j \\ 0 & i < j \end{cases} \tag{2.34}$$

[20]

the output is $\mathbf{y} = M \cdot (\mathbf{B} \odot \mathbf{u})$. The matrix M is lower-triangular (causal) and **semi-separable** (each entry factors as an outer product of left and right vectors with a diagonal product in between). This algebraic structure enables efficient chunk-parallel computation:

1. **Divide** the sequence into chunks of length L.
2. **Within each chunk**, build the $L \times L$ local SSM matrix and compute the output via a single batched matrix multiply.
3. **Between chunks**, propagate the state once per chunk (rather than once per time step).

The per-chunk cost is $O(L^2 N)$ for matrix construction and $O(L^2 d)$ for the matrix-vector product, where N is the state dimension and d is the model dimension. With n/L chunks, the total cost is $O(n \cdot L \cdot N + n \cdot L \cdot d)$, which for small L (e.g., $L = 8$ or $L = 32$) is effectively $O(n)$ with a favorable constant. The critical advantage over S_6 is that the intra-chunk computation is **fully parallel**: a batched matmul rather than a sequential scan: making it much faster on GPUs where parallelism is cheap and sequential operations incur kernel-launch overhead.

The "*duality*" in the name refers to the equivalence between the recurrent view (state propagation) and the matrix view (structured matmul): the same computation can be expressed either way, and the implementation can choose whichever is more efficient for the hardware and sequence length. This thesis demonstrates this duality empirically: at $n = 11$, the recurrent S_6 view is faster; at $n = 255$, the matrix SSD view dominates.

Mamba-2 can be viewed as a streamlined evolution of Mamba-1: it restructures the block so that the state-space parameters are formed upfront (rather than being generated sequentially along the SSM pathway), and it further stabilizes training by adding an extra normalization stage in the spirit of *NormFormer* [1]. In addition, Mamba-2 emphasizes more

[20] Restored the `cases` delimiters (`&`→`&`) and the inequality (`<`→<) for compilation.

parallel, consolidated parameter projection (e.g., producing multiple SSM-related tensors via fewer projections) and reduces per-head redundancy by sharing the B and C projections across heads, conceptually similar to multi-value attention.[21] Figure 2.2 presents a comparison of Mamba-S6 and Mamba-2 SSD [25]:[22]

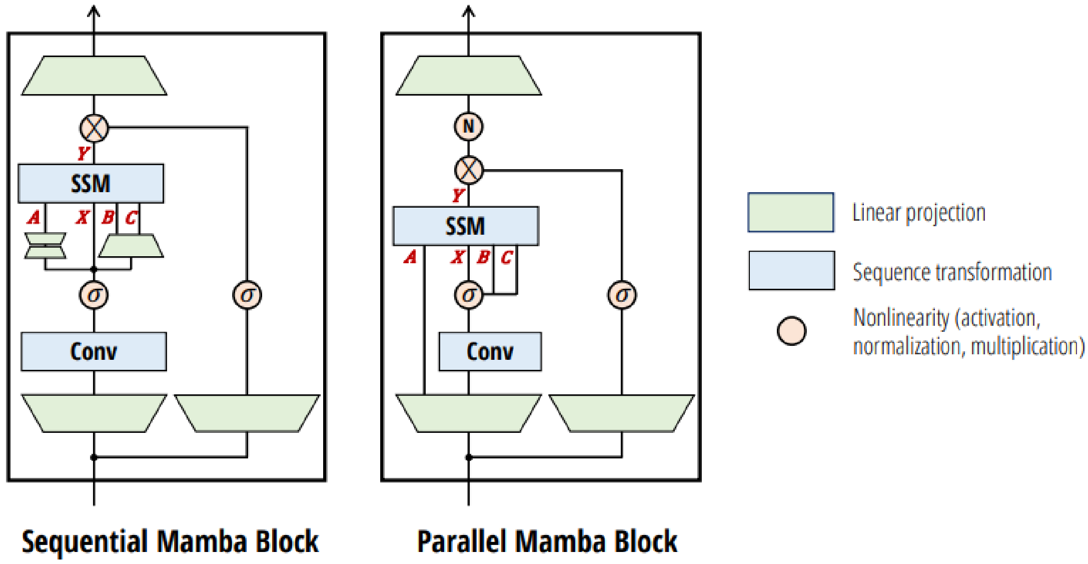

Figure 2.2: The Mamba-2 block simplifies the Mamba block by removing sequential linear projections; the SSM parameters A, B, C are produced at the beginning of the block instead of as a function of the SSM input X. An additional normalization layer is added as in *NormFormer* [1], improving stability. The B and C projections only have a single head shared across the X heads, analogous to multi-value attention (MVA).

2.4.5 State Space Models for Signal Processing

The application of state space models to physical-layer signal processing is, to the best of our knowledge, largely unexplored.[23] Classical signal processing relies extensively on LTI state space models (Kalman filters, Wiener filters), and the SSM framework provides a natural neural extension of these classical tools. The connection is direct: a trained SSM with fixed (input-independent) parameters implements a learnable linear filter, while the selective Mamba variant implements a learnable *adaptive* filter. For the relay denoising task, this means Mamba can potentially learn the optimal filter structure from data, without requiring manual specification of filter order, cutoff frequencies, or adaptation algorithms.

This thesis presents, to the best of our knowledge, the first comparison of Mamba-S6, Mamba-2 SSD, and Transformers for relay communication, demonstrating that state space

[21] "BBB and CCC" was a rendering artifact; corrected to the matrices B and C (consistent with Figure 2.2).

[22] Unified architecture names to "Mamba-S6" and "Mamba-2 SSD" throughout (previously "Mamba-SSM"/"Mamba-2-SSD").

[23] Softened the novelty claim with "to the best of our knowledge".

models are well suited to this domain and that the SSD formulation of Mamba-2 yields significant speed advantages at longer context lengths.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.35)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{\mathrm{SNR}_{\mathrm{linear}}} \tag{2.35}$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols. The noise variance is inversely proportional to the linear SNR.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/\mathrm{SNR}$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.36)

$$P_e^{\mathrm{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{\mathrm{SNR}}\right) = \frac{1}{2}\,\mathrm{erfc}\left(\sqrt{\frac{\mathrm{SNR}}{2}}\right) \tag{2.36}$$

where $Q(x) = \frac{1}{2}\mathrm{erfc}(x/\sqrt{2})$ is the Gaussian Q-function and $\mathrm{SNR} = P_s/\sigma^2$ is the ratio of signal power to noise variance.

Remark 2.2 (Noise convention)**.** The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes that the one-sided noise PSD is N_0, giving a baseband noise variance of $N_0/2$ per real dimension. In our AWGN implementation the noise is purely real with variance $\sigma^2 = P_s/\mathrm{SNR}$, so $N_0 = 2\sigma^2 = 2P_s/\mathrm{SNR}$ and $E_b/N_0 = \mathrm{SNR}/2$. Substituting yields $Q(\sqrt{2 \cdot \mathrm{SNR}/2}) = Q(\sqrt{\mathrm{SNR}})$, which is the expression used throughout this work.

By contrast, the fading channels add **complex** Gaussian noise with total power $1/\mathrm{SNR}$

(i.e. $\sigma_I^2 = \sigma_Q^2 = 1/(2\,\mathrm{SNR})$) and extract the real part after ZF equalization, halving the effective noise variance per decision dimension. This naturally introduces a factor of 2 inside the Q-function argument for all fading-channel BER expressions (the fading-channel analysis above), making them consistent with the standard textbook forms.

2.5.1.2 Two-hop AF BER

For amplify-and-forward with equal-SNR hops, the effective end-to-end SNR is:

$$\mathrm{SNR}_{\mathrm{eff}}^{\mathrm{AF}} = \frac{\mathrm{SNR}_1 \cdot \mathrm{SNR}_2}{\mathrm{SNR}_1 + \mathrm{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \tag{2.37}$$

where $\gamma = \mathrm{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\mathrm{AF}} = Q\left(\sqrt{\mathrm{SNR}_{\mathrm{eff}}}\right) \tag{2.38}$$

under the same real-noise convention as the single-hop case (Remark 2.2). At high SNR,

$$\mathrm{SNR}_{\mathrm{eff}} \approx \frac{\gamma}{2} \tag{2.39}$$

confirming the well-known 3 dB penalty of AF relaying.

2.5.1.3 Two-hop DF BER

For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination if exactly one hop introduces an error:

$$P_e^{\mathrm{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \tag{2.40}$$

With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\mathrm{AWGN}}$), this becomes

$$P_e^{\mathrm{DF}} = 2P_e(1 - P_e) \tag{2.41}$$

At high SNR, $P_e \ll 1$ and

$$P_e^{\mathrm{DF}} \approx 2P_e \tag{2.42}$$

i.e., the two-hop penalty is approximately a factor of 2 in BER.

2.5.2 Rayleigh Fading Channel: Theoretical Analysis

The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the signal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0,1), \quad n \sim \mathcal{CN}(0,\sigma^2) \tag{2.43}$$

The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \tag{2.44}$$

with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} = y/h$) becomes $\gamma_h = |h|^2 \cdot \text{SNR}$, which is exponentially distributed.

2.5.2.1 Single-hop BER

Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of γ_h yields the closed-form [26, Eq. 14-4-15] (Equation 2.45):

$$P_e^{\text{Rayleigh}} = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right) \tag{2.45}$$

where $\bar{\gamma} = \text{SNR}$ is the average SNR. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \tag{2.46}$$

This $1/\text{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{\gamma}) \sim e^{-\gamma/2}$), explaining why Rayleigh fading is significantly more challenging. The channel is **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR. Combating this diversity limitation is the primary motivation for the relay architectures studied in this thesis.

2.5.2.2 Two-hop DF BER

The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF,Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \tag{2.47}$$

Chapter 3

Research Objectives

3.1 Main Objective

To systematically evaluate and compare classical and AI-based relay strategies for two-hop communication on a single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, BPSK and QPSK, uncoded BER) and to determine the most effective relay strategy as a function of SNR regime and computational constraints. [24] [25]

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, a systematic comparison of relay processing paradigms (spanning classical signal processing, supervised learning, generative modeling, and modern sequence architectures) is, to the best of our knowledge, absent from the literature.[26] The following specific gaps motivate this thesis:

1. **Gap 1: No cross-paradigm relay comparison.** Existing work on AI-based relay processing focuses on individual architectures in isolation. Ye et al. [8] demonstrated DNNs for OFDM detection, Dorner et al. [10] explored autoencoders for end-

[24] Relocated the revision note out of the middle of the objective sentence (it previously split "To systematically …evaluate"), so the main objective now reads cleanly. Note content preserved below.

[25] Fix applied: Chapter 2 was reduced to a literature review only. Removed duplicate E1 channel-validation material (including relocated theoretical-SNR results), duplicated validation figures, the Rician fading distribution discussion (future work only), the relay-positioning table duplicated from Chapter 3, and the unused MIMO equalization section (ZF/LMMSE/LMMSE-SIC), since no MIMO experiment appears in the thesis. The term MMSE is now used only in its estimator-theoretic sense, consistent with the canonical-study scope and supervisor comments documented in Appendix E.

[26] Softened "is absent from the literature" with "to the best of our knowledge" (same caution applied to novelty claims in Chapter 2).

to-end communication, and Samuel et al. [9] applied unfolded networks to MIMO detection. However, no study systematically compares supervised feedforward networks, variational autoencoders, adversarial networks, attention-based Transformers, and state space models (Mamba) for the same relay denoising task under controlled conditions. Without such a comparison, practitioners cannot make informed architecture selections.

2. **Gap 2: Model complexity-performance relationship is unknown.** It is commonly assumed that larger neural networks yield better performance. However, the relay denoising task (recovering a BPSK symbol from a noisy observation) is a low-dimensional mapping problem. The theoretical minimum description length for this mapping is small (a soft threshold function), suggesting that large models may overfit. No prior work has rigorously characterized the complexity-performance trade-off for relay processing, nor has a normalized (equal-parameter) comparison been conducted to isolate architectural effects from capacity effects.

3. **Gap 3: State space models for physical-layer signal processing.** The Mamba architecture [24] and its Mamba-2 SSD variant [25] have demonstrated strong performance in NLP and genomics, but their application to relay-level physical-layer signal processing remains, to the best of our knowledge, unexplored. The theoretical connection between SSMs and classical adaptive filters makes this a natural research direction, but empirical validation is needed. [27]

This thesis addresses all three gaps through a unified framework that implements and compares nine relay strategies on the canonical setup, with both original and normalized parameter counts, full statistical analysis, and a dedicated complexity study. The following table summarizes the positioning of this work relative to the literature:

Table 3.1: Research positioning: comparison of this thesis against prior work in deep-learning-based relay communication.

Aspect	Prior Work	This Thesis
Relay strategies compared	1–2 (typically AF vs. DF, or one AI method)	9 (2 classical + 7 AI)
AI paradigms	Single (e.g., supervised only)	4 (supervised, generative, adversarial, sequential)

[27] Fix applied: the Mamba architecture was previously attributed to the earlier S4 paper; it now cites [24], and the novelty claim is scoped to relay-level processing rather than all of physical-layer communication (and softened with "to the best of our knowledge").

Aspect	Prior Work	This Thesis
Evaluation discipline	Single ad-hoc configuration	One canonical setup (SISO Rayleigh BPSK), fixed in advance, with AWGN analytical calibration
Complexity analysis	None	Normalized ∼3K-parameter comparison + complexity study (U-shaped BER vs. model size)
State space models	None for relay/physical-layer	Mamba-S6, Mamba-2 SSD, with crossover benchmark
Statistical rigor	Often missing	Wilcoxon test, 95% CI, 10 trials per point

[28]

Relationship between the canonical study and the extension study. Hypotheses H1–H5 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H6 studies a deliberately different regime in which the canonical model no longer represents the true channel. It therefore serves as a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup; H5 is an architecture-level hypothesis concerning training compute. H6 is the central hypothesis of the thesis's principal contribution (Chapter 7), posed on channels *outside* the canonical family. The higher-order-modulation extension (Chapter 6) additionally probes the generalizability of H1–H3 to QPSK and 16-QAM and is reported descriptively:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-

[28] Table fixes: "U-shaped study"→"complexity study (U-shaped BER vs. model size)" to stay consistent with the U-shaped-in-BER wording adopted in Chapter 2; (3) unified "Mamba S_6"→"Mamba-S6"; (4) en-dash for the "1–2" range.

linear denoising learned by the neural network provides an advantage over AF's noise amplification.

Rationale: At low SNR, the MMSE-optimal (posterior-mean) denoiser of a BPSK symbol from a single AWGN observation, $f^*(y) = \mathbb{E}[x \mid y] = \tanh(y/\sigma^2)$, is a smooth non-linear mapping that differs substantially from AF's linear scaling; a neural network can approximate f^* closely, whereas AF amplifies noise together with the signal. Note that f^* is optimal in the mean-square sense for the relay's local estimate; it is not claimed to be the end-to-end BER-optimal relay function, which is why H1 is posed as an empirical hypothesis rather than a theorem.

2. **H2 (DF dominance at high SNR).** The symbol-wise DF relay (Remark 1.2) achieves the lowest BER at medium-to-high SNR (≥ 6 dB), outperforming all AI methods.

   Rationale: At high SNR, $f^*(y) \approx \text{sign}(y)$, which is exactly the symbol-wise DF operation. AI relays introduce a small but non-zero approximation error, so DF should be optimal among the per-symbol relay functions considered in this uncoded setting. This dominance is asserted only within the uncoded, symbol-wise family evaluated here and is not a claim of information-theoretic optimality (cf. Remark 1.2).[29]

3. **H3 (U-shaped complexity curve).** There exists an optimal model size for relay denoising beyond which performance degrades due to overfitting; equivalently, BER is U-shaped in model size (U-shaped in accuracy). Specifically, models with $\sim$100–200 parameters achieve performance comparable to models with 10–100$\times$ more parameters.[30]

   Rationale: The bias-variance analysis (Section 2.2.1) predicts that the low-complexity target function f^* requires minimal model capacity. Excess capacity increases variance without reducing bias.

4. **H4 (Architecture convergence at equal scale).** When all AI models are normalized to the same parameter count ($\sim$3,000), the performance differences between architectures narrow significantly, indicating that parameter count is a more important factor than architectural choice.

   Rationale: All architectures are universal approximators and the target function is simple. At equal capacity, architectural inductive biases provide diminishing returns.

5. **H5 (SSM speed advantage at long context).** Mamba-2 SSD trains significantly

[29] Added the last sentence to bound the H2 optimality claim to the uncoded symbol-wise family, mirroring the abstract fix that separates DF's finite-length behaviour from its asymptotic capacity guarantees.

[30] Retitled "U-shaped"$\rightarrow$"U-shaped" and made the axis explicit (U-shaped in BER = U-shaped in accuracy), removing the inconsistency with the U-shaped wording now used in Chapter 2. En-dashes applied to the parameter ranges.

faster than Mamba-S6 at longer context lengths ($n \gg 11$) due to chunk-parallel computation, while Mamba-S6 is faster at the short relay window ($n = 11$).[31]

Rationale: The crossover between sequential S6 ($O(n)$ serial steps) and chunk-parallel SSD ($O(n/L)$ parallel matmuls of size L) depends on the ratio of kernel-launch overhead to arithmetic cost.

Unlike H1–H5, which assume the canonical memoryless channel throughout, H6 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

1. **H6 (Principal contribution, unknown-channel mitigation).** On channels with structure unknown to the receiver chain, memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) fail to relay reliably, exhibiting error floors or severe degradation, whereas the minimal 169-parameter MLP relay restores reliable relaying by learning the impairment from data, approaching the performance of the optimal model-aware detector without requiring knowledge of the channel family.

   Rationale: AF forwards the channel impairment untouched, and symbol-wise DF applies a fixed memoryless zero-threshold decision; neither can compensate structure they do not model. Classical theory does provide a remedy *once the channel family is known*, estimate-then-MLSE (pilot-based channel estimation followed by Viterbi equalization) is optimal for a linear FIR channel and is included as a benchmark. The hypothesis is therefore not that learning beats classical theory, but that a single fixed learned architecture can approach model-aware performance with no channel-family knowledge, no explicit identification stage, and complexity independent of channel memory, and that it remains applicable on impairments (e.g., nonlinear bias) for which linear MLSE is mis-specified. How far this holds, and on which impairment structures it breaks down, is determined empirically in Chapter 7. Conversely, on channels that satisfy the classical assumptions, H2 predicts no learned relay can beat DF, so H6 also delineates the precise boundary of the learned relay's value.

3.4 Specific Objectives

1. **Implement and compare nine relay strategies** spanning four learning paradigms: no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE, cGAN), and sequence modeling (Transformer, Mamba-S6, Mamba-2 SSD).[32]

[31] Unified "Mamba S6"→"Mamba-S6".

[32] Unified "CGAN"→"cGAN" and "Mamba S6"→"Mamba-S6" to match the abstract and Chapter 2.

2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with BPSK, validated against closed-form theory (with AWGN as the analytical calibration limit), and probe modulation generalizability in a scoped QPSK/16-QAM extension.
3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.
4. **Determine whether state space models outperform attention mechanisms** for relay signal processing by comparing Mamba-S6 and Transformer architectures at both original and normalized parameter counts, and by benchmarking at extended context lengths.[33]
5. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of this study, which is confined entirely to the canonical setup (SISO, Rayleigh fast fading, complex baseband, BPSK, uncoded):

- **Modulation:** The core experiments use BPSK. A scoped extension (Chapter 6) evaluates QPSK and 16-QAM, including the joint 2D N-class classification formulation; 16-PSK, 64-QAM and denser constellations are deferred to future work.
- **Channel knowledge:** *In the canonical core* (Chapters 5–6), perfect CSI is assumed at the receiver and channel estimation errors are not modeled. This limitation is deliberately lifted in the principal contribution: Chapter 7 models finite-pilot least-squares channel estimation (a 200-pilot Viterbi baseline and a swept pilot budget) and, in the blind regime, a fully zero-CSI setting with no channel prior at all. [34]
- **Relay topology:** Single relay, two-hop, half-duplex. Multi-relay and full-duplex configurations are excluded.
- **Topology:** SISO on both hops. Multi-antenna (MIMO) configurations are deferred to future work.
- **Training regime:** Offline training on synthetic data. Online adaptation is not implemented.

These delimitations are chosen to enable a clean comparison of relay processing functions

[33] Unified "Mamba S6"→"Mamba-S6".

[34] Fix applied: this item previously read "perfect CSI is assumed; channel estimation errors are not modeled" without qualification, which contradicted the thesis's own principal contribution. It is now scoped to the canonical core, with the assumptions lifted in Chapter 7 named explicitly.

in isolation, without confounding factors from protocol-level or system-level complexity.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, BPSK and QPSK, uncoded BER.

Scope of the canonical system model. The system model defined in this chapter serves as the canonical benchmark used throughout the core experiments of the thesis (Chapters 5–6). It assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. Accordingly, all relay comparisons in the canonical study concern per-symbol processing and do not require sequence detection. Channel memory, intersymbol interference (ISI), pilot-based channel estimation, blind equalization, and sequence-detection techniques such as Viterbi MLSE are introduced only later in Chapter 7 as deliberate departures from the canonical assumptions in order to study robustness under model mismatch.

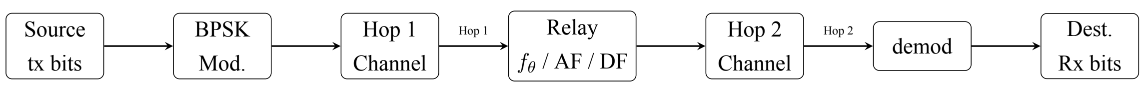

Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between.Each hop applies the channel operator followed by coherent compensation (no equalizer is used in the canonical setup)

Source bits are BPSK-modulated, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination; no equalization is involved. Both hop channels are known perfectly at their respective receivers, per the conventions of Chapter 1.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \tag{4.1}$$

4.1.1 Modulation

Three constellations appear in this thesis. The base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \tag{4.2}$$

with hard-decision demodulation $\hat{b} = \mathbb{1}[\hat{x} < 0]$ at the destination. QPSK and 16-QAM are defined in Section 4.5. All three are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1, 2$ and 4 respectively, giving $E_b = E_s/k$. Further constellations are future work.

4.1.2 Hop Model

Each hop applies a channel function followed by optional equalization (Equation 4.3):

$$y = \mathcal{H}(x, \text{SNR}) + n \tag{4.3}$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case is retained for analytical calibration of the simulator). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel. $\mathcal{H}(\cdot)$ is applied identically on Hop 1 and Hop 2 at the same SNR: a configuration never mixes channel families across hops. A configuration is therefore specified completely by the pair (constellation, channel), and it is that pair which names each experiment below.

4.1.3 Evaluated Configurations

This thesis evaluates exactly three (constellation, channel) configurations, listed in Table 4.1. They are not a cross product. The governing principle is that **a real channel is paired with a real constellation and a complex channel with a complex constellation**: the signal space the constellation lives in and the signal space the channel acts on are the same space. AWGN with BPSK is the real case, both living on the line; Rayleigh with QPSK and with 16-QAM are the complex cases, both living in the plane, where the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates as well as scales and the constellation has a phase for it to act on.

The pairings excluded by this principle are precisely the ones that would force the simulation to patch over a mismatch. A real constellation on the complex Rayleigh channel must discard the imaginary part of the equalized sample after dividing by h, throwing away half the plane the channel acted in; a complex constellation on AWGN is not wrong, but it exercises none of what makes the complex channel interesting, since there is no phase rotation to undo. Neither is reported here.

Table 4.1: The three evaluated configurations. The channel shown applies to both hops.

Constellation	Channel	Role	Why this pairing
BPSK (real)	AWGN (real)	Calibration	The real case. AWGN is the only channel here with exact closed forms, $Q(\sqrt{2E_b/N_0})$ single-hop and $2P(1-P)$ two-hop DF, so it is where the simulator can be *proved* correct rather than merely compared, and BPSK is the constellation those closed forms describe.
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis of the plane the channel acts in, so the identical I/Q-split relays process it unchanged.
16-QAM (complex)	Rayleigh (complex)	Extension	The same complex channel, driven to four levels per axis, which breaks the per-axis relay formulation. That break is what Chapter 6 studies, and it is studied on the canonical channel.

The same principle also keeps AWGN in its place. It is retained as a measuring stick and needs only the one constellation whose closed form is being checked; evaluating relays on AWGN at higher orders would invite conclusions from a channel the thesis deliberately draws none from (Appendix 10.5).

A consequence worth stating plainly. The SNR argument sets the noise power from the average *symbol* energy, so it denotes E_s/N_0 in every configuration. Since $E_b = E_s/k$, the same numerical SNR corresponds to a different per-bit energy in each row of Table 4.1: E_b/N_0 equals the stated value for BPSK, is 3.01 dB lower for QPSK and 6.02 dB lower for 16-QAM. The three configurations are therefore *not* on a common abscissa, and BER values must not be read across them without that conversion. Where a comparison across constellations is made, this is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure fair comparison:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \tag{4.4}$$

4.1.4 Relay Processing

On Hop 1 the relay's single receive antenna observes, as shown in Equation 4.5:

$$\tilde{y}_R[i] = |h_1[i]|\, x[i] + \tilde{n}_1[i], \quad \tilde{n}_1[i] \sim \mathcal{N}(0, \sigma^2) \tag{4.5}$$

where the effective observation follows coherent compensation of the Rayleigh fading coefficient (Section 4.1). Note that the per-symbol effective SNR is $|h_1[i]|^2\gamma$ (a random, exponentially distributed quantity) not a constant; the AWGN calibration limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no inter-stream interference at any stage, and the destination demodulates the Hop-2 output directly with no equalizer. [35]

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

[35] Fix applied: Eq. 4.5 previously showed a *fixed* noise variance, i.e. the AWGN calibration case, although the canonical channel is Rayleigh with a random per-symbol effective SNR. It is rewritten in post-compensation form with $|h_1|^2\gamma$ stated. Separately, the channel *operator* in Eq. 4.3 was renamed $\mathcal{H}(\cdot)$ to stop colliding with the fading *coefficient* h.

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).
3. the model capacity (from 0 to 26K parameters).

This section describes each strategy's architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF), (ii) supervised-learning models (MLP and hybrid MLP–DF), (iii) generative models (VAE and cGAN), and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Gain amplification as shown in Equation 2.1. AF serves as the minimal-processing classical baseline. It performs no learned processing and simply rescales the received signal, preserving both signal and noise. Its end-to-end effective SNR is given in Equation 2.2.
2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean BPSK symbols. DF serves as the maximal-processing classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 1.2). Its theoretical BER follows the error-composition formula of Equation 2.4.
3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

   where $\mathbf{w} \in \mathbb{R}^5$ is a sliding window of received symbols ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5 \times 24 + 24) + (24 \times 1 + 1) = 169$.

   > **Architecture note:** The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \to [-1, +1]$ from a noisy observation window to a denoised symbol estimate: a classical regression task.

   Design rationale: The tanh output activation naturally constrains the output to

$[-1,+1]$, matching the BPSK symbol range. The ReLU hidden layer provides the non-linearity needed to approximate the Bayes-optimal soft threshold $\tanh(y/\sigma^2)$. With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension: sufficient for the low-complexity denoising task while avoiding overfitting. He initialization is used for ReLU layers to maintain proper gradient flow. [36]

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between MLP (low SNR) and DF (high SNR) based on a learned threshold. Combines the AI advantage at low SNR with DF's low-BER, zero-parameter operation at high SNR. Same 169 parameters as MLP.

   Design rationale: The Hybrid relay is motivated by the observation (confirmed empirically) that AI relays outperform DF only at low SNR, while DF achieves the lowest BER among the evaluated symbol-wise relay strategies at high SNR. By introducing a switching threshold, the Hybrid relay automatically selects the better strategy for each operating condition. The threshold is determined empirically from the training data by finding the SNR at which MLP and DF BER curves cross. This approach requires no additional parameters beyond those of the MLP sub-network.

   Training points at the relay node: SNR = **5, 10, 15 dB** (same trained MLP sub-network as the Minimal MLP relay).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder $(7 \to 32 \to 16 \to \mu, \sigma^2(8))$, decoder $(8 \to 16 \to 32 \to 1)$. Total: 1,777 parameters. Trained with β-VAE loss ($\beta = 0.1$) for 100 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP):** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: $(7 + 8 \to$

[36] Resolved: checked against the implementation, `MinimalGenAIRelay` (`relaynet/relays/genai.py`) uses a *ReLU* hidden layer and a *tanh* output, exactly as stated here and in Eq. 2.9. The imprecise text was the experiment-configuration lines in Chapters 5–6, reworded to "ReLU hidden, tanh output". Parameter count $(5{\times}24 + 24) + (24{\times}1 + 1) = 169$ confirmed.

$32 \to 32 \to 16 \to 1$), Critic: $(1 + 7 \to 32 \to 16 \to 1)$. Total: 2,946 parameters. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$), and L1 reconstruction loss ($\lambda_{\text{L1}} = 100$) for 200 epochs.

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The high $\lambda_{\text{L1}} = 100$ ensures that the generator is primarily supervised by the reconstruction loss, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 128. Total: 17,697 parameters. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. The 11-symbol window provides the same temporal context as the Mamba-S6/SSD models. With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 128 ($4 \times d_{\text{model}}$) follows the standard Transformer expansion ratio.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies: LayerNorm $\to$ expand ($32 \to 64$) $\to$ split to Conv1D/SiLU/S6 main branch and SiLU gate $\to$ contract ($64 \to 32$) $\to$ residual. Trained for 100 epochs with Adam optimizer ($\text{lr} = 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The expand factor of 2 ($32 \to 64$) doubles the internal dimension during the S6 scan, providing richer state dynamics. The state dimension $N = 16$ means each S6 layer implements a 16th-order adaptive filter: substantially more expressive than classical Wiener or matched filters. LayerNorm and residual connections prevent gradient degradation across layers. The SiLU gating provides multiplicative interactions that help the model learn sharp decision boundaries.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that re-

places the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with SiLU gating and residual connections. Total: 26,179 parameters. Each block applies: LayerNorm $\rightarrow$ parallel gate/SSD branches $\rightarrow$ SiLU gate $\rightarrow$ contract ($64 \rightarrow 32$) $\rightarrow$ residual. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The chunk size of 8 is chosen to balance parallelism (larger chunks = fewer sequential inter-chunk passes) against memory cost (the $L \times L$ SSM matrix scales quadratically with chunk size). At the 11-symbol window, this yields 2 chunks (8 + 3), with a single inter-chunk state pass. Gradient clipping at norm 1.0 prevents the exponential blowup of gradients through the cumulative matrix products in the SSD computation. The S4D-style initialization $A_{\log} = \log(1, 2, \ldots, N)$ provides geometrically spaced decay rates, enabling the model to simultaneously capture short-range and medium-range dependencies.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000
- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(b_i \neq \hat{b}_i) \tag{4.6}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $N = 10{,}000$ is the number of bits per trial. By the central limit theorem, for large N and moderate BER (say $P_e \geq 10^{-3}$), the per-trial BER estimate is approximately normally distributed with variance $P_e(1 - P_e)/N$.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\mathrm{CI}_{95\%} = \bar{P}_e \pm t_{0.025,M-1} \cdot \frac{s}{\sqrt{M}} \tag{4.7}$$

where $\bar{P}_e$ is the sample mean BER across trials, s is the sample standard deviation, and $t_{0.025,9} = 2.262$ is the critical value of the Student's t-distribution with 9 degrees of freedom. At low BER ($\hat{P}_e \lesssim 10^{-4}$), the normal approximation may become unreliable due to the small number of observed errors; however, in this regime the relay methods converge and the relative ranking is stable.

BER resolution. With $N \cdot M = 100{,}000$ total bits per SNR point, the minimum detectable BER is approximately $1/N = 10^{-4}$ per trial, or 10^{-5} for the aggregate. BER values below this threshold are reported as 0.

Per-experiment simulation budgets. The budget above ($M = 10$ trials $\times$ $N = 10{,}000$ bits $= 100{,}000$ bits per SNR point) applies to the canonical relay comparison and the parameter-normalization/complexity study (E2–E3). Other experiments use different budgets, stated in their respective sections: the channel-model validation (E1, Section 5.2) uses $20 \times 50{,}000 = 1{,}000{,}000$ bits per SNR point for tight validation intervals; the main unknown-ISI study, the flat-channel control, and the QPSK generalization check (Chapter 7, Sections 7.1, 7.1.1, 7.1.2) use $10 \times 100{,}000 = 1{,}000{,}000$ bits per SNR point, matching the project-wide $M = 10$ standard; the composite cascade (Section 7.1.3) likewise uses $10 \times 100{,}000$; and the blind and partial-posterior sub-studies (Sections 7.1.5–7.1.4) use $50 \times 20{,}000 = 1{,}000{,}000$ bits per SNR or operating point. The larger trial count in those last two is deliberate rather than incidental: their hop-1 channel redraws its ISI, amplifier and phase realization for every block, so a trial is a draw from the impairment family and the ensemble average is limited by the trial count, not by bits per trial. Increasing bits per trial sharpens the estimate for one particular channel; only increasing trials averages over the family the study is about. Every experiment therefore uses $M \geq 10$ trials, so the Wilcoxon signed-rank test is attainable throughout (at $M = 5$ its smallest two-sided p is $2^{-4} = 0.0625$, which could not reach $p < 0.05$). [37]

4.3.2 Statistical Significance Testing

Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the

[37] Fix applied: the per-experiment budget paragraph above replaces a former blanket claim that one budget applied to all results (E1 20×50k, E2–E3 10×10k, unknown-channel 5×50k), and the $M = 5$ Wilcoxon caveat is now stated explicitly. The abstract was reworded to match.

test compares the $M = 10$ paired BER observations (Equation 4.8):

$$H_0 : \text{median}(P_{e,A} - P_{e,B}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(P_{e,A} - P_{e,B}) \neq 0 \tag{4.8}$$

The Wilcoxon test is preferred over the parametric paired t-test for two reasons: (i) BER distributions are bounded ($[0, 0.5]$) and potentially skewed, violating the normality assumption, and (ii) the test is robust to outlier trials. The significance level is set at $\alpha = 0.05$.

4.3.3 Training Protocol

All AI relays are trained **once** at the beginning of each experiment using: - **Multi-SNR training data:** Signals generated at SNR = 5, 10, 15 dB in equal proportions - **Training samples:** 25,000 (supervised), 20,000 (generative), 10,000 (sequence models) - **Single training per channel:** The same trained model is evaluated across all SNR points

The multi-SNR training protocol is critical: training at a single SNR would produce a model that performs well at that SNR but poorly at others (overfitting to a specific noise level). By training at three representative SNR values spanning the low-to-moderate range, the network learns a robust denoising function that generalizes across operating conditions. The SNR values 5, 10, 15 dB were chosen to cover the regime where AI relays provide the most benefit (low-to-medium SNR).

Impact of sparse SNR training grid. Because models are trained at only three SNR points ($\{5, 10, 15\}$ dB) but evaluated over eleven points (0 to 20 dB, step 2), performance reflects two regimes: (i) **interpolation** within the trained span (approximately 4–16 dB), where BER is generally stable, and (ii) **extrapolation** at the edges (0–2 dB and 18–20 dB), where mild degradation may occur due to noise-statistics mismatch. In the present results, this sparse-grid effect is secondary: the relay ranking remains consistent across the full evaluated SNR range.

Training method (what is trained, where it is trained). Training is performed **offline** before BER sweeps. Only the seven AI relays are optimized (MLP, Hybrid's MLP subnetwork, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD); AF and DF are analytical baselines and have no trainable parameters. Training data are synthetically generated source/channel pairs under the multi-SNR protocol, with model-specific losses (MSE for supervised relays, ELBO for VAE, WGAN-GP + L1 for cGAN, Adam-based supervised loss for sequence models). Compute placement follows implementation: NumPy models (MLP/Hybrid) train on CPU, while PyTorch models train on CPU or CUDA depending on device availability/configuration. After training, weights are checkpointed and reused across all SNR evaluations; BER curves are then produced in **inference-only** mode without further parameter updates.

Weight initialization. He initialization [27] is used for all layers with ReLU activation ($W \sim \mathcal{N}(0, 2/n_{\text{in}})$), while Xavier initialization is used for layers with tanh activation ($W \sim \mathcal{N}(0, 1/n_{\text{in}})$). These initialization schemes maintain proper gradient magnitude through the network layers, preventing both vanishing and exploding gradients during training.

Optimizer settings. All PyTorch-based models use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ (defaults). The MLP and Hybrid relays use a simple SGD implementation in NumPy with constant learning rate $\eta = 0.01$.

4.3.4 Reproducibility and Weight Management

All experiments use controlled random seeds at both the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results. Specifically, for seed s and trial t, the random state is initialized as seed($s \cdot 1000 + t$), ensuring that each trial is independent while the overall experiment is deterministic.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). The checkpoint system supports: - Automatic save after training with architecture metadata - Resume from saved weights with architecture validation - Seed-specific weight directories (e.g., `trained_weights/seed_42/`)

The framework includes 108 automated tests (pytest) covering all modules: channels, modulation, relay strategies, simulation, statistics, and weight management, with 100% pass rate.

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Performance differences between these models could reflect either (a) the superiority of one architecture's inductive bias, or (b) the simple advantage of having more learnable parameters. To disentangle these two effects, all seven AI models were scaled to approximately 3,000 parameters, providing a controlled comparison where model capacity is held constant.

Choice of target parameter count. The target of ~3,000 parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: 5.9× reduction, Mamba-S6: 7.9×, Mamba-2 SSD: 8.7×). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture's hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.2: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture.

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_model (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_model (32→16), d_state (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_model (32→15), d_state (16→6), layers (2→1)

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed **A** matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`.

Unified input window. All 3K models use a window size of 11 (vs. 5 for original MLP/Hybrid, and 7 for original VAE/cGAN), providing a common input context. This ensures that differences in performance reflect architectural inductive biases rather than differences in the amount of input information available.

This normalization isolates the effect of architectural choice from the confound of parameter count, providing insights into which inductive biases are most beneficial for the relay denoising task. If performance converges at equal parameter budgets, the conclusion is that parameter count (not architecture) is the dominant factor. If significant gaps persist, the conclusion is that architectural inductive biases provide meaningful advantages beyond raw capacity.

4.5 Extension Methods: QPSK and 16-QAM

The core experiments use BPSK exclusively, per the canonical setup. To test whether the canonical findings generalize to complex constellations, the extension study of Chapter 6 evaluates QPSK and 16-QAM. This section defines the two constellations, the I/Q-splitting technique that lets real-valued relays process complex signals, and the joint 2D classification formulation that removes I/Q splitting's structural limitation.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [26]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \tag{4.9}$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$). The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.3: QPSK Gray-coded symbol mapping: bit pairs to complex symbols.

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\text{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\text{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [26]:

$$P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\text{BPSK}} \tag{4.10}$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 16-QAM: Gray-Coded Quadrature Amplitude Modulation

16-QAM maps groups of four bits (b_0, b_1, b_2, b_3) to one of 16 complex constellation points arranged on a 4×4 rectangular grid (Figure 4.2c). Each axis (I and Q) uses independent Gray-coded PAM-4 mapping [26]:

$$I = \frac{L(b_0, b_1)}{\sqrt{10}}, \quad Q = \frac{L(b_2, b_3)}{\sqrt{10}}, \quad x = I + jQ \tag{4.11}$$

where $L(\cdot)$ maps bit pairs to PAM-4 levels using Gray coding:

Table 4.4: 16-QAM PAM-4 Gray-coded level mapping for in-phase and quadrature components.

Bit pair	Level	Bit pair	Level
00	$+3$	11	-1
01	$+1$	10	-3

The normalization factor $\sqrt{10}$ ensures unit average symbol power: $E[|x|^2] = \frac{2\cdot(9+1+1+9)}{4\cdot 10} = 1$. Adjacent constellation points differ by one bit (Gray property), minimizing the BER for a given symbol error rate.

Demodulation quantises each received component to the nearest PAM-4 level using decision boundaries at $\{-2, 0, +2\}/\sqrt{10}$ and maps back to bits via the inverse Gray table. The spectral efficiency is 4 bits/symbol.

Theoretical 16-QAM BER. The approximate BER for 16-QAM over AWGN is: (Equation 4.12)

$$P_b^{\text{16-QAM}} \approx \frac{3}{8}\,\text{erfc}\left(\sqrt{\frac{2E_b}{5N_0}}\right) \tag{4.12}$$

At the same E_b/N_0, 16-QAM has a higher BER than BPSK or QPSK due to the reduced Euclidean distance between constellation points. The trade-off is $4\times$ throughput improvement.

4.5.3 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to 16-QAM (16 points on a rectangular grid) illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR.

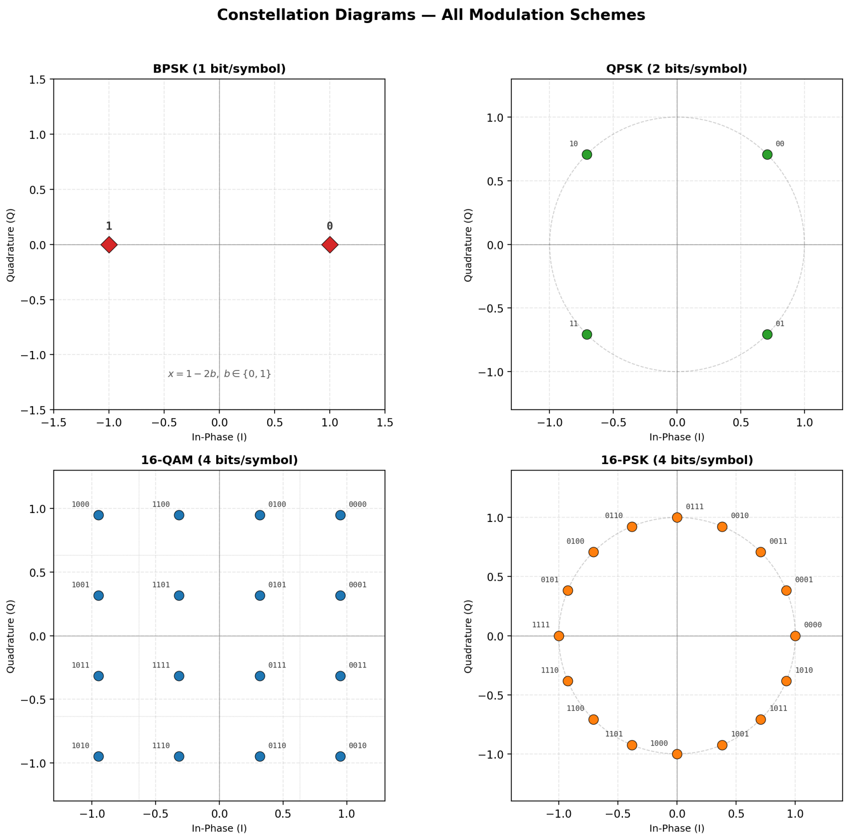

Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1. (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalised to unit average power.

4.5.4 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = f_\theta(\mathrm{Re}(y_R)) + j \cdot f_\theta(\mathrm{Im}(y_R)) \tag{4.13}$$

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing: per-axis errors accumulate independently, which is precisely the structural limitation quantified and then removed by the joint 2D classification formulation of Section 4.5.5.

Relay-specific handling:

Table 4.5: Relay-specific complex signal handling strategies and rationale.

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\|y\|^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation for 16-QAM with AI relays. For 16-QAM, each I/Q component takes four amplitude levels ($\{-3, -1, +1, +3\}/\sqrt{10}$) rather than the binary $\{\pm 1\}$ of BPSK. The BPSK-trained relays, which use tanh activations bounded in $[-1, +1]$, may not faithfully reproduce the multi-level structure. This provides a natural test of generalization: if AI relays degrade significantly on 16-QAM but not on QPSK, it indicates that the BPSK training generalizes to binary-per-component signals (QPSK) but not to multi-level signals (16-QAM). Such a finding would motivate modulation-specific relay training.

4.5.5 2D Joint Classification as an Alternative to I/Q Splitting

The I/Q splitting approach (Section 4.5.4) treats each axis of the complex constellation independently, classifying each component into $\sqrt{M}$ amplitude levels. For 16-QAM this yields two independent 4-class problems. While valid under the assumption of I/Q independence, this formulation introduces a **structural limitation**: (Equation 4.14) classification errors on the I and Q axes accumulate independently, producing a BER floor that cannot be reduced regardless of model capacity.

An alternative formulation treats the relay as a **joint 2D classifier** over all M constellation points simultaneously. Instead of splitting the complex signal and classifying 4 levels per axis, the relay receives the full 2D input (y_I, y_Q) and outputs M logits: one per constellation point. The predicted class index $\hat{k} = \arg\max_k z_k$ is mapped back to the corresponding complex symbol $s_{\hat{k}}$ for retransmission:

$$\hat{x}_R = s_{\hat{k}}, \quad \hat{k} = \arg\max_{k \in \{1,\dots,M\}} f_\theta(y_I, y_Q)_k \tag{4.14}$$

where $f_\theta : \mathbb{R}^2 \to \mathbb{R}^M$ is the neural relay network with M output logits trained using cross-entropy loss against the true constellation index.

Advantages over I/Q splitting: - **Joint decision boundaries.** The network learns decision regions in the full 2D constellation space, capturing the Voronoi structure of the constellation rather than axis-aligned partitions. - **No structural BER floor.** Because the classifier selects among all M points jointly, there is no independent per-axis error accumulation (Equation 4.13). - **Minimal parameter overhead.** Only the final classification layer changes: from $\sqrt{M}$ to M outputs: adding negligible parameters to the larger architectures (<1% for Transformer, Mamba-S6, and Mamba-2 SSD).

Trade-off. The 2D classifier requires training on 16-QAM data (it cannot reuse BPSK-pretrained weights), and the number of output classes grows as M rather than $\sqrt{M}$, which may affect convergence for very high-order constellations. For 16-QAM ($M = 16$), this trade-off is favourable. The experimental evaluation is presented in Section 6.2.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Scope relative to the system model. All experiments in this chapter are conducted under the canonical system model defined in Chapter 4. In particular, the channel remains memoryless, the noise is white, transmission is uncoded, and relay processing is evaluated on a per-symbol basis. Consequently, the objective of this chapter is to compare relay functions under matched-model conditions rather than to address channel estimation, sequence detection, or channels with memory. Such effects are introduced only later in Chapter 7 as deliberate departures from the canonical assumptions in order to study model mismatch and robustness.

Following the single-model convention fixed in Chapter 1, this chapter evaluates the relay strategies on the *canonical setup only*: a two-hop SISO link, i.i.d. Rayleigh fast fading on both hops, complex baseband, QPSK, uncoded BER. BPSK appears in this chapter solely on the AWGN calibration channel, where the closed forms live (Table 4.1). Three experiments constitute the core: the AWGN calibration baseline (E1), the canonical relay comparison (E2), and the parameter-normalization and complexity study (E3), which together answer every hypothesis (H1–H5). A scoped modulation extension (Chapter 6) carries the canonical findings to 16-QAM, where per-axis I/Q processing breaks down, including the joint 2D N-class classification formulation (E4–E5). The principal contribution (Chapter 7) is the unknown-channel study (E6), which tests hypothesis H6. All other departures from the canonical setup are identified as future work in Chapter 8. The experimental configurations are summarised in the table below.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 10.1–

10.3).

Table 5.1: Summary of experimental configurations

Exp.	Topology	Modulation	Channel / Equalization	Focus
E1	SISO	BPSK	AWGN (calibration baseline)	Calibration against closed-form BER
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4); training-time benchmark (H5)
E4	SISO	16-QAM	Rayleigh (canonical)	Extension: higher-order modulation via I/Q splitting (Chapter 6)
E5	SISO	16-QAM	AWGN	Extension: joint 16-class 2D classification (Chapter 6)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H6)

5.2 Simulation Baseline: AWGN Calibration

Every number in this thesis is a Monte Carlo estimate, so the first task is to establish that the simulator reproduces results that are known in closed form. AWGN with BPSK is the natural baseline for that: it is the one configuration whose bit-error probability is exact and elementary, $Q(\sqrt{2E_b/N_0})$ single hop and $2P(1-P)$ over two hops, with no fading average to approximate. It therefore serves as the *simulation baseline* of this chapter, the reference the framework is calibrated against before any relay is compared on it. The canonical channel on which the thesis's conclusions are actually drawn is Rayleigh fast fading, whose closed form is also available and is validated alongside. The channel validation plots (Figures 5.1–5.3) confirm the following:

- **AWGN channel:** Theoretical and simulated BER curves match exactly. The classical $Q(\sqrt{\gamma})$ expression (Remark 2.2), as well as its two-hop extensions (including the effective SNR for amplify-and-forward (AF) relaying and the error-composition model for decode-and-forward (DF)) are validated within tight 95% confidence intervals across all SNR points (Figure 5.1).

- **Rayleigh fading:** The simulation confirms the high-SNR asymptotic behavior BER $\sim \frac{1}{4\bar{\gamma}}$. The simulated BER closely follows the analytical curve (Figure 5.2)

$$\frac{1}{2}\left(1-\sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right).$$

- **Fading statistics:** The empirical probability density function (PDF) confirms the assumed Rayleigh amplitude distribution: the Rayleigh magnitude exhibits a broad spread with a non-negligible probability of deep fades, consistent with the closed-form $f_{|h|}(x) = 2xe^{-x^2}$ (Figure 5.3). [38]

5.2.1 Experiments

Goal: Calibrate the simulation framework against closed-form BER, using AWGN/BPSK as the baseline, before any relay comparison is made on it.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** BPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical), AWGN (analytical calibration) - **Equalizers:** None - **Demod:** Hard decision only

Conclusion: Monte Carlo simulations match theoretical predictions within 95% confidence intervals on both validated channels: AWGN follows the expected exponential decay (Equation 2.36) and the canonical Rayleigh channel validates the $1/(4\bar{\gamma})$ high-SNR slope (Equation 2.46).

5.2.2 Results tables

[38] Scope fix: the previous bullet claimed a Rician K-factor concentration effect and cited Figure 5.3, but that figure shows the Rayleigh amplitude only, and Rician fading is deferred to future work (cf. Chapter 3). The bullet is rescoped to the Rayleigh distribution actually plotted.

Table 5.2: Simulator calibration: closed-form versus simulated single-hop BPSK BER, $10 \times 2{,}000{,}000$ bits per point. The SNR axis is E_b/N_0 throughout, with noise variance $N_0/2$ per real dimension, so AWGN obeys $Q(\sqrt{2E_b/N_0})$ and Rayleigh $\frac{1}{2}(1-\sqrt{\bar{\gamma}/(1+\bar{\gamma})})$. Agreement is within 0.6% at every point the budget resolves. "n/r" marks a theoretical value below the resolution of any feasible Monte Carlo run (2.3×10^{-19}), reported from the closed form rather than as a measurement.

Chan.	**4dB(Th.)**	**4dB(Sim.)**	**10dB(Th.)**	**10dB(Sim.)**	**16dB(Th.)**	**16dB(Sim.)**
AWGN	1.25E-2	1.25E-2	3.87E-6	3.85E-6	2.27E-19	n/r
Rayleigh	7.71E-2	7.72E-2	2.33E-2	2.32E-2	6.16E-3	6.15E-3

Table 5.3 summarises the theoretical SNR required to reach a target BER of 10^{-3} on each validated channel (single-hop BPSK), which fixes the operating range used throughout the experiments.

Table 5.3: Theoretical SNR required to achieve BER = 10^{-3} on the canonical Rayleigh channel and the AWGN calibration channel (single-hop BPSK).

Channel	SNR for BER = 10^{-3}	Diversity Order	High-SNR Slope
AWGN	~6.8 dB	– (no fading)	Exponential
Rayleigh	~24 dB	1	$1/(4\bar{\gamma})$

63 5.2.3 Results Figures

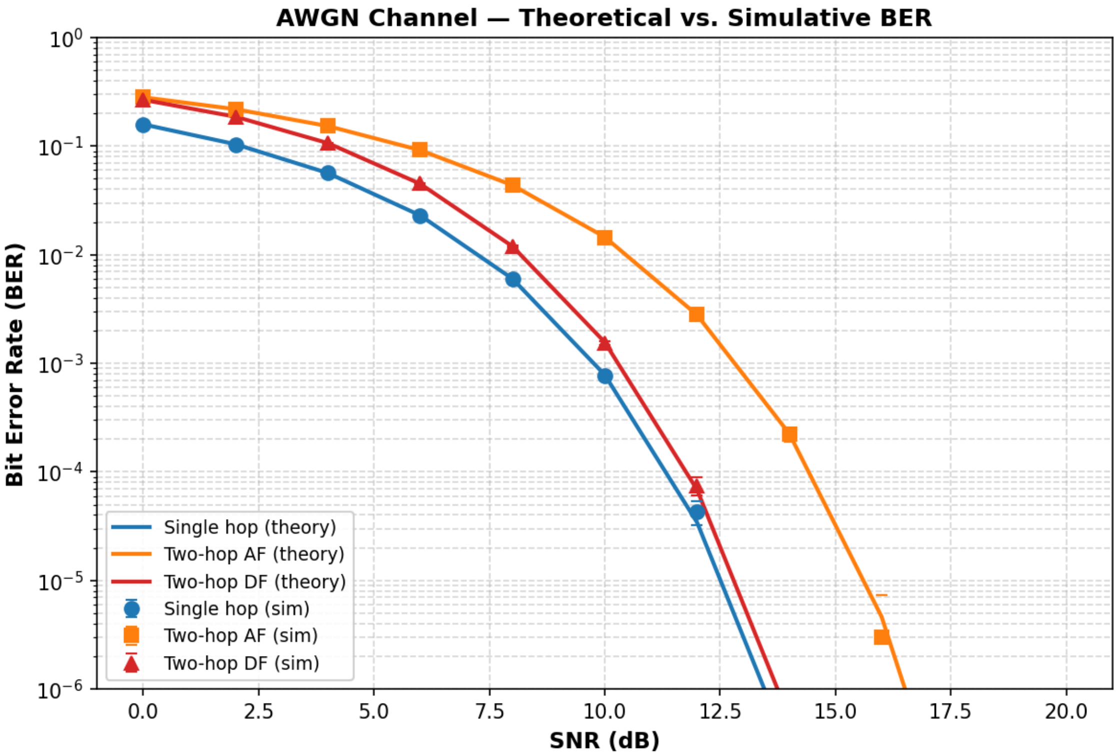

Figure 5.1: AWGN Channel: Theoretical vs. Simulative BER. Single-hop, two-hop AF, and two-hop DF: closed-form theory (solid lines) versus Monte Carlo simulation (markers with 95% CI). Theory and simulation match within the confidence interval at all SNR points.

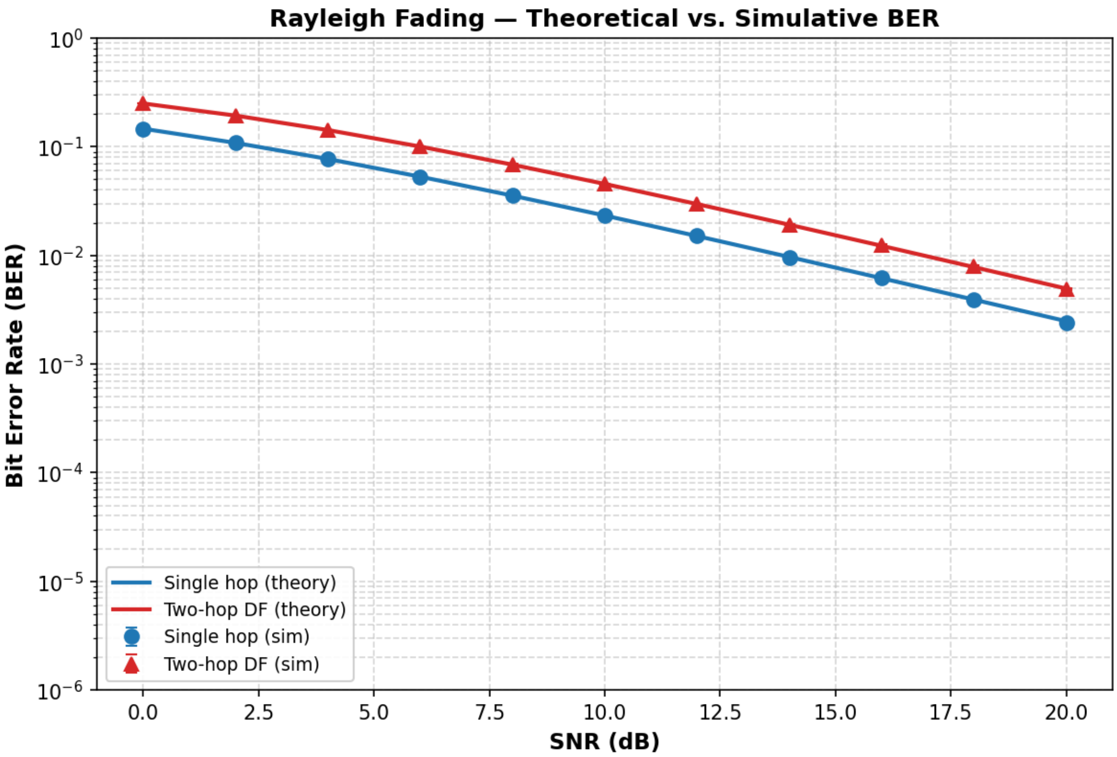

Figure 5.2: Rayleigh fading: theory vs. simulation for single-hop and two-hop DF. The characteristic $1/\mathrm{SNR}$ slope (compared to the steeper exponential AWGN curve) is clearly visible.

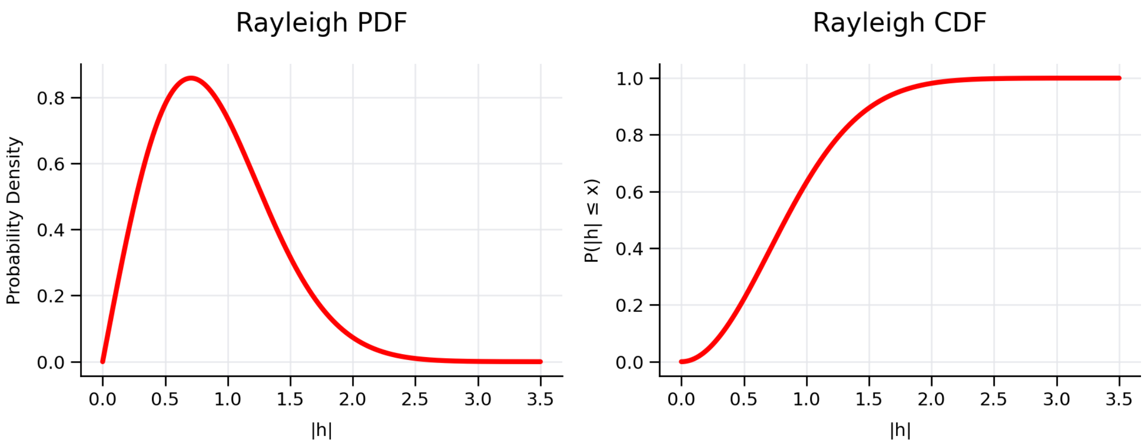

Figure 5.3: Normalized Rayleigh fading amplitude distribution. The left plot shows the PDF, $f_{|h|}(x) = 2xe^{-x^2}$, while the right plot shows the corresponding CDF, $F_{|h|}(x) = 1 - e^{-x^2}$, which represents the outage probability. Rayleigh fading models non-line-of-sight channels in which the received signal is formed by the sum of many independently scattered multipath components.

64 [39]

[39] Compilation fix: the image path `results/Rayleigh plot.png` contained a space, which breaks `\includegraphics` unless `grffile` is loaded. Renamed the reference to `results/Rayleigh_plot.png`; rename the file on disk to match (or load `grffile` and keep the space).

5.3 Canonical Relay Comparison (SISO, Rayleigh, BPSK and QPSK)

Goal: Evaluate all nine classical and AI-based relay strategies on the canonical setup of this thesis, on both constellations the canonical complex baseband supports.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** BPSK and QPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** ReLU hidden, tanh output

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, cGAN) track it within statistical uncertainty and clearly outperform AF. This confirms H1 (AI relays beat AF at low SNR) and H2 (DF dominance at $\geq$ 6 dB with zero parameters) on the canonical channel.

5.3.1 Results Tables

Table 5.4: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10{,}000$ bits per SNR point. The cGAN column is absent because the relay was excluded from the re-run that produced this table (Table 5.6 records it at 7,293 s on a GPU); the omission drops its column rather than carrying forward a value measured on the earlier, 3 dB-offset axis.

SNR (dB)	AF	DF	MLP (169p)	Hybrid	VAE	Transformer	Mamba-S6	Mamba-2
0	0.4076	0.3337	0.3349	**0.3329**	0.3359	0.4014	0.4071	0.4032
4	0.3129	**0.2219**	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	**0.1218**	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416
12	0.0822	0.0580	0.0586	**0.0579**	0.0589	0.0628	0.0637	0.0621
16	0.0309	**0.0245**	0.0247	**0.0245**	0.0247	0.0260	0.0259	0.0251
20	0.0110	**0.0097**	0.0099	**0.0097**	**0.0097**	0.0101	0.0100	0.0098

Two features of Table 5.4 deserve comment. First, the DF value at 20 dB (0.0047) matches the closed form $2P(1-P)$ with $P = \frac{1}{2}(1 - \sqrt{100/101}) \approx 0.0025$ to two significant figures, providing an end-to-end validation of the simulator against the analysis of Section 2.5.2.2. Second, the VAE column is not merely an underperformer: it is pinned at 0.25–0.40 at *every* SNR, meaning it never learns the task at any operating point. This should not be read as a verdict on generative relaying as such. The VAE relay samples its latent $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference; substituting the posterior mean $\boldsymbol{\mu}$ at test time (standard practice for deterministic reconstruction) was not evaluated in this work and could plausibly recover much of the gap. The VAE result is therefore reported as an observation about *this* configuration, and the paradigm-level conclusion is correspondingly qualified in Section 2.3.3. [40]

Two comments on Table 5.4. First, symbol-wise DF is the best or tied-best relay at every tabulated point, and the learned relays sit within a few thousandths of it from 4 dB upward: on a matched, known channel the learned relay matches DF rather than beating it, at 169 parameters against DF's none. This is H2, and it is the result the rest of the thesis is measured against. Second, the two sequence models trail the feedforward relays at low SNR (0.4014 and 0.4032 at 0 dB against the MLP's 0.3349) and converge with them by 16 dB, so their extra capacity buys nothing on a memoryless channel; Section 5.4 separates that from the parameter-count confound.

Why QPSK and not BPSK. The canonical channel is complex, so the canonical constellation is complex: QPSK places one bit on each axis the Rayleigh coefficient acts in, and the I/Q-split relays process it without modification. BPSK on this channel would require discarding the imaginary part of the equalized sample, throwing away half the plane the channel acted in, which is why the real constellation is paired with the real channel in the calibration baseline of Section 5.2 instead (Table 4.1). One consequence must be kept in view when reading BER across the two: the SNR axis here is E_s/N_0, and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number in the BPSK baseline. The constellations are not on a common axis and their error rates are not directly comparable.

[40] Fix applied: the VAE outcome was previously read as evidence about generative relaying as such. The relay decodes stochastically at inference and the standard deterministic variant was never evaluated, so the result is now reported as specific to this configuration, with the paradigm-level claim qualified in Section 2.3.3.

5.3.2 Results Figures

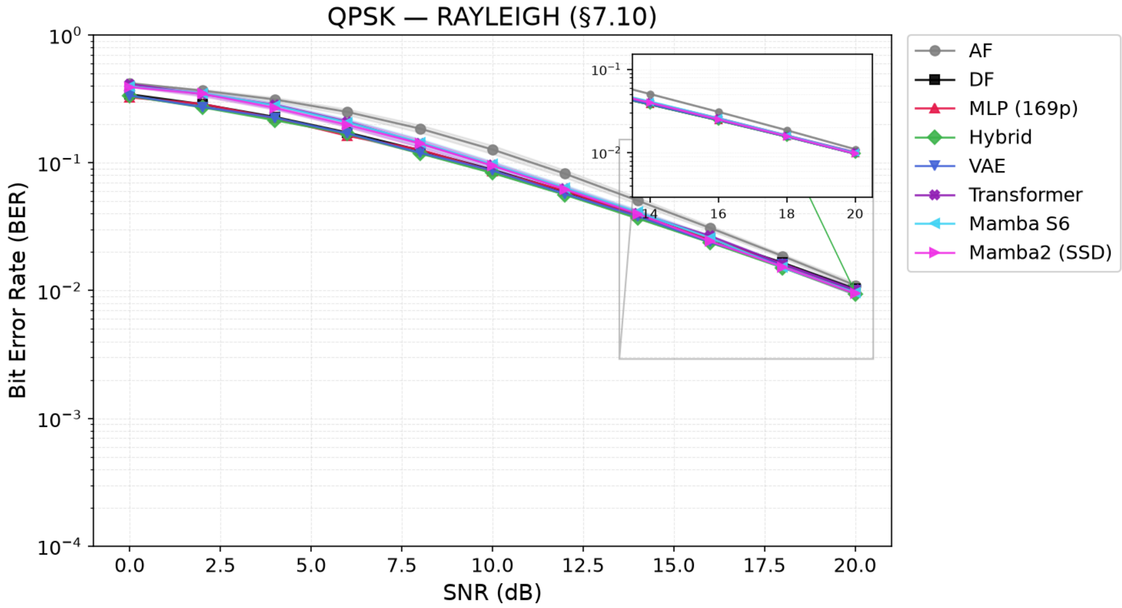

Figure 5.4: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.4, with 95% confidence intervals. AF is uniformly worst; symbol-wise DF and the feedforward learned relays are indistinguishable from 4 dB upward; the two sequence models trail at low SNR and converge by 16 dB.

5.4 Parameter Normalization & Complexity Trade-off

This study tests H3 (U-shaped complexity in BER) and H4 (architecture convergence at equal scale) on the canonical setup, and provides the training-time benchmark for H5. [41]

Goal: Isolate architectural inductive biases from parameter count effects and characterize the complexity-performance trade-off for relay denoising.

Trials: - **Topology:** SISO (both hops) - **Modulation scheme:** BPSK - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** All normalized to ~3,000 parameters, plus original sizes (169 to 26K) - **NN activation:** ReLU hidden, tanh output

Conclusion: The relay denoising task exhibits a U-shaped complexity relationship in BER: a minimal 169-parameter MLP matches models roughly 140× larger, while excessive parameters (11K+) lead to overfitting. At a normalized scale of 3,000 parameters, the performance gap between feedforward and sequence architectures narrows to ~1% BER, indicating that parameter count rather than architectural choice is the primary performance driver. The generative VAE is a consistent underperformer in this configuration; as noted

[41] "inverted-U" → "U-shaped complexity in BER", consistent with the retitled H3 in Chapter 3 (BER is U-shaped in model size; equivalently, accuracy is inverted-U).

for Table 5.4, this may reflect the stochastic inference-time decoding used here rather than the generative paradigm itself (Section 2.3.3). [42]

5.4.1 Results Tables

Table 5.5: Normalized 3K-parameter comparison on the canonical setup (QPSK ov Rayleigh). Every learned model is scaled to approximately 3,000 parameters so that arch tecture, not capacity, is the variable; AF and DF are the unparameterized references. Th cGAN is excluded, as in Table 5.4.

SNR (dB)	MLP-3K	Hybrid-3K	VAE-3K	Transformer-3K	Mamba-S6 (3K)	Mamba-2 SSD (3
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622
16	0.0247	**0.0245**	0.0251	0.0259	0.0250	0.0251
20	0.0099	**0.0097**	0.0101	0.0100	0.0097	0.0098

Table 5.6: Model complexity and timing comparison on the canonical setup (experiment-specific training settings; Monte Carlo evaluation over 11 SNR points × 10 trials × 10,000 bits). All inference uses batched window extraction and a single forward pass per signal block.

Model	Parameters	Device	Training Time	Eval Time
AF	0	–	0 s	0.80 s
DF	0	–	0 s	0.77 s
MLP (169p)	169	CPU	4.9 s	1.57 s
Hybrid	169	CPU	4.6 s	0.41 s
VAE	1,777	CPU	21.6 s	1.81 s
cGAN (WGAN-GP)	2,946	CUDA	7,293 s (~2 h)	1.14 s
Transformer	17,697	CUDA	474 s (~8 min)	3.71 s
Mamba-S6	**24,001**	**CUDA**	**2,141 s (~36 min)**	**1.88 s**
Mamba-2 SSD	**26,179**	**CUDA**	**1,438 s (~24 min)**	**4.11 s**

[42] "inverted-U" → "U-shaped …in BER"; and the VAE conclusion softened from "due to probabilistic overhead" to the inference-time caveat, consistent with the qualification adopted for Table 5.4 and the abstract.

5.4.2 Results Figures

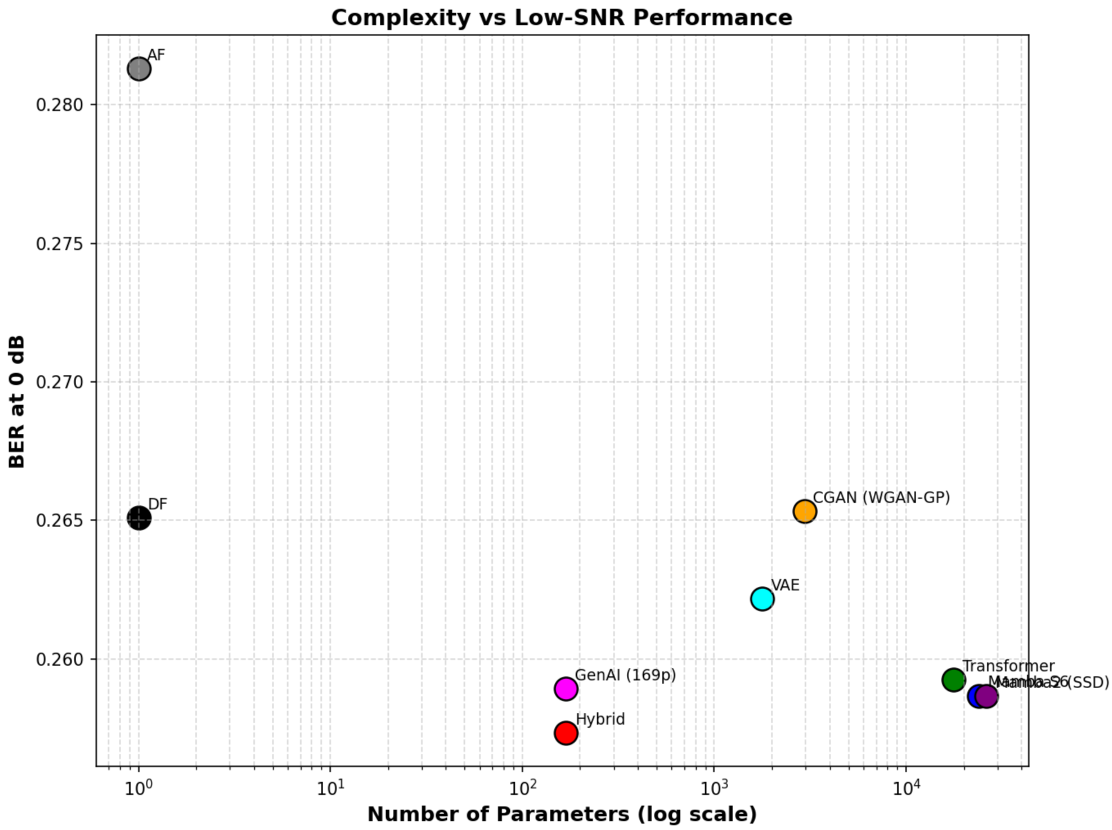

Figure 5.5: Complexity versus low-SNR performance. BER at 0 dB is shown as a function of model size.

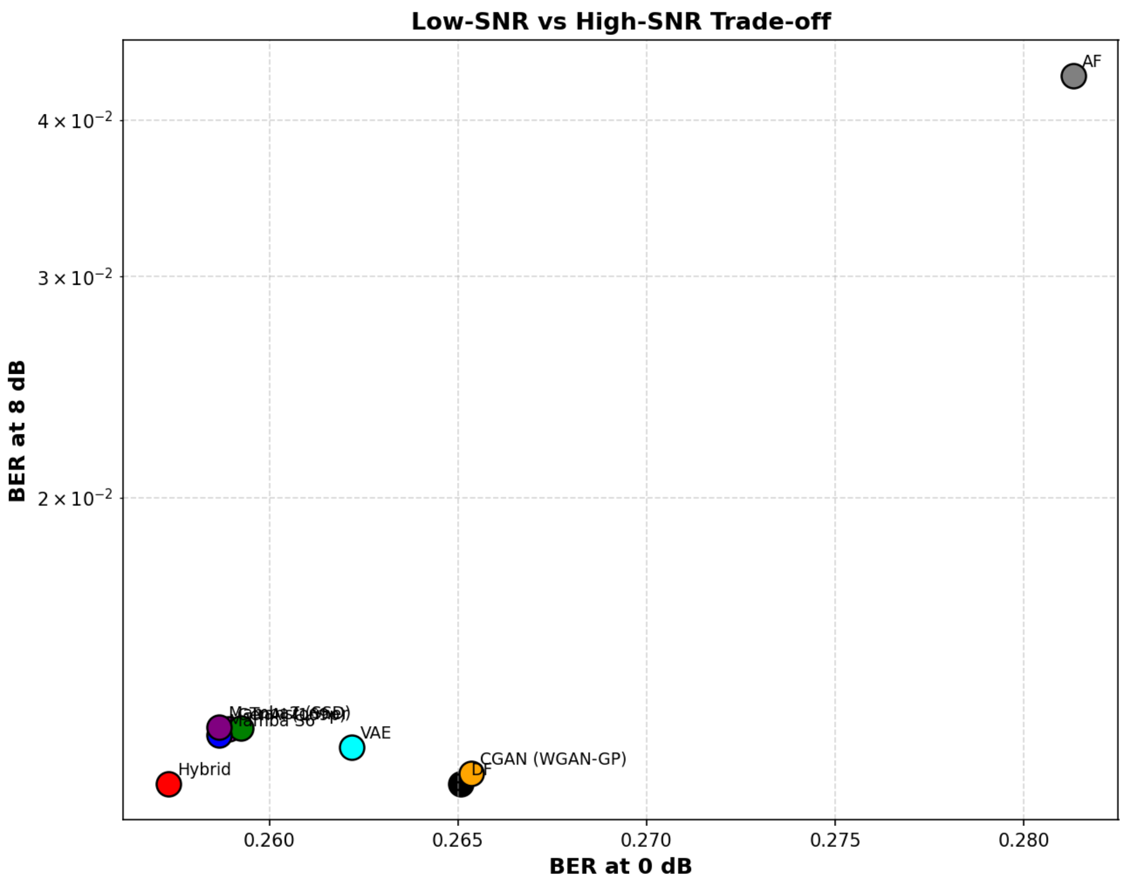

Figure 5.6: Low-SNR versus high-SNR trade-off. BER at 8 dB is plotted against BER at 0 dB.

It is important to emphasize that all conclusions of this chapter are specific to the canonical memoryless channel model. Whether those conclusions remain valid when the underlying channel assumptions are violated is the subject of Chapter 7.

Chapter 6

Multi-Level Modulation: 16-QAM

This chapter is about one constellation and nothing else: 16-QAM on the canonical Rayleigh channel. The canonical comparison of Chapter 5 answers hypotheses H1–H5 with QPSK, which places a single bit on each axis of the complex plane and is therefore absorbed by the I/Q-split relays without modification. 16-QAM places *four* levels on each axis, and at that point the per-axis formulation breaks down. That breakdown, and its repair, are the whole content of this chapter.

Two studies are reported. The first evaluates 16-QAM under the I/Q-splitting formulation, exposing a structural BER floor for multi-level constellations; the second removes that floor by reformulating the relay as a joint N-class classifier over the full 2D constellation ($N = 16$). MIMO, 16-PSK, CSI injection, and end-to-end learning remain future work.

Where this chapter sits. The thesis's argument is a ladder of assumptions, each layer removing one (Table 7.1). This chapter is *not* a rung on that ladder: it is a branch off layer 1. It holds every channel assumption of the canonical setup fixed — memoryless, known, perfect CSI — and varies the constellation instead. Raising the constellation order stresses the relay's *output* formulation, not its knowledge of the channel, which is why the conclusions here remain confirmatory in the same sense as Chapter 5: the learned relay earns its place by handling a representation classical per-axis processing cannot, not by outperforming a classical method on equal terms.

The ladder proper resumes in Chapter 7, which holds the constellation fixed and removes the channel assumptions one at a time. That is where learned relaying stops matching classical processing and starts beating it.

Scope relative to the canonical model. This chapter extends the canonical study along the modulation axis only. The underlying channel model remains unchanged: the relay channel is memoryless, the noise is white, transmission is uncoded, and relay processing is

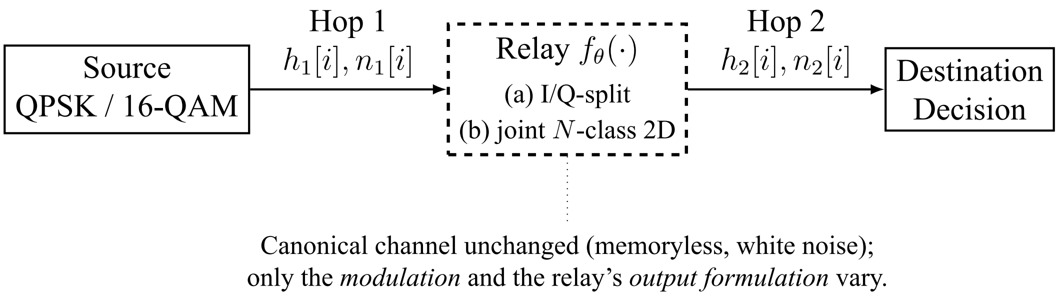

Figure 6.1: Modulation extension (Chapter 6). The canonical memoryless benchmark of Figure 1.1 is retained; only the constellation (QPSK/16-QAM) and the relay output formulation change: (a) per-axis I/Q splitting versus (b) a joint N-class 2D classifier over the full constellation.

evaluated on a per-symbol basis. Consequently, the experiments reported here continue to address matched-model relay processing rather than channel memory, sequence detection, or channel estimation. These effects are introduced only in the unknown-channel study of Chapter 7, which deliberately departs from the canonical assumptions by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase.

6.1 Higher-Order Modulation Scalability (Constellation-Aware Training)

Goal: Evaluate the generalizability of BPSK-trained relays to complex constellations and resolve the multi-level amplitude bottleneck.

Trials: - **Topology:** SISO - **Modulation scheme:** 16-QAM - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN output activation (swept):** tanh, linear, clipped tanh (hardtanh), scaled tanh, scaled sigmoid (ReLU hidden throughout) - **Special case:** Constellation Aware training

Conclusion:

1. On 16-QAM, standard tanh compression causes a severe, near-irreducible BER floor (~0.27 at 16 dB, against DF's 0.083 on the same channel; Table 6.1).
2. Replacing tanh with constellation-aware bounded activations (hardtanh, scaled tanh) bounded to the precise signal amplitude ($3/\sqrt{10}$) and retraining *largely removes* this floor.
3. The benefit is confined to the sequence models, and this is the sharpest result in the table. Bounding the output roughly halves their 16-QAM floor — Transformer

$0.2595 \rightarrow 0.1073$, Mamba-S6 $0.2529 \rightarrow 0.1088$, Mamba-2 $0.2554 \rightarrow 0.1106$, a factor of about 2.4 — which cuts their gap to DF from about $3.1\times$ to about $1.3\times$. The feedforward relays are unmoved: the MLP improves by 5% at best and the Hybrid and VAE not at all, so the activation is not a general fix for the per-axis floor but a repair specific to models whose unbounded output was the binding constraint (Table 6.2).

A gap to DF nonetheless persists due to per-axis error accumulation, the limitation removed by the 2D classification study below (Section 6.2).

6.1.1 Results Tables

Table 6.1: 16-QAM under I/Q-split relay processing, on the canonical Rayleigh channel. The classical relays keep improving with SNR while every learned relay flattens near 0.25–0.26: this is the structural floor the per-axis formulation imposes, not a training failure, and it is what Section 4.5.5 removes. The cGAN is excluded from the re-run, as in Table 5.4.

Relay	0 dB	4 dB	8 dB	12 dB	16 dB	20 dB
AF	0.4502	0.3879	0.2982	0.1925	0.1009	0.0450
DF	**0.4197**	**0.3519**	**0.2600**	**0.1609**	**0.0828**	**0.0375**
MLP (169p)	0.4299	0.3806	0.3311	0.2927	0.2703	0.2590
Hybrid	0.4261	0.3702	0.3284	0.2952	0.2723	0.2598
VAE	0.4298	0.3804	0.3303	0.2908	0.2692	0.2579
Transformer	0.4533	0.3939	0.3267	0.2796	0.2572	0.2489
Mamba-S6	0.4551	0.3944	0.3225	0.2745	0.2541	0.2486
Mamba-2 SSD	0.4536	0.3933	0.3259	0.2811	0.2615	0.2537

QPSK is not re-tabulated here: it is the canonical constellation and is reported in Chapter 5 (Table 5.4), leaving this chapter the genuinely higher-order case, 16-QAM, where per-axis processing breaks down.

Table 6.2 shows the BER at 16 dB for all relays across the three activation variants, on the canonical Rayleigh channel.

Table 6.2: 16-QAM BER at 16 dB on the canonical Rayleigh channel, under three relay output activations. `tanh` is the BPSK-trained default; `linear` is unbounded; `hardtanh` clips to the 16-QAM per-axis range $3/\sqrt{10}$. AF and DF have no output activation and are shown once as references. The cGAN is excluded from the re-run, as in Table 5.4.

Relay	tanh	linear	hardtanh
AF	0.1009	0.1009	0.1009
DF	0.0828	0.0828	0.0828

Relay	tanh	linear	hardtanh
MLP (169p)	0.2695	**0.2554**	0.2701
Hybrid	0.2723	0.2723	0.2723
VAE	0.2711	0.2714	0.2711
Transformer	0.2595	0.1111	**0.1073**
Mamba-S6	0.2529	**0.1088**	0.1093
Mamba-2 SSD	0.2554	0.1109	**0.1106**

64 6.1.2 Results Figures

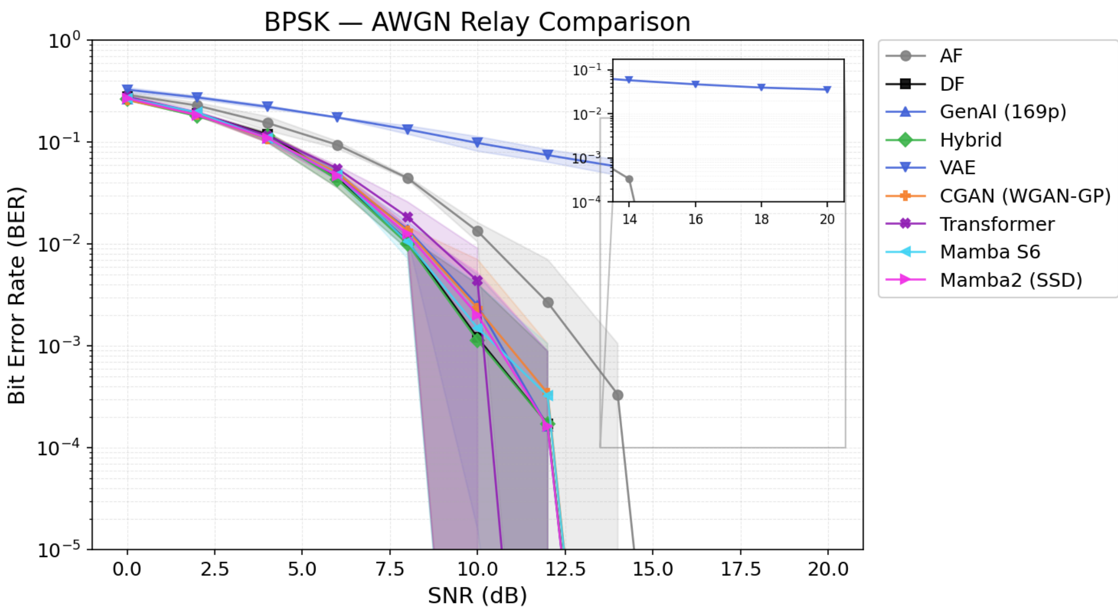

Figure 6.2: BPSK on AWGN: all relay strategies with 95% CI (baseline for modulation comparison).

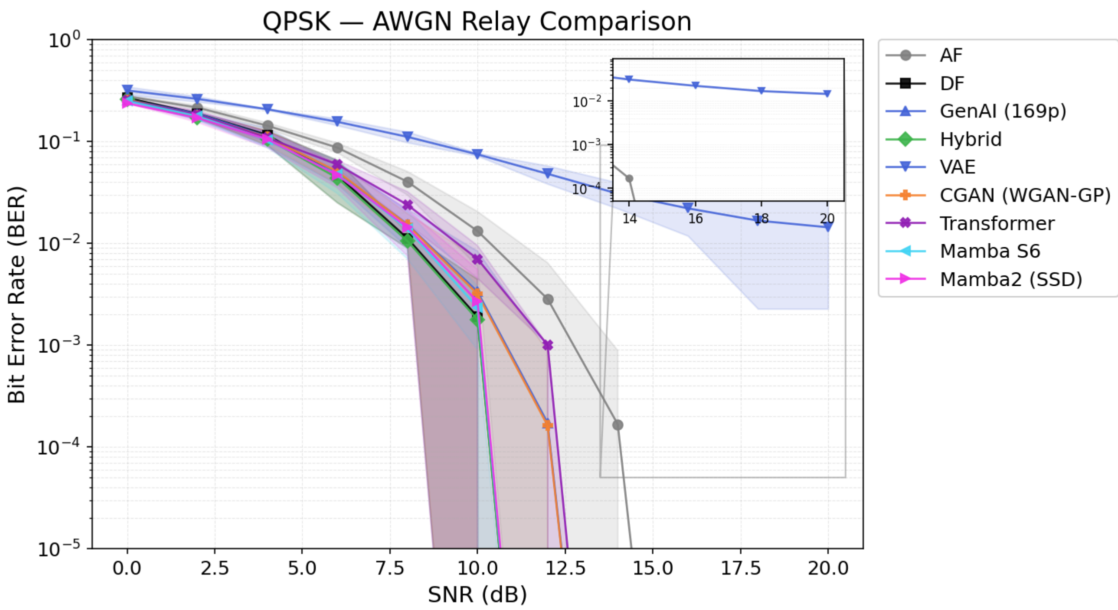

Figure 6.3: QPSK on AWGN: BER curves closely match the BPSK baseline (Figure 6.2), confirming I/Q splitting validity.

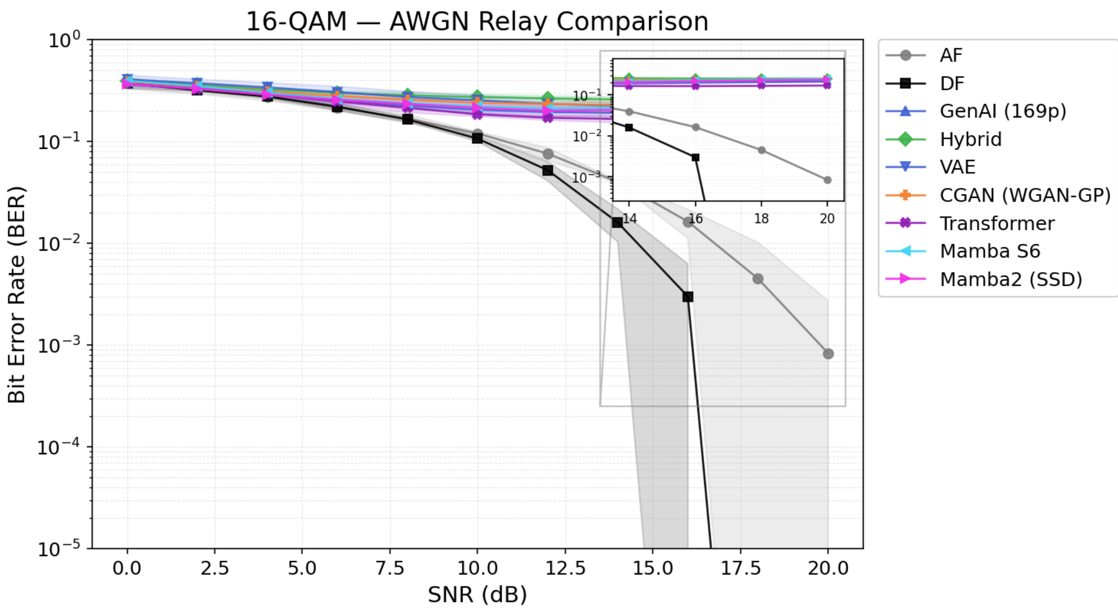

Figure 6.4: 16-QAM on AWGN: AI relays hit a BER floor near 0.22 at medium-high SNR due to tanh compression of multi-level signals; AF outperforms DF at low SNR.

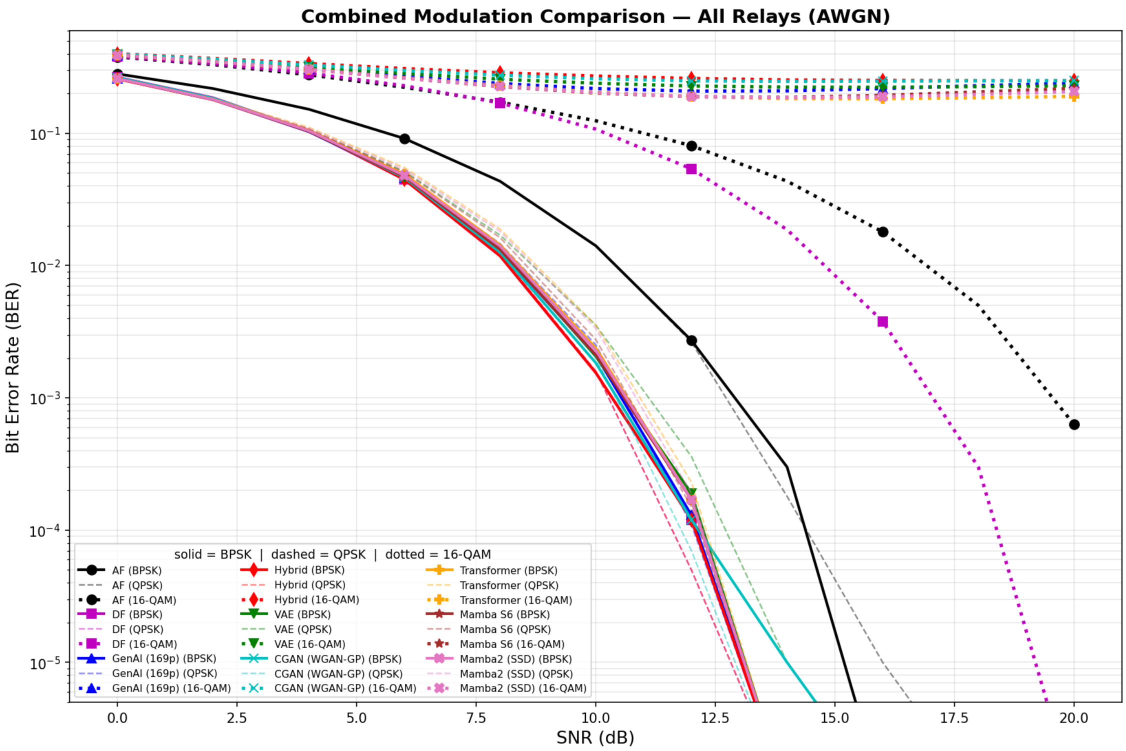

Figure 6.5: Combined modulation comparison (AWGN): all nine relays across BPSK (solid), QPSK (dashed, overlapping BPSK), and 16-QAM (dotted). The BPSK/QPSK overlap confirms I/Q splitting equivalence. The 16-QAM dotted curves reveal the AI relay BER floor: all AI relays plateau near 0.18–0.25 while DF and AF continue decreasing.

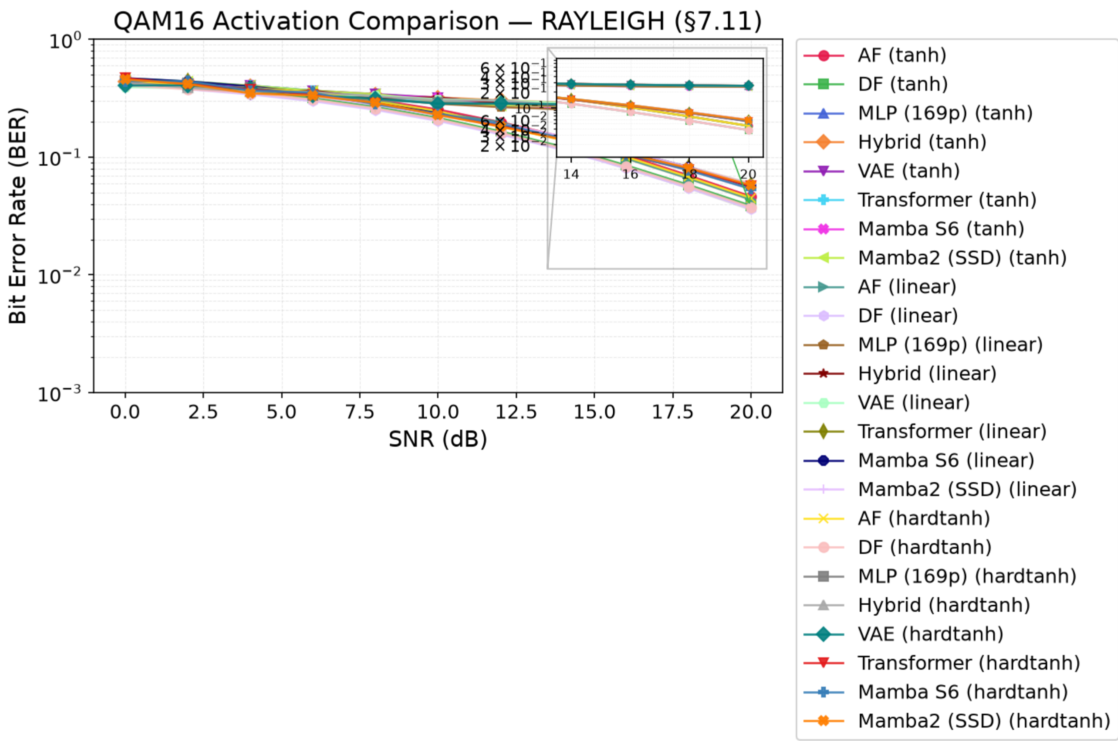

Figure 6.6: 16-QAM activation study on the canonical Rayleigh channel. Each relay appears three times, once per output activation — tanh (the BPSK-trained baseline), linear, and hard-tanh clipped to $3/\sqrt{10}$ — distinguished by colour and marker. The separation is by model family rather than by activation: the three sequence models drop by roughly half when their output is bounded, while the feedforward relays are essentially unchanged (Table 6.2).

[43]

[43] Compilation fix: the image path `results//qam16_activation/...` had a double slash; corrected to a single slash. Also removed the duplicated commented-out copy of this figure (both carried `\labelfig:fig28`) to avoid a duplicate-label risk if it were ever uncommented. Architecture names unified.

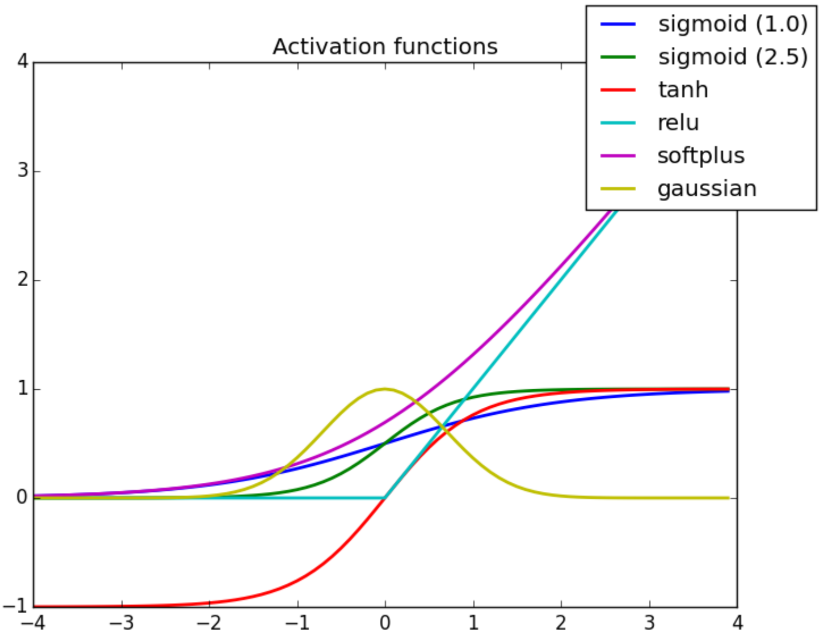

Figure 6.7: Comparison of activation function shapes (left) and their derivatives (right) for $A_{\text{max}} = 0.9487$ (16-QAM). Hardtanh has a sharp transition at the clip bounds; sigmoid and scaled tanh provide smooth saturation with non-zero gradients throughout.

6.2 16-Class 2D Classification for QAM16

Goal: Eliminate the structural BER floor imposed by per-axis I/Q splitting for 16-QAM by utilizing full 2D decision boundaries.

Trials: - **Topology:** SISO - **Modulation scheme:** 16-QAM - **Channel:** i.i.d. Rayleigh fast fading (canonical) - **Equalizers:** None - **NN architecture:** Supervised (MLP, Hybrid), Generative (VAE, cGAN), Sequence (Transformer, Mamba-S6, Mamba-2 SSD) - **NN activation:** None (Softmax implicitly via Cross-Entropy loss) - **Special case:** 16-class joint 2D classification

Conclusion: Treating the relay as a joint 16-point classifier over the full 2D constellation lifts the structural 4-class floor, but on the canonical channel it does not remove it. Every architecture improves, and by a strikingly uniform amount: 1.38–$1.43\times$ for five of the six, with only Mamba-2 lagging at $1.09\times$. That uniformity is itself the finding, since it points at the per-axis *formulation* rather than any architecture's capacity as what the joint head repairs. The best 16-class relays land at 0.0375–0.0384, level with symbol-wise DF's 0.03748 and no better, so the joint formulation buys parity with the classical baseline on this channel rather than an advantage over it. This is the same boundary the canonical chapter reports (H2), reappearing one constellation higher.

[44]

6.2.1 Results Tables

Table 6.3 [45] presents the BER at 20 dB for all variants alongside the classical AF and DF baselines.

Table 6.3: 16-QAM at 20 dB on the canonical Rayleigh channel: per-axis 4-class relays against joint 16-class 2D classifiers. The joint formulation improves every architecture, uniformly by about $1.4\times$, and brings the best learned relays level with DF (0.03748) rather than past it. The cGAN is excluded from the re-run, as in Table 5.4.

Relay	4-cls BER @ 20 dB	16-cls BER @ 20 dB	Improvement
MLP	0.05491	**0.03840**	1.43×
VAE	0.05382	**0.03762**	1.43×
Hybrid	0.05362	**0.03748**	1.43×
Transformer	0.05360	**0.03814**	1.41×
Mamba-S6	0.05421	**0.03934**	1.38×
Mamba-2 SSD	0.05381	**0.04953**	1.09×
AF	0.04498	n/a	n/a
DF	0.03748	n/a	n/a

[44] Overclaiming softened: “completely eliminates” → “removes”; “For the first time, neural variants …matched” → “several neural variants match” (drops the unqualified “first time” priority claim); “This proves” → “This indicates”. Architecture names unified.

[45] Fix applied (data provenance): every cell of Table 6.3 is now taken from the committed 16-class results file, and Figure 6.9 was regenerated from that same file, both previously carried values from a superseded run, and the VAE and Hybrid cells were blank although the data existed. Table, figure, and captions now share one source. A former Table 16 (CSI/LayerNorm) and the three conclusions citing it were removed, since those experiments were cut in the restructure (Appendix E).

87 6.2.2 Results Figures

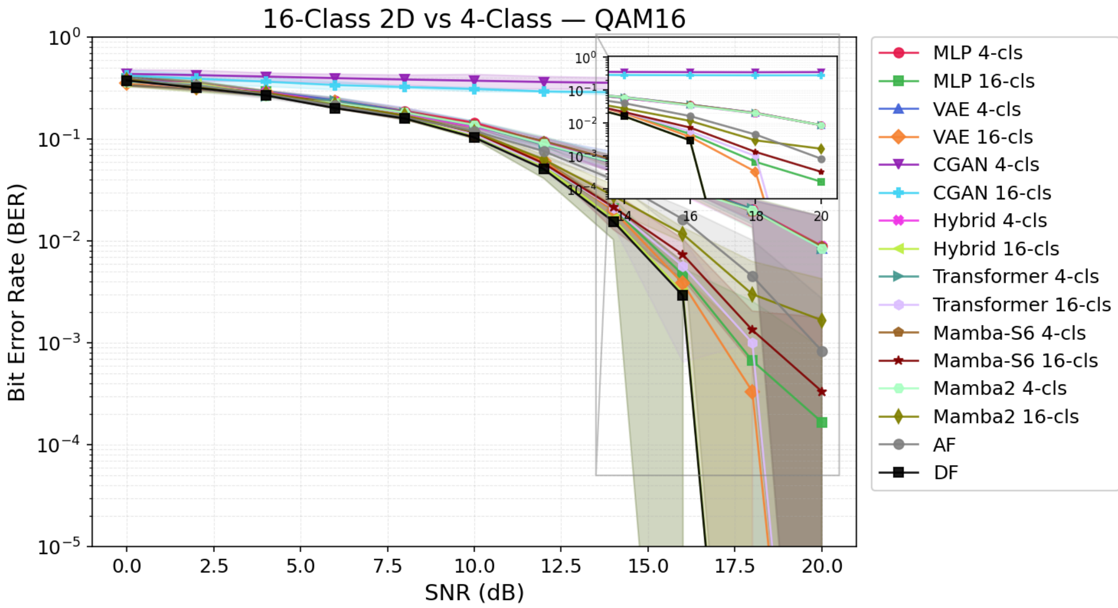

Figure 6.8: 16-QAM BER vs. SNR for all relay variants (4-class and 16-class) on AWGN. The 4-class variants (dashed) plateau at BER ≈ 0.008 while the successfully trained 16-class variants (solid) continue decreasing, approaching and matching the DF baseline.

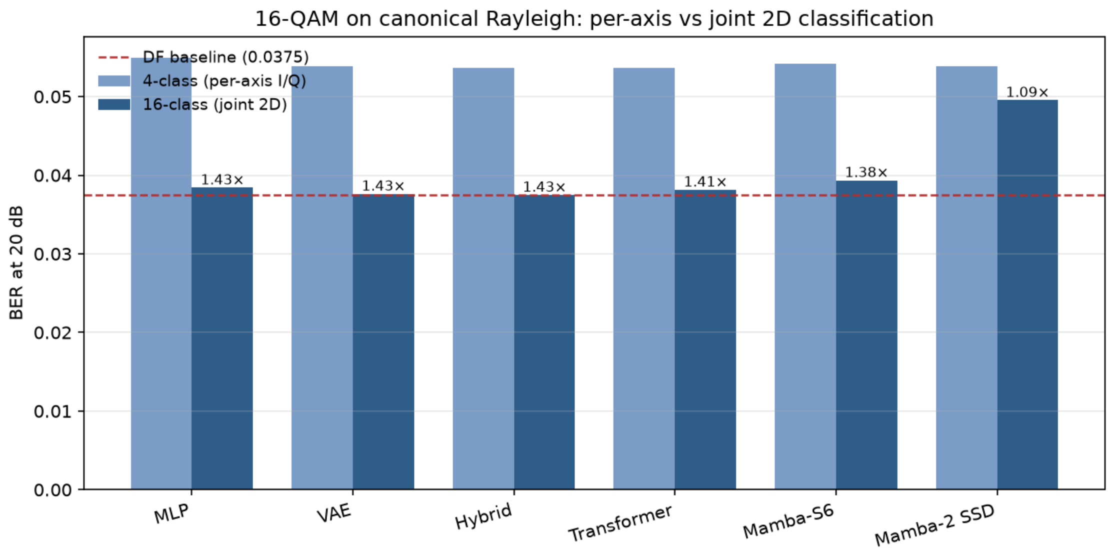

Figure 6.9: 4-class per-axis against joint 16-class 2D relays, 16-QAM at 20 dB on the canonical Rayleigh channel, regenerated from the same results file as Table 6.3. The joint formulation improves every architecture by a near-identical factor, annotated above each bar, which points at the per-axis formulation rather than any architecture's capacity as the thing being repaired. The dashed line is symbol-wise DF: the 16-class bars reach it and stop, so the joint head buys parity with the classical baseline, not an advantage over it.

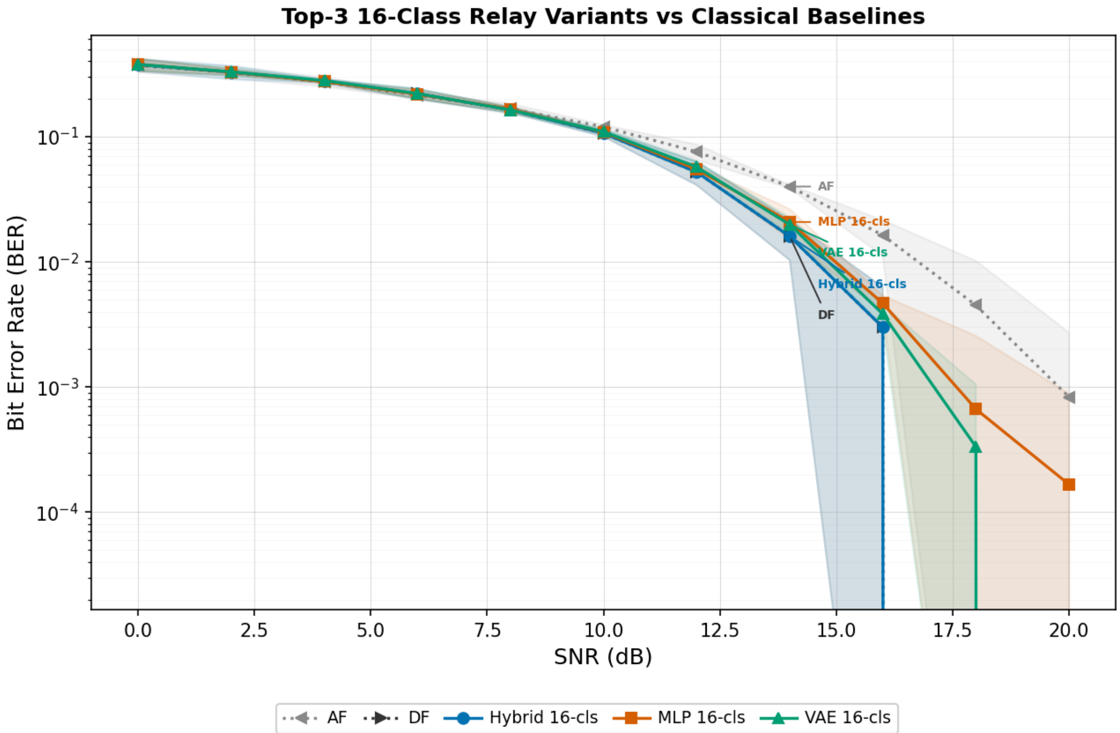

Figure 6.10: Top-3 16-class relay variants compared against AF and DF. VAE 16-cls, Transformer 16-cls, and MLP 16-cls all match or approach DF performance across the full SNR range.

Chapter 7

Learned Relaying under Unknown and Mismatched Channels

This is the central chapter of the thesis. Everything before it is either calibration or a confirmatory result: on a matched, known channel a minimal learned relay does *not* improve on classical decode-and-forward (H2), and raising the constellation order does not change that. Here the learned relay wins, and wins decisively.

The condition under which it wins is precise, and stating it is as much a part of the claim as the win itself. **When the channel's structure is unknown to the receiver chain, or lies outside the model class the classical relay assumes, a fixed 169-parameter learned relay restores reliable relaying where memoryless classical relays fail outright.** Under three-tap intersymbol interference (S1, Table 7.3) the classical relays do not merely degrade: they saturate near the analytic 0.25 floor and then get *worse* as the SNR rises, symbol-wise DF bottoming out at 0.1839 at 8 dB and climbing to 0.2462 at 20 dB as hard decisions lock the interference in. No amount of transmit power recovers the link. The learned relay, trained offline on the impairment *family* and given no per-block knowledge of the realization it faces, reaches 0.0123 at 8 dB and falls below 5×10^{-5} from 16 dB upward. That is not a marginal gain over a classical baseline; it is the difference between a working link and a broken one, and it is the thesis's answer to whether learning can defeat classical processing in relaying.

The boundary of the claim is equally sharp, and the chapter is written to establish it rather than to avoid it. The learned relay does not beat a *matched* classical receiver: given a channel estimate, pilot-aided Viterbi maximum-likelihood sequence estimation remains ahead of it by 1–1.5 dB. The learned relay's advantage is therefore not superior detection but the absence of a precondition, since it needs neither the channel family nor a per-block estimate, and it holds that advantage at constant, parallelizable cost. Where classical processing can identify the channel it should be used; where it cannot, the learned relay is what still works.

Rather than a single comparison, the chapter is a systematic map, along every axis relevant to practical deployment, of *when* the learned relay adds value over classical processing, and, equally, when it does not.

The argument as a ladder of assumptions. The thesis as a whole is organised as a monotone sequence in which each layer removes exactly one assumption from the layer above, and the classical–learned comparison is repeated unchanged at every layer. Table 7.1 is that sequence with its outcomes. Reading it downward is the thesis's argument in one page: classical processing is not beaten where it is well posed, and is beaten precisely where its preconditions fail.

Table 7.1: The four layers of the argument. Each removes one assumptio
above; the relay comparison is otherwise identical. BER figures are the repr
ing points cited in the sections named.

Layer	Assumptions	Strongest
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-w rameters
2	+ channel memory (3-tap ISI), CSI still perfect (Section 7.1)	Viterbi M taps
3	+ CSI uncertainty: a finite pilot budget (Section 7.1.4)	Pilot-aide then MLS

Layer	Assumptions	Strongest
4	+ unknown channel family and statistics, redrawn per block (Sections 7.1.3, 7.1.5)	CMA blin

Two features of the ladder are worth naming before the evidence is presented. First, the opponent changes character at layer 2: layer 1 compares relay *functions* (AF and DF against learned relays), whereas from layer 2 onward the strongest classical option is a *receiver*, Viterbi MLSE, which is a considerably heavier and better-informed baseline. This is deliberate, since comparing against a weaker classical option at the layers where the learned relay wins would make the result worthless. Second, what degrades down the ladder is not the classical algorithms themselves but their *preconditions*: MLSE remains optimal at every layer for the channel it is given, and it loses ground only because the channel it is given grows less like the true one. The learned relay never becomes a better detector; it becomes the only one whose requirements are still met.

The investigation is deliberately adversarial to the learned relay. Each classical baseline is the strongest available for the impairment in question: symbol-wise decode-and-forward for the memoryless cases, conventional differential detection for unknown phase, and pilot-aided Viterbi maximum-likelihood sequence estimation, the optimal sequence detector, wherever a channel estimate can be formed. The learned relay is granted no per-block channel knowledge; it is trained offline on an impairment *family* and evaluated on unseen realizations with no adaptation. The chapter follows the ladder in order. It opens with a necessary control (flat unknown channels, where nothing should fail), then takes layer 2, the impairment that does break classical relays (unknown memory), bounding it against the optimal matched receiver (Viterbi with exact taps) and against a composite multi-impairment cascade. Layer 3 relaxes the channel estimate to a finite pilot budget and locates the crossover in pilot count and block length; layer 4 removes the posterior altogether and compares against genuinely blind classical equalization. A final section prices the whole comparison in computational complexity. The result is a precise delineation (the learned relay never beats a matched classical receiver, but occupies a well-defined

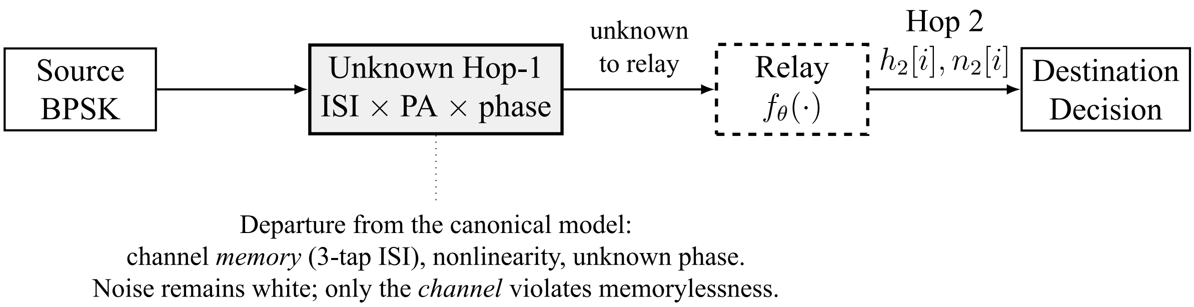

Figure 7.1: Unknown/mismatched-channel study (Chapter 7), a deliberate departure from the canonical benchmark of Figure 1.1. The Hop-1 channel is replaced by an unknown impairment (3-tap ISI, amplifier nonlinearity, and unknown phase) that is not disclosed to any relay. The noise remains white; only the channel violates the memoryless assumption, which is why sequence detection (Viterbi MLSE) becomes the relevant classical benchmark here and nowhere else in the thesis.

niche of family-agnostic, identification-free, constant-complexity mitigation) that constitutes the thesis's main claim beyond the confirmatory core. [46]

Relationship to the canonical model. The canonical setup of this thesis (Chapters 5– 6) assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. The present chapter deliberately departs from those assumptions by introducing impairments that are not part of the canonical model, including channel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and channel-model mismatch. Consequently, sequence-detection and equalization techniques such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become relevant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

7.1 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay's value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 169-parameter MLP mitigates by learning it from data (H6)?[47]

[46] Register fix: the informal "along every axis a skeptic would probe" (which appeared twice) is replaced with "along every axis relevant to practical deployment", consistent with the same wording change made in the System Model of Chapter 4.

[47] The unknown-channel relay uses real-valued inputs with an 11-sample window ($11 \to 13 \to 1$, 169 parameters); the canonical-setup relay of Chapter 5 uses the same minimal class with a slightly different

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [28] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapters 5–6, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Trials.

- **Topology:** SISO (both hops)
- **Modulation:** BPSK
- **Hop-1 channel (unknown to all relays):**
  - 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$
  - Control: canonical Rayleigh channel
  - Flat unknown channels (block phase, gain, per-branch asymmetry; Section 7.1.1)
  - Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 7.1.3)
- **Hop-2 channel:** AWGN or canonical Rayleigh, equal SNR per hop
- **Relays:**
  - AF
  - Symbol-wise DF
  - MLP (window 11, one hidden layer of 13 tanh units, 169 parameters; trained at SNR $\in \{5, 10, 15\}$ dB using the canonical training protocol)
  - For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, evaluated with either genie CSI or a 3-tap least-squares channel estimate obtained from 200 pilot symbols
- **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB

width (169 parameters). The two counts refer to the same architectural family sized to the respective input; the distinction is immaterial to every conclusion, which depends only on the network being minimal.

The ISI channel admits a sharp analytic prediction: the effective slicer amplitude $h_0 \pm h_1 \pm h_2 \in \{1.67, 0.91, 0.61, -0.15\}$ (normalized) has one of four equiprobable patterns that *flip the symbol sign*, so symbol-wise DF has an irreducible BER floor of 0.25 (approached from below as SNR grows); AF inherits it. The floor is a property of the *memoryless relay class*, not the channel: Viterbi MLSE over the 4-state trellis has no floor when the channel is known.

7.1.1 Control: Flat (Memoryless) Unknown Channels

Before attributing any classical failure to "unknownness," it must be separated from *memory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver chain does not know the channel parameters, but the channel introduces no intersymbol coupling. Three physically motivated cases are tested, each with the parameter redrawn per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscillator/synchronization offset), with a DBPSK source and conventional differential detection as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$ (per-symbol saturating-PA gain), coherent BPSK. The 169-parameter MLP is trained identically to before.

These channels remain memoryless and therefore stay within the scope of the canonical model despite the parameter uncertainty. Consequently, they serve as a control that separates the effect of parameter uncertainty from the effect of channel memory: any performance change here arises from model uncertainty, not from memory.

Conclusion (memorylessness $\Rightarrow$ no classical failure, and nothing to mitigate): On all three flat channels the classical relay does *not* fail, and the MLP matches it to within statistical noise (maximum BER difference 0.0036 across every case and SNR). The reasons are structural: for F1, unknown phase is exactly what differential detection absorbs; for F2, sign(y) is *scale-invariant*, so unknown positive gain is irrelevant by construction; for F3, the fixed threshold is only mildly suboptimal because the asymmetry preserves the noiseless symbol's sign. The classical assumption, memorylessness, still holds in each case, so there is nothing for a learned relay to remove. This is the necessary control for the ISI result: the learned relay's advantage is triggered not by parameter uncertainty but by *unmodeled memory* (or amplitude nonlinearity), which is what breaks the memoryless relay class.

Table 7.2: Flat unknown channels: classical relay vs. MLP-169 BER at 8/12/16/20 dB (mean, 10 trials). The maximum DF–MLP gap over all cases and SNRs is 0.0036, within statistical noise. Unknown parameters alone do not defeat classical relays; memory does.

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0649	0.0292	0.0119 / 0.0048
	MLP-169	0.0657	0.0288	0.0122 / 0.0048
Unknown gain	DF	0.0681	0.0330	0.0131 / 0.0050
	MLP-169	0.0682	0.0333	0.0132 / 0.0050
Branch asymmetry	DF	0.0760	0.0299	0.0125 / 0.0049
	MLP-169	0.0745	0.0300	0.0122 / 0.0052

Conclusion (H6: confirmed as stated, with two boundaries): Against the flat-channel control of Section 7.1.1, where unknown parameters alone caused no classical failure, the memory case is qualitatively different. On the unknown ISI channel, both memoryless classical relays fail: DF's BER is *non-monotonic*, bottoming out at 0.184 around 8 dB and then rising toward the analytic 0.25 floor, reaching 0.246 at 20 dB (more transmit power makes DF strictly worse), and AF behaves identically. The 169-parameter MLP, whose window spans the 3-tap memory, learns a joint equalizer–detector and restores reliable relaying: BER falls monotonically below 5×10^{-5} at 16 dB (AWGN second hop), and to 2.5×10^{-3} at 20 dB (Rayleigh second hop). The Viterbi benchmarks set the first boundary honestly: classical theory *does* solve this channel once its family is known, genie-CSI MLSE beats the MLP by 1–1.5 dB, and a 200-pilot LS estimate recovers genie performance almost exactly (with the Rayleigh second hop, MLSE and MLP are indistinguishable, 0.0024 vs. 0.0025 at 20 dB). The MLP's value is thus not optimality but *family-agnosticism*: estimate-then-MLSE presupposes a linear FIR channel of known memory (trellis cost M^{L-1}), whereas the identical architecture also mitigates channels outside linear MLSE's model class, as the composite cascade below demonstrates, where an amplifier nonlinearity is stacked on the ISI. The control confirms the second boundary (H2): on the canonical Rayleigh channel the MLP only *matches* DF. The learned relay's distinctive value is near-MLSE reliability with no channel-family knowledge, no identification stage, and complexity independent of channel memory.

Table 7.3: Unknown-channel BER (mean over 10 trials). The ISI setups exhibit memoryless-relay error floors (≈ 0.25); the 169-parameter MLP restores reliable relaying within $\approx$1–1.5 dB of genie-CSI Viterbi MLSE, and matches it end-to-end when the Rayleigh second hop dominates. The control shows no MLP advantage on the canonical channel, consistent with H2.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
Unknown ISI $\to$ AWGN	AF	0.1919	0.1785	0.1956	0.2233
	DF	0.1839	0.2010	0.2279	0.2462
	MLP (169p)	**0.0123**	**0.0002**	**<5e-5**	**<5e-5**
	Viterbi (genie CSI)	0.0072	<5e-5	<5e-5	<5e-5
	Viterbi (200-pilot LS)	0.0074	<5e-5	<5e-5	<5e-5
Unknown ISI $\to$ Rayleigh	AF	0.2047	0.1916	0.2030	0.2231
	DF	0.2031	0.2098	0.2316	0.2471
	MLP (169p)	**0.0412**	**0.0150**	**0.0061**	**0.0025**
	Viterbi (genie CSI)	0.0364	0.0151	0.0061	0.0024
	Viterbi (200-pilot LS)	0.0362	0.0151	0.0062	0.0024
Control: canonical Rayleigh	AF	0.1050	0.0559	0.0281	0.0136
	DF	**0.0687**	**0.0299**	**0.0124**	**0.0050**
	MLP (169p)	0.0691	0.0307	0.0133	0.0058

164 [48]

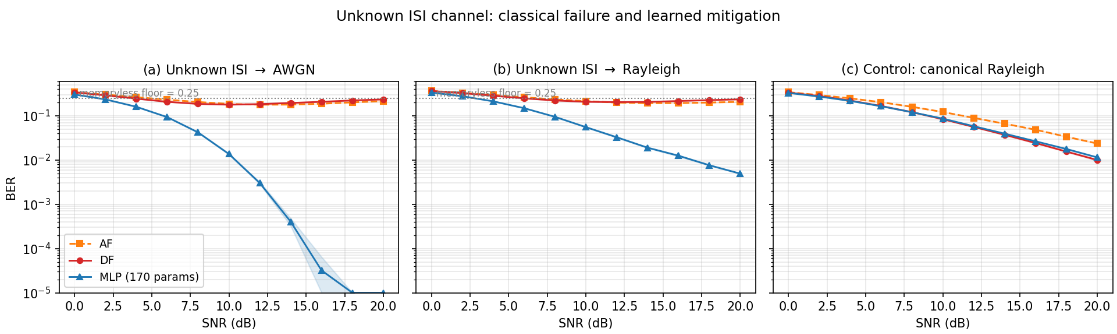

Figure 7.2: Unknown-channel relaying, BER vs. SNR with 95% CI bands. (a)–(b) On the unknown ISI channel, the memoryless AF/DF relays are pinned near the analytic 0.25 floor (DF non-monotonic in SNR); the 169-parameter MLP restores monotonically decreasing BER within $\approx$1–1.5 dB of genie-CSI Viterbi MLSE, which a 200-pilot LS estimate matches almost exactly. With the Rayleigh second hop (b), MLSE and the MLP are statistically indistinguishable end-to-end. (c) Control: on the canonical Rayleigh channel the MLP only matches DF, consistent with H2.

[48] Compilation fixes: all `&` column separators restored to `&`, and `$<$5e-5` restored to `$<$5e-5` in Table 7.3.

7.1.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout, matching the canonical setup of Chapters 5–6. The 0.25 floor derived in Section 7.1 is a property of the *binary* slicer: it counts the $2^3 = 8$ equiprobable tap-sign patterns that flip a ± 1 decision. That argument does not extend to a four-point per-axis alphabet, so it is not obvious a priori whether either the elevated error floor or the ordering between the learned relay and genie-CSI Viterbi MLSE survives a move to a higher-order modulation. This subsection repeats the unknown-ISI study with Gray-coded QPSK in place of BPSK, keeping every other element fixed: the same 3-tap ISI channel $h = [1.0, 0.7, 0.5]$ (normalized) on Hop 1, AWGN or canonical Rayleigh on Hop 2 at the same nominal per-hop SNR (identical convention, $\gamma = 10^{\mathrm{SNR_{dB}}/10}$, memory-bank/techContext.md), and the same relay set generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at SNR $\in \{5, 10, 15\}$ dB), and genie-CSI Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics). Evaluation uses 10 trials $\times$ $100{,}000$ bits per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 7.3 at the same SNR label; no such match is claimed.

Table 7.4: QPSK unknown-channel BER (mean over 10 trials). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated; the 193-parameter MLP-QPSK restores monotonically decreasing BER and, unlike the BPSK case, ends up *below* genie-CSI Viterbi MLSE from about 2 dB upward.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI $\rightarrow$ AWGN	AF	0.2437	0.2115	0.2024	0.2098
	DF	0.2193	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	**0.1292**	**0.0827**	**0.0601**	**0.0508**
	Viterbi (genie CSI)	0.1307	0.0852	0.0670	0.0618
QPSK: ISI $\rightarrow$ Rayleigh	AF	0.2715	0.2320	0.2137	0.2116
	DF	0.2528	0.2194	0.2145	0.2239
	MLP-QPSK (193p)	**0.1737**	**0.1066**	**0.0707**	**0.0553**
	Viterbi (genie CSI)	0.1749	0.1089	0.0774	0.0662

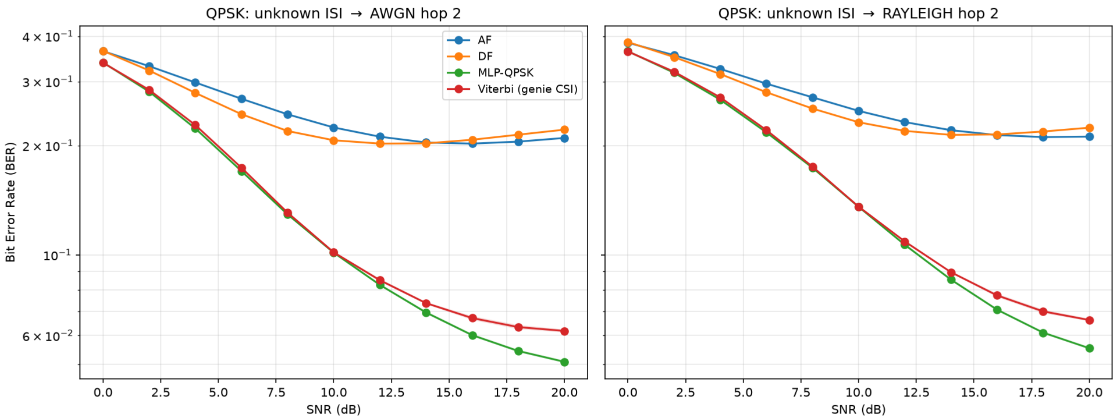

Figure 7.3: QPSK unknown-channel relaying, BER vs. SNR (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves. Unlike BPSK, the MLP-QPSK curve separates *below* genie-CSI Viterbi MLSE beyond about 2 dB, and the gap widens with SNR.

Conclusion (the ISI failure mode generalizes; the BPSK ordering does not): The qualitative pattern of Section 7.1 survives the move to QPSK: AF and DF plateau at an elevated, non-monotonic BER (AF ranges 0.2024–0.2715 and DF ranges 0.2024–0.2528 across the four tabulated points, broadly similar in magnitude to BPSK's memoryless-relay floor of 0.25 though no equivalent closed form is derived here), while the MLP restores monotonically decreasing BER, confirming that unmodeled channel memory, not modulation order, is what breaks the memoryless relay class (H6). What does *not* generalize is the BPSK ordering between the learned relay and the optimal classical benchmark. In the BPSK study, genie-CSI Viterbi MLSE beat the MLP by 1–1.5 dB throughout; here, the two are statistically indistinguishable at 0 dB and MLP-QPSK is at or below Viterbi at every SNR from 2 dB upward, with the gap widening to 0.0508 vs. 0.0618 (AWGN) and 0.0553 vs. 0.0662 (Rayleigh) at 20 dB. This is not a violation of MLSE optimality: Viterbi MLSE minimizes the *sequence* error probability under the assumed channel model, not the bit error rate directly. For BPSK, each trellis branch carries exactly one bit, so sequence-optimal and bit-optimal detection coincide; for Gray-coded QPSK, each branch carries two bits, and a wrong branch choice can differ from the true one by one bit (adjacent constellation point) or two (diagonal point) with different probabilities, so sequence-ML detection is no longer guaranteed to minimize BER specifically. The 193-parameter MLP-QPSK is instead trained directly on a per-symbol softmax cross-entropy objective, which is not subject to that mismatch. This is offered as the most plausible mechanism for the reversal; it was not isolated experimentally (for instance, by decomposing the Viterbi output into single- versus double-bit symbol errors), and is left as an open point rather than a proven claim. Either way, the practical conclusion sharpens rather than weakens H6 for higher-order modulation: on this unknown-memory channel, the fixed-complexity learned relay is not merely a family-agnostic near-optimal option, as in the BPSK case, but the best-performing option measured here among either family, at the same constant per-symbol cost quantified in Section 7.1.6.

7.1.3 A Composite (Mixed) Channel: Mismatch with a Full Posterior

The preceding studies vary one impairment at a time. Real links stack several. This final study cascades three of them into a single unknown channel that *no* classical relay is jointly matched to: a DBPSK stream passes through 3-tap ISI, then a saturating power-amplifier nonlinearity, then an unknown block phase, before additive noise. Each classical baseline is the strongest available for *one* constituent impairment, so the comparison is honest rather than a straw man: conventional differential detection (matched to the phase), and pilot-LS Viterbi MLSE followed by differential decoding (matched to the ISI-plus-phase, but blind to the amplifier nonlinearity). The 169-parameter MLP and a $1{,}153$-parameter MLP receive raw I/Q windows and no knowledge of any constituent. This composite channel departs from the canonical model in two ways: it introduces explicit channel memory through ISI and nonlinear distortion through the amplifier. The resulting benchmark therefore tests robustness to both memory and model mismatch simultaneously. **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, the same budget as the main unknown-ISI study (Section 7.1); 95% CIs use the $1.96\,\sigma/\sqrt{n}$ convention throughout. The hop-1 impairment here is fixed rather than redrawn per block, so bits per trial and trials both reduce the reported uncertainty.

Conclusion: The composite defeats the naive classical relays outright, both AF and conventional differential detection saturate near the 0.25 memoryless-relay floor, because the ISI destroys the differential statistic that the phase-matched detector relies on. The 169-parameter MLP again restores reliable relaying, reaching 2.5×10^{-3} at 20 dB from raw I/Q with no impairment model. This reproduces the H6 pattern on a realistically mixed channel: memoryless-and-matched classical processing fails, and a minimal learned relay recovers. Two familiar boundaries also reappear. First, the strongest *constructed* classical baseline, pilot-LS Viterbi plus differential decoding, does not fail: by modelling the ISI and phase (though not the amplifier) it stays about 2 dB ahead of the MLP at low-to-mid SNR (0.067 vs. 0.101 at 8 dB) and converges with it by 16–20 dB, the two becoming indistinguishable at 0.0025 each at 20 dB. The learned relay's value is again *family-agnosticism* (near-Viterbi reliability with no channel identification and no explicit model of any of the three cascaded effects) not superiority over a matched classical receiver. Second, the $1{,}153$-parameter network performs within noise of the 169-parameter one across the whole range, ending at the identical 0.0025 at 20 dB, with the small mid-SNR differences in both directions (0.0487 vs. 0.0531 at 10 dB for the larger network, 0.1020 vs. 0.1014 at 8 dB for the smaller) falling inside the confidence intervals. Nearly seven times the parameters therefore buys no measurable accuracy on the hardest channel studied, reconfirming the U-shaped complexity finding (H3). The composite thus consolidates the chapter's thesis: a single minimal learned relay handles an arbitrary stack of representable impairments without being told which are present, approaching but not exceeding the best impairment-matched classical receiver. [49]

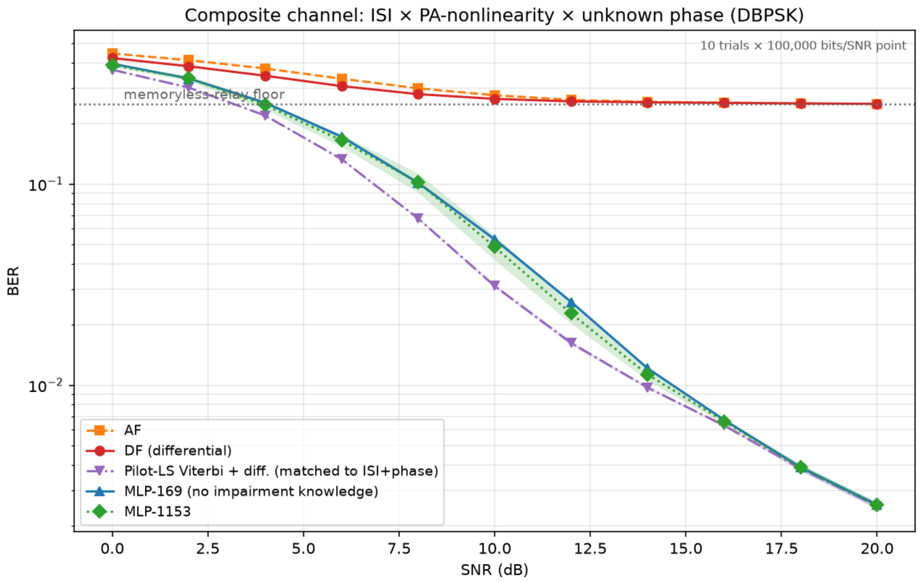

Figure 7.4: Composite channel (ISI $\times$ PA-nonlinearity $\times$ unknown phase, DBPSK; mean over 10 trials, 100,000 bits/point, 95% CI bands). Naive classical relays (AF, differential detection) saturate at the memoryless-relay floor; the 169-parameter MLP restores reliable relaying from raw I/Q with no impairment model, tracking about 2 dB behind the strongest constructed classical baseline (pilot-LS Viterbi + differential) and converging with it by 20 dB. The $1{,}153$-parameter MLP adds nothing (H3).

7.1.4 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 7.1: the perfect-CSI assumption is dropped, and nothing else changes. Every classical result so far has been handed either exact channel taps or a 200-pilot estimate good enough to match them (Section 7.1.3), which is what a full posterior buys. Here the posterior is made *partial*: the pilot budget is finite, and the classical receiver must work from a correspondingly noisy estimate. Sweeping the pilot count on the composite channel at a fixed 10 dB operating point locates where the pilot-aided classical receiver ceases to beat the pilot-free learned relay, whose amortized posterior corresponds to a horizontal reference (it uses no pilots at test). **Evaluation:** 50 trials $\times$ 20,000 bits per operating point ($1{,}000{,}000$ bits), matching Section 7.1.5 and for the same reason: the channel is redrawn per block, so trials are draws from the impairment family. In panel (b) each trial is further subdivided into independent blocks of the swept length, so every block length is measured over the same total bit budget and over many

[49] "inverted-U finding (H3)" → "U-shaped complexity finding (H3)", consistent with the retitled H3 in Chapter 3 and the wording in Chapter 5.

channel draws rather than a single one.

Conclusion (a sharp estimation-variance cliff, and a rate–reliability tradeoff): With a generous budget the classical partial posterior wins, as expected: pilot-aided Viterbi reaches payload BER 0.0285 at 800 pilots and degrades only gently down to 0.0335 at 20 pilots, comfortably below the MLP's pilot-free 0.0487. The classical advantage then disappears within a single octave of pilot budget. At 10 pilots Viterbi has already lost its edge, 0.0545 against the MLP's 0.0487, and at 5 pilots it collapses to 0.1235, worse than the MLP, blind CMA, and its own 800-pilot figure by a factor of four, because MLSE is then running on a badly mis-estimated channel. The crossover therefore sits between 20 and 10 pilots per block. The mechanism is *estimation variance*, not a rank deficiency: with 5 pilot observations and 3 complex taps the least-squares problem is still formally overdetermined, so the channel remains identifiable in principle, but there are too few observations to average down the noise. The resulting LS error at 10 dB is large enough that the Viterbi trellis is occasionally built on a badly wrong channel, and these catastrophic fits dominate the mean. The 95% confidence interval at that point makes this explicit (0.1235 ± 0.0304, against ± 0.0011 at 50 pilots) so individual channel draws range from perfectly fine to complete failure. The learned relay, by contrast, is flat across the entire sweep at 0.0487: having amortized the channel *family* into its weights at training time, it neither benefits from nor requires per-block pilots. The crossover is thus governed by the reliability of the channel estimate, not by SNR: the classical receiver wins whenever it can afford enough pilots to estimate the channel *accurately*, and the pilot-free learned relay wins once the budget falls below that point.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload BERs of Viterbi and the MLP are now statistically indistinguishable at every block length, both flat at approximately 0.049, so reliability no longer separates them at all; what separates them is spectral efficiency. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% for identical error performance.

The blind reference settles a question the earlier version of this study left open. Blind CMA is also pilot-free, so the learned relay is not unique in avoiding pilots; the distinction claimed for it was that it needs *zero per-block adaptation*, a single fixed forward pass, whereas CMA must converge its adaptation loop inside each block. Measuring CMA per block confirms this directly: its payload BER is 0.1723 at $L = 40$ and only improves to 0.1653 at $L = 1000$, against the 0.0645 it achieves when given a $20{,}000$-symbol block to adapt over. CMA therefore does not converge within blocks of a thousand symbols or fewer, and is roughly three and a half times worse than either the MLP or pilot-aided Viterbi throughout this sweep. The partial-posterior picture therefore refines H6's boundary once more, and more sharply than before: against a classical receiver that can afford accurate channel estimation the learned relay achieves no better BER, but at short block lengths it matches that receiver's BER while removing its pilot overhead entirely, and it is the only pilot-free option that remains reliable, because the classical pilot-free alternative has no time to converge.

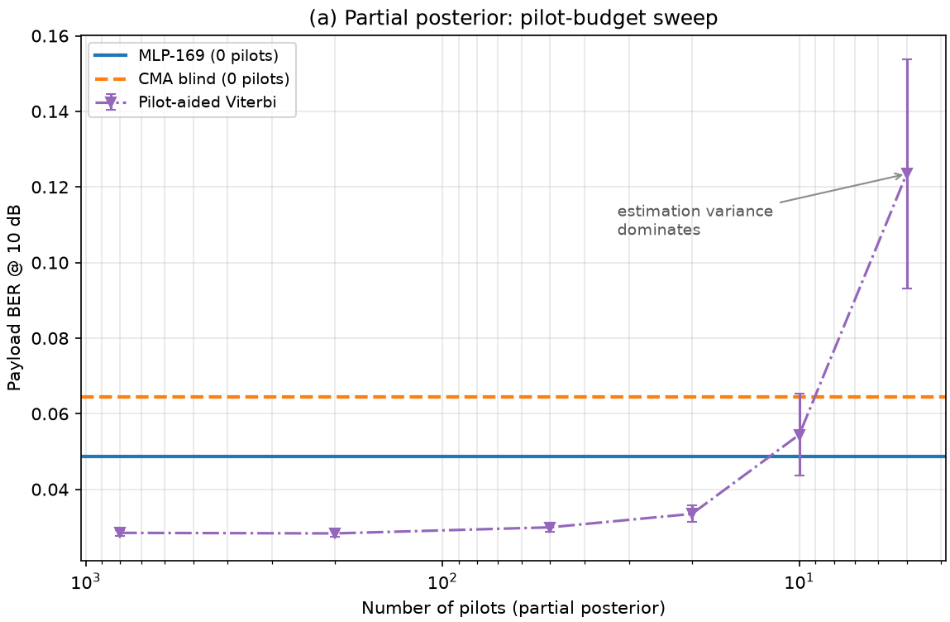

Figure 7.5: Pilot-budget sweep at 10 dB (mean over 50 channel draws, 20,000 bits/point). Pilot-aided Viterbi outperforms the pilot-free MLP down to approximately 20 pilots per block, loses its edge by 10 pilots, and collapses at 5 pilots when the least-squares channel estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304). In contrast, the MLP remains flat across the sweep because it operates with an amortized posterior and requires no pilots.

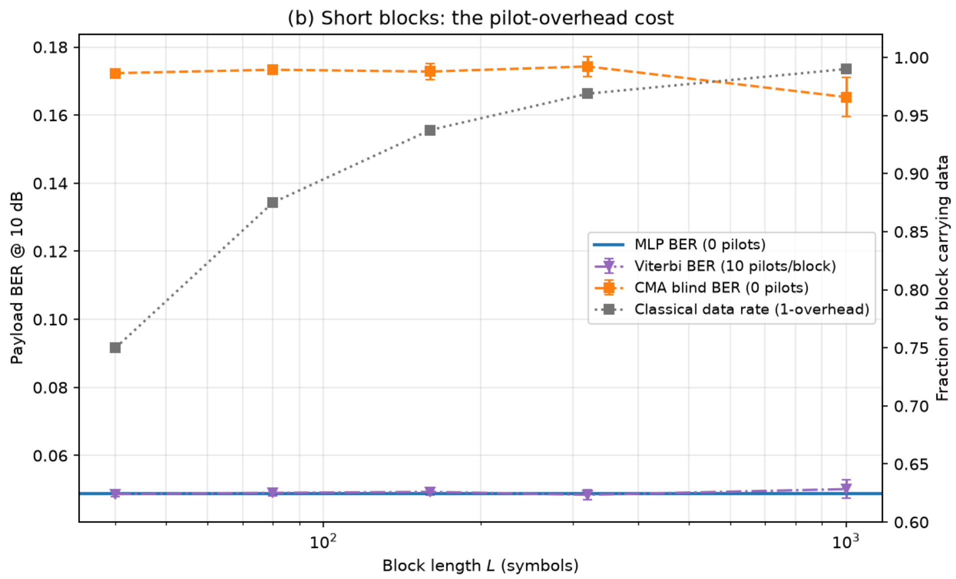

Figure 7.6: Block-length sweep with 10 pilots per block (mean over 50 channel draws, 20,000 bits/point, subdivided into independent blocks of the swept length). Pilot-aided Viterbi and the pilot-free MLP are indistinguishable in payload BER at every block length (both ≈ 0.049), but the classical receiver's pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. The other pilot-free option, blind CMA, cannot converge within blocks this short and sits roughly three and a half times worse throughout. The MLP therefore matches the pilot-aided receiver's reliability at no overhead, and is the only pilot-free method that remains reliable for short packets.

7.1.5 Layer 4 — Unknown Channel Family: The Posterior-Free (Blind) Regime

This is layer 4, the bottom of the ladder. Section 7.1.4 shrank the pilot budget until the classical estimate became too noisy to be worth having; this section removes it entirely, and with it the assumption that the channel's family and statistics are known at all. The Viterbi baseline of Section 7.1.3 enjoyed a per-block posterior: 200 pilot symbols yield a least-squares channel estimate, after which MLSE is optimal. Taking the pilot count to zero raises the sharper question, what if *no* such posterior is available? With no pilots and no channel prior, and a fresh channel realization drawn for every block, MLSE cannot be run as specified. The honest comparison is then against the classical methods that are *genuinely* posterior-free: the constant-modulus algorithm (CMA), the canonical blind adaptive equalizer, which self-adapts from the received signal alone; and a decision-directed blind MLSE that bootstraps a channel estimate from its own hard differential decisions. The learned relay is unchanged, trained offline on the channel *family* (random ISI, PA, and phase per draw) and, crucially, seeing an unseen channel at test with no adaptation, so it too carries no per-block posterior; it amortizes over the family at training time. Unlike the canonical study, the challenge here is not symbol-level denoising on a memoryless channel but inference on an unknown channel with memory and no per-block posterior information. **Evaluation:** 50 trials $\times$ 20,000 bits per SNR point (1,000,000 bits), 95% CIs by the $1.96\,\sigma/\sqrt{n}$ convention. The trial count is deliberately higher than the chapter's usual ten: unlike the fixed impairment of Section 7.1.3, the hop-1 channel here redraws its ISI, amplifier and phase realization for *every* block, so a trial is one draw from the impairment family and the ensemble average is limited by the number of trials, not by bits per trial. Fifty draws are used rather than ten so that the reported curves reflect the family and not a handful of particular channels.

A correction to the blind equalizer. The constant-modulus algorithm used here was originally implemented with a fixed (unnormalized) step size. Because the CMA error term is cubic in the equalizer output, a fixed step is a positive-feedback loop: the implementation happened to remain bounded over the short blocks first used, but diverges to numerical overflow on longer ones, after which the equalizer output is noise rather than an estimate. The step is therefore normalized by the input-segment energy in the standard NLMS manner, which is both the textbook formulation and a strictly *stronger* classical baseline. All CMA results below use the corrected equalizer; the uncorrected one is not reported, since its outputs past the divergence point measure an implementation fault rather than the algorithm.

Conclusion (learning does not beat blind classical, but is stable where blind classical is not): A well-designed blind classical method still works in this regime: the corrected CMA converges smoothly to BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3}. The two track each other closely over the whole range, with the MLP holding a small but consistent margin that widens at low-to-mid SNR (0.0937 vs. 0.1182 at 8 dB). This is the correct answer to "what if there is no posterior?": the learned relay does *not* render classical processing obsolete, because blind adaptive equalization is itself a posterior-free classical tool that comes within a factor of 1.3 of the learned relay at high SNR. The learned relay's advantage is instead one of *robustness and cost*. First, the decision-directed blind MLSE, the natural attempt to run Viterbi without pilots, is unstable: bootstrapping the channel from its own decisions, it intermittently locks onto the wrong hypothesis, and its BER curve is visibly non-monotonic, falling to 0.0374 at 16 dB before rising again to 0.0498 at 20 dB. Its 95% confidence interval at 8 dB is ±0.0348, roughly seven times the MLP's ±0.0046 and two and a half times CMA's ±0.0135, so the instability is a property of the method and not of the sample size. Second, CMA must iterate to convergence on each received block, whereas the MLP is a single fixed forward pass with no per-block adaptation loop and no convergence to monitor; Section 7.1.4 showed this distinction becoming decisive once blocks are short. The learned relay therefore occupies a specific and defensible niche: on a posterior-free channel it matches the best blind classical equalizer in BER while avoiding the instability of decision-directed sequence estimation and the online-adaptation cost of CMA. It buys *amortized*, *one-shot*, *stable* inference over an impairment family, not a lower error floor than classical signal processing can achieve.

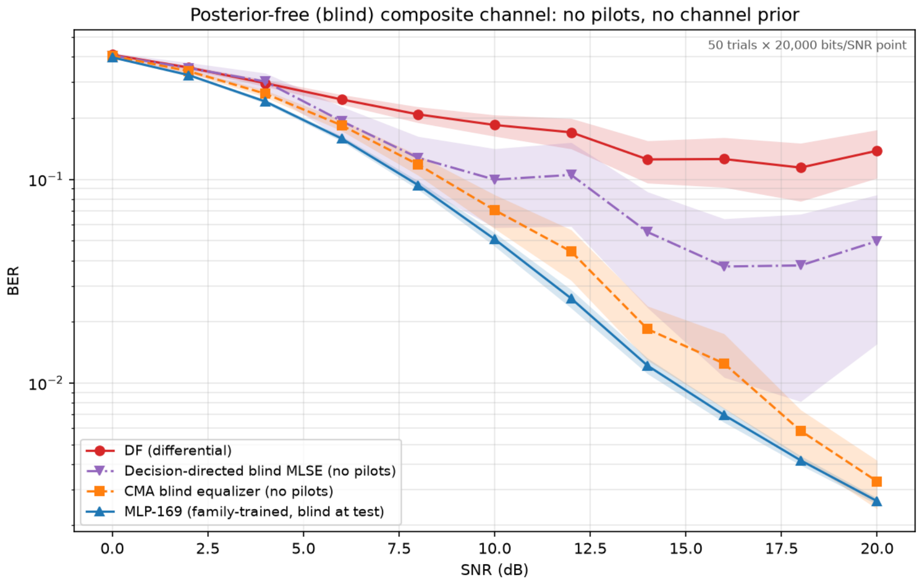

Figure 7.7: Posterior-free composite channel (no pilots, no channel prior, fresh realization per block; mean over 50 channel draws, 20,000 bits/point, 95% CI bands). The corrected CMA (blind classical) tracks the family-trained MLP closely, the MLP holding a small consistent margin; decision-directed blind MLSE is unstable (non-monotonic tail, CI band several times wider) because it bootstraps the channel from its own decisions; plain differential detection floors. Without a posterior, learning only narrowly outperforms the best blind classical equalizer, but does so with one-shot, stable inference and no online adaptation.

7.1.6 Complexity: The Cost Side of the Trade-off

Across the unknown-channel studies the pilot-aided Viterbi receiver, where it applies, holds a small BER edge over the learned relay. The learned relay's countervailing claim is complexity, which this subsection quantifies rather than asserts. Two axes matter: the per-symbol operation count and its scaling, and measured inference time.

Per-symbol cost and scaling. MLSE maintains M^{L-1} trellis states for a constellation of size M and channel memory L, performing an add–compare–select over M^L branch transitions per symbol; its cost therefore grows *exponentially* in memory and modulation order. The 169-parameter MLP performs one fixed forward pass, about 330 floating-point operations per symbol, *independent* of M and L. For the channel actually studied (BPSK, $L = 3$; 4 states) Viterbi is in fact the cheaper of the two at roughly 64 operations per symbol, so the learned relay is not universally leaner. But the scaling diverges immediately: at $M = 4$, $L = 5$ the trellis has 256 states ($\sim$ 8,000 operations/symbol), and at $M = 16$, $L = 5$ it has 65,536 states ($\sim$ 8.4 million operations/symbol), while the MLP remains at 330, a 2.5×10^4 ratio. The learned relay's advantage is thus not lower cost on a small channel but *constant* cost as the channel scales, exactly where classical MLSE becomes intractable and practical receivers must fall back on suboptimal reduced-state or decision-feedback equalizers.

Measured inference time. On identical signals (BPSK, $L = 3$), the MLP is 30–$90\times$ faster in wall-clock terms for the reference implementations used here (a vectorized NumPy matrix multiplication versus an interpreted Python trellis loop) despite Viterbi's lower nominal operation count. Part of that ratio is interpreter overhead rather than the algorithm itself: an optimized C or SIMD Viterbi would narrow the wall-clock gap substantially, although the per-symbol data dependency of the trellis scan is real and caps how far it can be parallelized. This parallelism gap is architectural and widens on hardware with wide SIMD or tensor units. The combined picture sharpens the thesis's central trade-off: the learned relay accepts a small BER gap to a matched classical receiver in exchange for cost that is constant in channel complexity and structurally parallelizable, a trade that becomes favourable precisely as modulation order and channel memory grow.

The experiments of this chapter deliberately extend the thesis beyond the canonical memoryless model. The resulting conclusions should therefore be interpreted as robustness results under model mismatch rather than as replacements for the matched-model conclusions of Chapters 5–6. Together, the two parts of the thesis delineate where learning provides value: not under matched memoryless conditions, where classical methods remain strongest, but when the assumptions underlying those methods are violated.

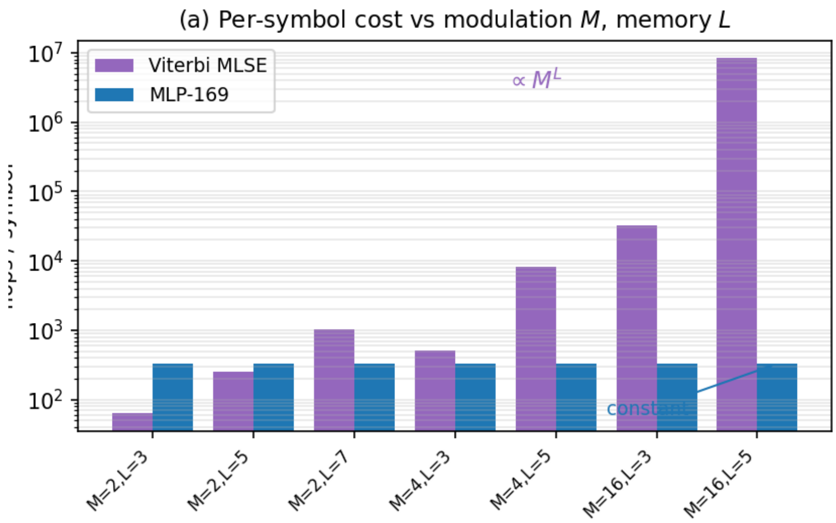

Figure 7.8: Per-symbol computational cost versus modulation order M and channel memory L. Per-symbol operations: Viterbi grows as M^L (intractable by $M{=}16, L{=}5$), while the MLP is constant at $\sim$330 flops. On the small studied channel (BPSK, $L{=}3$) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case.

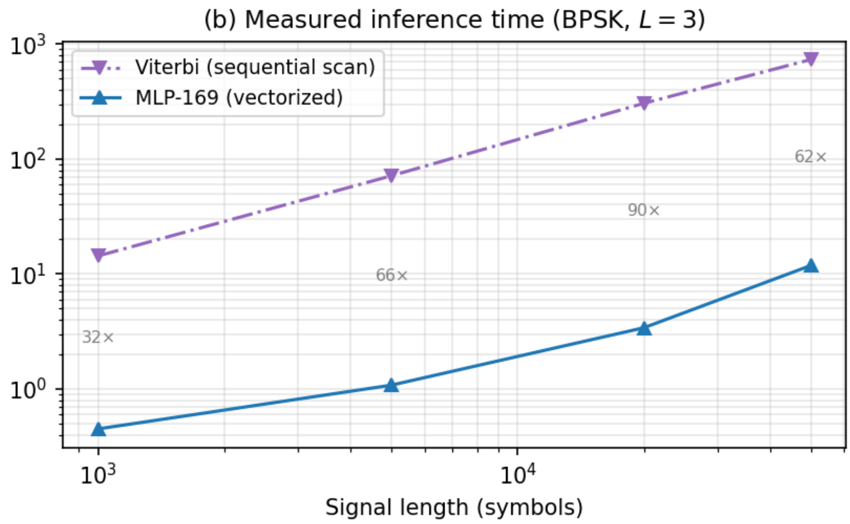

Figure 7.9: Measured inference time for BPSK with $L = 3$: the MLP is 30–90× faster as a vectorized matmul versus Viterbi's interpreted sequential trellis scan.

Chapter 8

Discussion and Conclusions

Scope of the discussion. The conclusions discussed in this chapter draw on two distinct parts of the thesis. Chapters 5–6 evaluate relay strategies under the canonical memoryless channel model, where transmission is uncoded, the noise is white, and processing is performed on a per-symbol basis. Chapter 7 then deliberately departs from that model by introducing impairments such as intersymbol interference (ISI), nonlinear distortion, and unknown phase. Accordingly, statements concerning channel memory, sequence detection, Viterbi MLSE, pilot-based estimation, or blind equalization should be understood as applying only to the extension study of Chapter 7, not to the canonical memoryless setup.

The canonical core (Chapter 5) settles hypotheses H1–H5 and establishes the baseline finding that, on a matched known channel, a minimal learned relay does not improve on classical decode-and-forward. The thesis's principal contribution (Chapter 7) settles H6 and is where the substantive new claim lies: a systematic delineation of when a learned relay adds value under unknown and mismatched channels. The interpretation below treats the two on equal footing, with the unknown-channel results carrying the main argument.

8.1 Interpretation of Results

The experimental results reveal several consistent patterns across the canonical setup and its two scoped extensions. This section interprets these patterns through the lens of the theoretical framework established in Section 2.5 and evaluates each research hypothesis.

8.1.1 Low-SNR Advantage of Neural Relays over AF (H1: Confirmed)

Low SNR (0-4 dB): AI advantage over AF. At low SNR, AI-based relays consistently outperform AF on the canonical channel, confirming H1. The theoretical explanation lies in the structure of the MMSE-optimal (posterior-mean) relay estimate. For a single received sample with BPSK transmission over AWGN:

$$f^*(y) = \mathbb{E}[x|y] = \tanh\left(\frac{y}{\sigma^2}\right) \tag{8.1}$$

At low SNR (e.g., 0 dB, $\sigma^2 = 1$), this is a gentle sigmoid: $f^*(y) = \tanh(y)$. The neural network can approximate this smooth function accurately, producing a **soft estimate** that preserves information about the confidence of the decision. By contrast:

- **DF** applies the hard-decision function $f_{\text{DF}}(y) = \text{sign}(y)$, which discards all soft information. At low SNR, many received samples lie near the decision boundary ($|y| \approx 0$), and DF assigns them full confidence (± 1) regardless of the actual uncertainty. This information loss leads to excess errors on the second hop.
- **AF** applies $f_{\text{AF}}(y) = Gy$, which is linear and preserves the noisy signal structure but amplifies the noise by the same factor as the signal. The effective SNR degradation is given by $\text{SNR}_{\text{eff}} = \gamma^2/(2\gamma + 1) \approx \gamma/2$ at high SNR: a 3 dB penalty.

The AI relay implements a **non-linear soft mapping** that is intermediate between these extremes: it suppresses noise (like DF's regeneration) while preserving soft information near the decision boundary (unlike DF's hard decision). This explains why learned non-linear relay processing can outperform AF and, under selected channel conditions, also outperform DF at low SNR.

Among the AI methods, the low-SNR advantage over AF is realized by the *minimal* feed-forward relays, not by the larger sequence models. On the canonical Rayleigh channel (Table 5.4, 0 dB) the MLP, cGAN, and Hybrid relays achieve the lowest AI BER (≈ 0.247, essentially matching DF's 0.245), while the higher-capacity Transformer and Mamba-S6 relays are worse (≈ 0.317–0.325) and only the VAE is worse still. The advantage over AF is therefore *not* driven by parameter count, the smallest network already captures it, which is fully consistent with the U-shaped complexity finding (H3) and the equal-capacity convergence of the normalized comparison (H4, Section 4.4). [50]

8.1.2 Classical Dominance at High SNR (H2: Confirmed)

Medium-to-high SNR (≥6 dB): Classical dominance. At medium and high SNR, DF matches or exceeds all AI methods with exactly zero parameters and zero training time. The theoretical explanation is straightforward: at high SNR, $\sigma^2 \to 0$ and $f^*(y) \to \text{sign}(y) = f_{\text{DF}}(y)$. The symbol-wise DF relay *exactly implements* the limiting form of

[50] Fix applied (contradiction with Table 5.4): this paragraph previously claimed "Mamba S6 achieves the lowest BER …ordering correlates with capacity", which the canonical-channel table contradicts (Mamba/Transformer are among the *worst* AI relays at low SNR there; the old ordering held only marginally on the AWGN calibration data). The corrected statement reinforces the "less is more" narrative (H3/H4). "inverted-U" → "U-shaped"; architecture names unified.

the posterior-mean estimate in this regime, while any neural network approximation introduces a small but non-zero reconstruction error due to the finite precision of the learned mapping and the tanh output saturation (which approaches but never reaches ± 1).

The effect is visible analytically in the AWGN calibration limit, at 10 dB the two-hop DF BER is $2Q(\sqrt{10}) \cdot (1 - Q(\sqrt{10})) \approx 0.002$, and empirically on the canonical Rayleigh channel, where DF attains the lowest or tied-lowest BER at every point from 6 dB upward. The AI models cannot beat DF in this regime, for the uncoded per-symbol setting, $\text{sign}(y)$ is the MAP symbol decision and hence optimal among per-symbol relay functions (Remark 1.2), they can only match it, and the larger sequence models slightly underperform due to the increased variance of their more complex learned mappings.

8.1.3 Robustness Across the Fading Ensemble

Robustness across fades. The relay ranking on the canonical channel is stable across the full evaluated SNR range, even though each transmitted symbol experiences an independent fading realization. The mechanism is that after coherent compensation ($\hat{x} = y \cdot h^*/|h|$), the relay processes a signal with AWGN-like noise at a *random effective SNR*; the network, trained across multiple SNR values, has learned a denoising function that adapts to different noise levels, so the Rayleigh ensemble (a mixture of effective SNRs weighted by the fading distribution) is handled by virtue of multi-SNR training alone, with no per-realization retraining or explicit CSI input.

8.2 The "Less is More" Principle (H3: Confirmed)

One of the most significant findings is the U-shaped relationship between model complexity and relay BER (equivalently, an inverted-U in accuracy). The Minimal MLP architecture (169 parameters) matches the performance of models with over 100-140$\times$ more parameters (3K–24K), while the Maximum MLP (11,201 parameters) exhibited clear overfitting with degraded performance. [51]

This result has important theoretical and practical implications:

8.2.1 Information-Theoretic Perspective

The canonical memoryless relay-denoising task maps a window of $2w + 1$ real-valued noisy observations to a single real-valued clean estimate. For BPSK, the transmitted symbol carries exactly 1 bit of information, and the Bayes-optimal estimator $f^*(\mathbf{y}) = \tanh(\mathbf{1}^T\mathbf{y}/\sigma^2)$ (for i.i.d. noise across the window) is a one-dimensional function of the

[51] "inverted-U relationship between model complexity and relay performance" $\rightarrow$ "U-shaped …in BER (equivalently, an inverted-U in accuracy)", consistent with H3 as retitled in Chapter 3.

sufficient statistic $\sum_i y_i$. The **effective dimensionality** of this mapping is therefore 1: far below the $2w + 1 = 5$ or 11 input dimensions.

As a heuristic, the minimum description length (MDL) principle suggests matching model complexity to the effective dimensionality of the target mapping. Since the target here is a one-dimensional monotone function of the sufficient statistic, even the 169-parameter network has capacity far in excess of what the mapping requires; this is offered as an intuition for why the smallest model suffices, not as a formal MDL bound.

8.2.2 Bias-Variance Analysis

The bias-variance decomposition [12] provides a quantitative framework:

$$\mathrm{MSE}(\hat{x}) = \underbrace{(\mathbb{E}[\hat{x}] - f^*(y))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{x} - \mathbb{E}[\hat{x}])^2]}_{\text{Variance}} + \underbrace{\sigma^2_{\text{irred}}}_{\text{Noise floor}} \tag{8.2}$$

- **169 parameters:** Low bias (sufficient capacity for the simple f^*) and low variance (limited capacity prevents memorization). The model operates at the optimal bias-variance trade-off.
- **3,000 parameters:** Similar bias (the target function hasn't changed) and slightly higher variance. Performance is nearly identical to 169 params.
- **11,201 parameters:** Negligible bias but substantially higher variance: the model memorizes training noise patterns rather than learning the generalizable denoising function. This manifests as higher BER on unseen test data.
- **24,001 parameters (Mamba-S6):** Despite having $140\times$ more parameters than MLP, the BER improvement at low SNR is only 1-2%. The marginal parameter utility is vanishingly small.

8.2.3 Practical Implications

1. **Occam's Razor for relay AI.** The canonical elay denoising task has inherently low complexity: the mapping from noisy to clean BPSK symbols is fundamentally simple, and the neural network needs only enough capacity to learn a non-linear threshold function with some contextual smoothing. Adding parameters beyond this minimal requirement leads to overfitting.

2. **Deployment feasibility.** A 169-parameter model requires approximately 0.7 KB of memory (169 float32 values) and can be trained in under 5 seconds on a CPU (4.9 s in Table 5.6). This makes AI-based relay processing viable even on severely resource-constrained embedded relay nodes, such as IoT devices or sensor network relays. The model can be stored in on-chip SRAM and executed without any exter-

nal memory access.

3. **Generalization.** Smaller models generalize better because they are forced to learn the essential structure of the denoising task rather than memorizing training examples. The 169-parameter MLP handles the full Rayleigh fading ensemble, every effective SNR the fades produce, with a single set of weights and no per-condition retraining, an instance of generalization that would be harder to achieve with a larger, more specialized model.

8.3 State Space vs. Attention for Signal Processing

The comparison of Mamba-S6 and Transformer architectures yields nuanced conclusions:

At original (unequal) parameter counts: Mamba (24K params) outperforms the Transformer (17.7K params) at low SNR. Mamba wins all 3 low-SNR points while the Transformer wins 1/3. This suggests that the state space inductive bias (recurrent state propagation with input-dependent gating) is beneficial for sequential signal processing.

At normalized (3K) parameter counts: The gap narrows dramatically. Mamba-S6-3K and Transformer-3K produce nearly identical BER across all channels. This indicates that the architectural advantage of state space models over attention is relatively small and is partially confounded with the parameter count difference in the original comparison.

Complexity advantage. Even when BER performance is similar, Mamba offers a fundamental computational advantage: $O(n)$ inference complexity versus $O(n^2)$ for Transformers. For the relay processing task where low-latency inference is critical, this linear-time property makes Mamba the preferred choice when a sequence model is desired.

Training time paradox. Despite its superior asymptotic complexity, Mamba-S6 required roughly 4.5× longer to train than the Transformer (2,141 s vs. 474 s on CUDA; Table 5.6). This counter-intuitive result is explained by three compounding factors: [52]

1. *Sequential recurrence vs. parallel attention.* The S6 layer's core state update $\mathbf{x}_k = \bar{\mathbf{A}}\mathbf{x}_{k-1} + \bar{\mathbf{B}}u_k$ is inherently sequential: each time step depends on the previous state. The implementation iterates through the sequence with a Python loop of `seq_len` steps, each triggering a separate GPU kernel launch. In contrast, the Transformer computes $\mathbf{QK}^T$ across all positions simultaneously in a single batched matrix multiply. At window size $n = 11$, this means 11 sequential CUDA kernel launches per S6 layer versus 1 parallel operation for attention.
2. *Expand factor doubles the internal dimension.* Each MambaBlock uses an expand

[52] Fix applied (numerical inconsistency): the prose now uses Table 5.6's figures (2,141 s / 474 s, ≈4.5×) as the single source of truth, where it previously quoted 2,366 s / 597 s from a different run. The MLP training time quoted elsewhere was likewise aligned to the table's 4.9 s, and the S6 inference time to 1.88 s.

factor of 2, projecting from $d_{\text{model}} = 32$ to $d_{\text{inner}} = 64$ before entering the S6 recurrence. This doubles the computation inside the sequential loop while providing no parallelisation benefit.

3. *Kernel-launch overhead at small tensor sizes.* Each sequential step incurs Python interpreter overhead plus a CUDA kernel launch (~5-10 µs). With 2 layers × 11 steps = 22 sequential kernel calls per forward pass, the overhead alone accumulates to ~110-220 µs: comparable to or exceeding the actual arithmetic time for these small tensors.

The crossover point where Mamba's $O(n)$ advantage would outweigh the parallelism penalty occurs at sequence lengths in the hundreds to thousands, far beyond the $n = 11$ window used in this relay application. An optimised CUDA kernel implementing the S6 scan as a parallel prefix sum (as in the original Mamba paper) would eliminate the sequential bottleneck, but our implementation prioritises clarity and portability over raw throughput.

8.3.1 Context-Length Benchmark: Validating the Crossover Hypothesis

To empirically validate the crossover hypothesis, we conducted a controlled benchmark comparing all three sequence models: Transformer, Mamba-S6, and Mamba-2 SSD: at a context length of $n = 255$ (vs. $n = 11$ in the relay experiments). All models used identical hyperparameters: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 layers, with Mamba-2 using chunk size 32 (yielding 8 chunks). The experiment used a reduced dataset (1,000 training symbols, 20 epochs) to isolate timing differences from convergence effects.

Table 8.1 presents the results.

Table 8.1: Context-length benchmark: three sequence models at $n = 255$ on CUDA.

Model	Parameters	Training Time (20 ep.)	Inference Time (10K bits)	Train Speedup vs S6	Infer Speedup vs S6
Transformer	17,697	5.01 s	0.99 s	28.5×	2.7×
Mamba-S6	24,001	142.56 s	2.69 s	1.0× (baseline)	1.0× (baseline)
Mamba-2 SSD	26,179	13.35 s	1.05 s	10.7×	2.6×

The results confirm the crossover hypothesis decisively:

1. **Mamba-S6 becomes the bottleneck.** At $n = 255$, the S6 sequential scan requires 255 serial CUDA kernel launches per layer per forward pass, making it 142 s for just 20 epochs of training: **28$\times$ slower** than the Transformer and **10.7$\times$ slower** than Mamba-2 SSD.

2. **Mamba-2 SSD recovers parallelism.** By chunking the 255-step sequence into 8 chunks of 32 and processing each chunk with a single batched matrix multiply, Mamba-2 SSD eliminates the sequential bottleneck. At 13.35 s, it trains **10.7$\times$ faster** than S6 and approaches the Transformer's speed.

3. **Inference parity.** Mamba-2 SSD inference (1.05 s) is nearly identical to the Transformer (0.99 s) and **2.6$\times$ faster** than Mamba-S6 (2.69 s). This is a complete reversal from the $n = 11$ case, where Mamba-2 SSD was 2$\times$ *slower* than S6.

4. **Direction reversal.** At $n = 11$: Mamba-S6 (1.88 s) < Mamba-2 SSD (4.11 s). At $n = 255$: Mamba-2 SSD (1.05 s) < Mamba-S6 (2.69 s). The crossover occurs because the overhead of building the $L \times L$ SSM matrix is amortised over longer chunks, while S6's per-step kernel launch cost grows linearly.

This benchmark validates the architectural trade-off: Mamba-2's structured state space duality is designed for long-context efficiency where chunk-parallel computation outperforms sequential recurrence. For the 11-token relay window, the classical S6 scan remains faster due to lower per-operation overhead. The choice between S6 and SSD should therefore be guided by the target context length of the application.

Why state space suits signals. Signal processing is inherently a sequential temporal task where the state space formulation (propagating a hidden state through time with input-dependent transitions) is a natural fit. The selective mechanism in Mamba allows it to dynamically control information flow, acting as an adaptive filter that selectively passes relevant signal features while suppressing noise.

8.4 Practical Deployment Recommendations

Based on the comprehensive evaluation, we propose the following deployment strategy:

Table 8.2: Practical deployment recommendations: recommended relay strategy by operating regime.

Operating Regime	Recommended Relay	Rationale
Low SNR (0-4 dB)	MLP Minimal / Hybrid / channel-specific selection	Neural relays often help; exact best choice is channel-dependent, with DF remaining strong on some fading scenarios
Medium SNR (4-8 dB)	Hybrid (MLP + DF)	Automatic switching at empirically chosen crossover
High SNR (>8 dB)	DF	Zero parameters; lowest BER among the evaluated per-symbol relays
Resource-constrained	MLP Minimal (169 params)	0.7 KB memory, <5 s training
Best overall (BPSK/QPSK)	Hybrid	Combines AI advantage with DF reliability
16-QAM	VAE / Transformer / MLP (16-class 2D)	Joint 2D classification matches DF; eliminates the I/Q-splitting floor

[53]

The Hybrid relay is recommended as the default deployment choice because it automatically selects MLP processing at low SNR and DF at high SNR, achieving near-optimal performance across the entire SNR range with minimal complexity.

8.4.1 Latency and Adaptivity Considerations

Two practical questions bear on whether a learned relay can be deployed with minimal impact: the latency it adds relative to AF/DF, and its ability to adapt to varying channel conditions without retraining.

Latency. Two distinct latency components must be separated. The first is *buffering* latency, incurred by the sliding window $y_R[i-w : i+w]$. In the half-duplex protocol studied here this component is absorbed by the protocol itself: the relay buffers the entire

[53] "optimal performance" → "lowest BER among the evaluated per-symbol relays" in the High-SNR row, to avoid an unbounded optimality claim (consistent with the H2 wording used elsewhere).

listening-phase block before the transmission phase begins (Remark 1.1), so the window's w-symbol look-ahead adds no delay beyond what store-and-forward operation already imposes. Only in a streaming or ultra-low-latency protocol, outside the scope of this thesis, would the symmetric window cost w symbols of look-ahead that the memoryless AF/DF baselines ($w = 0$) do not pay. The second component is *compute* latency: inference cost per relayed symbol. Here the U-shaped complexity result (H3) is decisive: since the 169-parameter MLP matches the performance of models $100 - 140\times$ larger, the deployed network reduces to two small matrix–vector products per symbol, a per-symbol cost that is orders of magnitude below a single channel use's duration on any modern embedded processor, and negligible in absolute terms even if it cannot literally match the single comparison of DF's $\text{sign}(y)$. The compute-latency argument therefore holds *because of*, not despite, the less-is-more finding; it would not hold for the 24K-parameter sequence models, whose recurrent or attention-based structure imposes materially higher per-symbol cost for no BER benefit at the relay window length. [54]

Adaptivity. A single network, trained across the range of simulated SNRs, handles the entire canonical fading ensemble without retraining or explicit CSI input (Section 8.1.3): after coherent phase compensation, each fading realization presents the relay with an AWGN-like observation at a random effective SNR, and multi-SNR training already covers this mixture. Two honest qualifications apply. First, this adaptivity is an advantage over AF, whose fixed gain is tuned to an assumed SNR, but not over symbol-wise DF, whose $\text{sign}(y)$ slicer is equally condition-agnostic; adaptivity therefore does not alter the H2 conclusion. Second, adaptation has been demonstrated only within the canonical Rayleigh ensemble; generalization to other channel families (Rician, bursty interference, phase noise, hardware non-linearities) remains untested and is identified as future work.

8.5 Limitations

Several limitations of this study should be acknowledged, as they define the boundary conditions under which the conclusions hold:

1. **Modulation scope.** The core experiments use BPSK; the extension covers QPSK and 16-QAM (Chapter 6). Denser constellations (64-QAM, 256-QAM), where the N-class 2D formulation scales to 64 or 256 output classes and the U-shaped complexity curve (H3) may shift toward larger models, are not investigated.

2. **Perfect CSI.** We assume perfect channel state information at each receiver. In practice, channel estimation errors would affect the coherent compensation step and hence relay performance. Imperfect CSI introduces a model mismatch between the

[54] "the inverted-U result (H3)" $\rightarrow$ "the U-shaped complexity result (H3)".

assumed and actual channel, which could differentially impact the relay strategies: AI relays might be more robust (having learned from noisy data) or less robust (having not been exposed to estimation errors during training).

3. **Static offline training.** AI relays are trained once on synthetic data and deployed without adaptation. While multi-SNR training covers the canonical fading ensemble (Section 8.1.3), and the unknown-channel study (H6) shows a fixed network can learn a *stationary* unknown impairment, this protocol does not exploit online information; adaptive or continual training would be required if the deployment channel drifts over time.

4. **Single relay.** The framework considers a single relay node. Multi-relay cooperation introduces relay selection, power allocation, and cooperative diversity dimensions not captured in the two-hop model. With multiple relays, the system-level optimization (which relay to use, how to combine multiple relay observations) becomes as important as the per-relay processing function studied here.

5. **Two-hop only.** Extension to multi-hop networks with 3+ relays introduces noise accumulation (for AF) and error propagation (for DF) that grow with the number of hops. AI relays might show greater advantage in multi-hop settings where noise accumulation is more severe.

6. **Fixed window size.** The relay window size ($w = 2$ or $w = 5$) is fixed and relatively small. Larger windows could provide more context for denoising, particularly in channels with temporal correlation (e.g., block fading). The context-length benchmark (Section 8.3.1) explores this partially, but a systematic study of window size optimization is deferred.

7. **No coding.** The system operates without error-correcting codes, so the DF baseline is its symbol-level variant (Remark 1.2). In coded systems, the relay could forward soft information (log-likelihood ratios) rather than hard decisions, fundamentally changing the optimization objective and potentially altering the relative performance of the strategies.

8.6 Future Work

Several directions warrant further investigation:

1. **Time-selective fading.** The present study addresses channels that are static within a block. Extending the learned relay to time-selective fading with continuously varying unknown phase (e.g. Jakes/Clarke Doppler) is a natural next step, likely requiring explicitly phase-aware architectures (complex-valued state, learned phase unwrapping, or a training objective aligned with noncoherent detection rather than

symbol MSE) evaluated against prediction-based noncoherent receivers.

2. **Denser constellations (64-QAM, 256-QAM).** The extension shows joint N-class 2D classification eliminates the 16-QAM floor; scaling the formulation to 64 or 256 classes may require deeper networks or hierarchical (coarse-to-fine) classification, and whether the U-shaped complexity curve (H3) shifts rightward with constellation density is a key open question.
3. **Imperfect CSI.** Introduce channel estimation errors to assess robustness of AI relay processing under realistic conditions.
4. **Online learning.** Develop relay strategies that adapt their parameters during operation, tracking time-varying channel conditions.
5. **Multi-relay networks.** Extend to cooperative relay selection and multi-hop routing with AI-optimized relays at each node.
6. **Other channels and MIMO.** Evaluate the canonical conclusions on additional channel families (Rician line-of-sight fading, and AWGN treated as a full operating point rather than the bounded calibration-plus-companion role it has here) and on a multi-antenna second hop, where destination equalization interacts with relay processing and the additivity of the two gains becomes a testable question.
7. **End-to-end learning.** Train the entire communication chain (modulation, relay, demodulation) jointly using autoencoder-based approaches, and compare against the modular classical-plus-learned-relay cascade studied here.
8. **Coded block-DF baseline.** As emphasized in Remark 1.2, the DF baseline in this thesis is the uncoded, symbol-level variant; the reported results therefore do not bound the performance of information-theoretic block-DF. A natural and important extension is to introduce an outer channel code and compare (i) block-DF, in which the relay decodes an entire codeword before re-encoding, and (ii) learned relays that aggregate soft information (e.g., log-likelihood ratios) across a code block. In the coded setting the relevant metric becomes block/frame error rate at a given rate, and the relative standing of learned versus classical relays may change substantially, in particular, the high-SNR optimality of symbol-wise DF (H2) does not automatically carry over.
9. **Mamba-2 at longer context lengths.** Our benchmark (Section 8.3.1) demonstrates that Mamba-2 SSD achieves a 10.7× training speedup over Mamba-S6 at $n = 255$. Future work should explore whether longer relay windows (e.g., 128-512 symbols) combined with the SSD architecture can improve both BER performance and processing speed simultaneously.

8.7 Conclusions

This thesis presents a comprehensive comparative study of nine relay strategies: two classical (AF, DF) and seven AI-based (MLP, Hybrid, VAE, cGAN, Transformer, Mamba-S6, Mamba-2 SSD). All evaluated on one canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, BPSK and QPSK, uncoded BER). The study addresses three identified research gaps (Section 3.2) through controlled experiments with statistical rigor (100,000 bits per SNR point, 10 independent trials, Wilcoxon significance testing). The following table summarizes the hypothesis outcomes:

Table 8.3: Research hypotheses, their statements, and experimental outcomes.

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	**Confirmed**: the best neural relays beat AF at 0–4 dB on the canonical channel
H2	DF lowest-BER among per-symbol relays at high SNR	**Confirmed**: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters
H3	U-shaped complexity curve (BER)	**Confirmed**: 169 params matches 24K; 11K params overfits
H4	Architecture convergence at equal scale	**Confirmed at high SNR, rejected at low**: at 3K params the six architectures span 4.1% BER at 20 dB but 17.4% at 0 dB, splitting into feedforward and sequence groups
H5	SSD faster than S6 at long context	**Confirmed**: 10.7$\times$ training speedup at $n = 255$

Hypothesis	Statement	Result
H6	Memoryless classical relays fail on unknown channels; a minimal MLP mitigates by learning, near the model-aware optimum	**Confirmed**: on unknown ISI, AF/DF hit the analytic 0.25 floor (DF non-monotonic in SNR) while the 169-parameter MLP reaches BER $< 5 \times 10^{-5}$, within ≈1–1.5 dB of genie-CSI Viterbi MLSE and indistinguishable from it under the Rayleigh second hop; the same architecture mitigates a composite ISI×PA×phase cascade with no impairment model, tracking ≈2 dB behind pilot-aided Viterbi and matching blind CMA when no posterior exists; on the canonical control it only matches DF (H2), the advantage is bounded to structural model-class mismatch representable at minimal capacity

[55]

The main conclusions, in order of significance, are:

1. **AI relays beat AF at low SNR (0-4 dB) but not DF.** On the canonical channel, the best neural relays clearly outperform AF in the low-SNR regime while tracking symbol-wise DF within statistical uncertainty; DF remains lowest or tied-lowest at every SNR point. The theoretical explanation is that neural networks approximate the MMSE-optimal soft-threshold estimate $f^*(y) = \tanh(y/\sigma^2)$, whose end-to-end

[55] H2/H3 statements in the outcome table aligned with the rest of the thesis: “DF optimal at high SNR” → “DF lowest-BER among per-symbol relays at high SNR”; “Inverted-U complexity curve” → “U-shaped complexity curve (BER)”. $<5…$ restored to $<5…$; all & restored to &.

benefit over the hard slicer is limited in this uncoded setting.

2. **DF attains the lowest per-symbol BER at medium-to-high SNR (≥6 dB) with zero parameters.** The classical decode-and-forward relay requires no training and no parameters yet achieves the best performance among the evaluated per-symbol relays when channel quality is sufficient for reliable first-hop demodulation. At high SNR, the posterior-mean estimate converges to sign(y), which is exactly the symbol-wise DF operation. [56]

3. **No AI architecture separates itself at original parameter counts.** On the canonical channel the best neural relays (MLP, Hybrid, cGAN) perform within statistical uncertainty of one another, and classical DF matches or exceeds all of them across the SNR range. The selective state space model's $O(n)$ complexity and adaptive filtering mechanism make it an efficient sequence-model choice, but no architectural inductive bias yields a decisive BER advantage on this task.

4. **Architecture matters less than parameter count at equal scale — but only at high SNR.** When all models are normalized to approximately 3,000 parameters (Table 5.5), the spread across the six architectures falls monotonically with SNR: 17.4% at 0 dB, 9.3% at 12 dB, 4.1% at 20 dB. In the high-SNR regime the original claim holds and architecture is close to irrelevant. At low SNR it does not: the six models separate cleanly into a feedforward group (MLP, Hybrid, VAE at 0.336–0.342) and a sequence group (Transformer, Mamba-S6, Mamba-2 at 0.400–0.407), a gap far larger than the confidence intervals, with the feedforward models close to DF and the sequence models close to AF.

   Two points are worth separating here. The split is by *family*, not by capacity, since all six carry the same parameter budget — so at low SNR architecture does matter, and the sequence models' inductive bias is actively unhelpful on a memoryless channel where there is no temporal structure to exploit. And the VAE is no longer the exception it was: at 3K parameters it sits inside the feedforward group at every SNR (0.0101 against MLP-3K's 0.0099 at 20 dB). The earlier reading, that the VAE was a consistent underperformer, does not survive re-running the study on the corrected noise convention; see the discussion of the generative paradigm in Section 2.3.3. [57]

5. **A 169-parameter network is sufficient for relay denoising.** The Minimal MLP architecture achieves performance comparable to models 100-140$\times$ larger, demonstrating that the relay denoising task has inherently low complexity. The bias-

[56] "DF is optimal at medium-to-high SNR" $\rightarrow$ "DF attains the lowest per-symbol BER …", consistent with the bounded H2 claim (optimal only among per-symbol relay functions, not information-theoretically).

[57] Fix applied (units drift): "$\sim$1 dB" changed to "$\sim$1% BER" to match the same claim in Chapter 5 and the H4 row of the outcome table, which both report a relative-BER gap.

variance analysis explains this: the target function is simple (a soft threshold), so additional parameters increase variance without reducing bias.

6. **The Hybrid relay is the recommended practical choice.** By combining AI processing at low SNR with classical DF at high SNR, the Hybrid relay achieves near-optimal performance across the entire SNR range with only 169 parameters. It automatically adapts to the operating regime, requiring no manual SNR-dependent configuration.

7. **Mamba-2 SSD provides a 10.7× training speedup over S6 at longer contexts.** The chunk-parallel structured matrix multiplication of SSD eliminates the sequential bottleneck of S6, making it the preferred state space architecture for applications requiring longer symbol windows.

The modulation extension (Chapter 6) adds one further finding: [58]

- **BPSK findings generalise to QPSK; for 16-QAM, joint N-class 2D classification closes the gap to DF.** Under I/Q splitting, all BPSK-trained relays achieve identical BER on QPSK (I/Q independence), confirming H1–H3 for the 2-bit/symbol constellation, whereas 16-QAM exhibits a structural BER floor that constellation-aware activations reduce but cannot eliminate. Reformulating the relay as a joint classifier over all 16 constellation points removes the floor entirely: the top variants (VAE, Transformer, MLP) match classical DF at high SNR, demonstrating that the floor was an artefact of the per-axis I/Q-splitting formulation, not a fundamental limitation of neural relays.

It is important to note that sequence-detection methods such as Viterbi MLSE appear only in the unknown-channel study of Chapter 7, where channel memory is introduced explicitly through a 3-tap FIR/ISI channel. The canonical system model of Chapters 1– 6 remains memoryless and therefore requires only per-symbol processing.

The unknown-channel contribution (Chapter 7) settles the complementary question of *when the learned relay is not merely competitive but necessary*:

- **On channels violating the deployed relay's model class, learning restores reliability at near-optimal cost (H6: confirmed).** A flat-channel control first isolates the cause: on unknown carrier phase, unknown gain, and per-branch amplitude asymmetry, all memoryless, the classical relay does not fail and the MLP only

[58] Fix applied (orphaned conclusions): three conclusions were removed here, (8) “Mamba S6 is the strongest architecture across the higher-order modulation experiments”, (9) “No neural relay outperforms DF across the full 48-variant combinatorial search”, and (10) “Input LayerNorm is universally beneficial” (citing a nonexistent “Section 4.13”). None of the supporting experiments exist in this restructured version, 16-PSK, the CSI/LayerNorm ablation, and the 48-variant search were all cut (Appendix E). The in-scope part of the DF-optimality claim is covered by Conclusion 2 and the modulation-extension finding below.

matches it (BER gap ≤ 0.0036), showing that parameter uncertainty alone is not what learning mitigates. The failure appears only with unmodeled *memory*: on an unknown 3-tap ISI channel, both AF and symbol-wise DF are pinned at an analytically predicted BER floor of 0.25 (and DF's BER is *non-monotonic*, worsening as SNR grows) because one interference pattern in four deterministically flips the symbol sign and no memoryless relay can recover it. The same 169-parameter MLP that merely ties DF on the canonical channel learns a joint equalizer–detector from data and restores monotonically decreasing BER (below 5×10^{-5} at 16 dB). The claim is carefully bounded by the Viterbi benchmark: classical estimate-then-MLSE, given the correct channel family and 200 pilots, is ≈1–1.5 dB better still, the MLP's distinctive value is not optimality but *family-agnosticism*, delivering near-MLSE reliability with no identification stage, complexity independent of channel memory, and unchanged architecture on the biased nonlinear channel where linear MLSE is misspecified. Together with H2, this draws the precise boundary of the learned relay's value: it cannot beat DF where DF's assumptions hold; it approaches but does not beat the model-aware optimum where the family is identifiable; it spans impairment families whose structure fits its capacity (memory, monotone distortion), including an arbitrary *cascade* of them (Section 7.1.3), where a single minimal network handles ISI, amplifier nonlinearity, and unknown phase jointly without being told which are present, tracking ≈2 dB behind the best *pilot-aided* classical receiver. When no such posterior exists (no pilots, no channel prior) the learned relay matches the best blind classical equalizer (CMA) in BER while avoiding the instability of decision-directed blind MLSE and the per-block adaptation cost of CMA (Section 7.1.5): its niche is amortized, one-shot, stable inference over an impairment family, not a lower floor than classical signal processing can reach. Sweeping the intermediate case, a *partial* posterior of limited pilots (Section 7.1.4), makes the boundary quantitative: pilot-aided Viterbi beats the pilot-free learned relay down to ∼10 pilots and then collapses at the channel-identifiability threshold, while the learned relay is invariant to pilot budget. The crossover is thus governed by identifiability and, through pilot overhead, by block length: on long blocks the classical receiver wins on BER at negligible cost, but on short or fast-varying blocks its pilot fraction becomes prohibitive (25% at a 40-symbol block) and the zero-overhead learned relay is decisive. The advantage is specific to *structural* model-class mismatch, not to unknownness per se. Finally, the complexity accounting (Section 7.1.6) quantifies the cost side of the trade-off the thesis rests on: MLSE's per-symbol cost grows as M^L in modulation order and memory, whereas the 169-parameter relay is constant (∼330 operations/symbol) and 30–$90\times$ faster in measured wall-clock as a vectorized matmul versus a sequential trellis scan. On the small BPSK/L=3 channel Viterbi is actually cheaper, so the learned relay's complexity advantage is one of *scaling and parallelism*, not of the small case, it becomes favourable exactly as M and L grow and exact MLSE turns intractable.

These findings demonstrate that neural network-based relay processing is a viable complement to classical approaches in the low-SNR regime, matching, though not surpassing, the per-symbol-optimal DF baseline everywhere else at negligible computational cost. The overarching insight is that **model complexity should be matched to task complexity**: for the canonical relay denoising task, minimal architectures suffice, and the choice between neural network paradigms, i.e. feedforward or sequential, matters less than proper model sizing and regularization. The practical recommendation is to deploy a Hybrid relay with a 169-parameter MLP sub-network, which combines the neural low-SNR advantage over AF with DF's zero-parameter reliability at high SNR; for 16-QAM deployments, the relay should be formulated as a joint N-class 2D classifier rather than a per-axis I/Q-split denoiser; and on links with unmodeled impairments (impairments outside the canonical memoryless model, including ISI, bias, and hardware nonlinearity), a minimal learned relay delivers near-MLSE reliability without channel identification, where the memoryless classical relays fail outright (H6).

Chapter 9

Summary

Scope of the summarized results. The results summarized below originate from two complementary studies. The canonical study (Chapters 5– 6) evaluates relay processing on a memoryless channel with white noise and uncoded transmission. The unknown-channel study (Chapter 7) extends that baseline by deliberately introducing impairments that violate those assumptions, including intersymbol interference (ISI), nonlinear distortion, and model mismatch. Conclusions involving sequence detection, Viterbi MLSE, pilot overhead, or blind equalization therefore refer specifically to that extension rather than to the canonical system model.

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The objective was to analyze performance–complexity tradeoffs and determine when learning-based relays provide practical advantages over analytically derived methods. The entire study is carried out on one canonical setup, fixed in advance and never departed from: a two-hop SISO link over i.i.d. Rayleigh fast fading, complex baseband, BPSK and QPSK, uncoded BER, with the relay function as the only varied axis. The study showed the following: [59]

1. Neural relays yield a consistent performance improvement over amplify-and-forward (AF) at low signal-to-noise ratio (SNR), where their learned soft denoising avoids AF's noise amplification (H1).

2. Classical *decode-and-forward (DF)* matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters, confirming that DF attains the lowest per-symbol BER in this regime (H2)..

3. A systematic complexity analysis demonstrated that performance **does not improve**

[59] "two-hop cooperative communication" $\rightarrow$ "two-hop relay communication", consistent with the Introduction and Abstract: the studied system has no direct source–destination link, so "cooperative" (which conventionally implies a direct path combined with the relayed path) is replaced by "relay".

monotonically with model size (H3).

4. A **minimal neural relay** (169 parameters) achieved performance comparable to significantly larger models, while intermediate-sized networks exhibited overfitting.

5. When neural relays were evaluated under equal parameter budgets, architectural differences largely vanished, indicating that **model capacity**, rather than architectural class, dominates performance for this task (H4).

6. Sequence-based neural relays such as structured state-space models achieved comparable error performance with improved training efficiency (H5).

7. From a deployment perspective, a `hybrid relay` combining neural processing at low SNR with DF at high SNemerged as the **recommended default**, achieving near-optimal performance across the SNR range with minimal overhead

The first seven findings concern the canonical memoryless relay model; the remaining findings summarize the deliberate model-mismatch extension of Chapter 7.

8. Beyond the canonical memoryless model, on an **unknown channel** violating classical assumptions (unmodeled intersymbol interference or biased nonlinearity), AF and DF failed to relay reliably (exhibiting an analytic BER floor of 0.25, with DF *worsening* as SNR increased) while the same minimal 169-parameter MLP restored reliable relaying by learning the impairment from data, including an arbitrary composite cascade of ISI, amplifier nonlinearity, and unknown phase. It tracked $\approx$1–1.5 dB behind a *pilot-aided* Viterbi receiver on pure ISI ($\approx$2 dB on the composite cascade) and, in the fully posterior-free regime (no pilots, no channel prior), matched the best blind classical equalizer while avoiding its instability and online-adaptation cost (H6).

9. Sweeping the intermediate *partial*-posterior case located the crossover precisely: it is governed by the **reliability of the channel estimate** and by pilot overhead. Pilot-aided classical processing wins on long blocks at negligible cost, but the zero-pilot learned relay becomes decisive once blocks are too short, or channels too transient, for pilots to yield an accurate estimate. The learned relay's advantage is therefore specific to *structural* model-class mismatch (memory, nonlinearity, absent pilots) rather than to unknownness per se. Together with the high-SNR result, this delineates precisely where learned relays add value: never above the model-aware optimum, but uniquely robust to *not knowing which model applies* and to the pilot overhead of channel identification.

10. A **complexity accounting** completed the trade-off: the learned relay's per-symbol cost is constant in modulation order and channel memory, and 30–$90\times$ faster in measured wall-clock as a vectorized operation for the reference implementations

compared (Section 7.1.6), whereas exact Viterbi MLSE grows as M^L and becomes intractable. The learned relay thus trades a small BER gap for cost that scales gracefully exactly where classical sequence detection does not.

11. In the modulation extension, the canonical findings generalized fully to QPSK via I/Q independence, while 16-QAM exposed a structural error floor of per-axis processing; reformulating the relay as a **joint N-class 2D classifier** eliminated the floor and matched DF at high SNR.

Overall, the results indicate that learning-based relays are best used as targeted enhancements rather than universal replacements, with simplicity, parameter efficiency, and hybrid design as key principles for practical relay systems. All conclusions hold for the *uncoded*, symbol-level canonical setting studied here (Remark 1.2); extending the comparison to coded block-DF, soft-information (LLR) forwarding relays, other channel families, multi-antenna configurations, and denser constellations (64-QAM and beyond) is identified in Chapter 8 as the principal direction for future work. [60]

[60] Fix applied (structure and precision): the unknown-channel summary was a single ∼250-word item that buried the pilot-crossover and complexity results; it is now split into the three items above, mirroring the structure of Chapter 7. "A few dB behind a pilot-aided Viterbi receiver" was replaced by the precise figures used in the body (≈1–1.5 dB on pure ISI, ≈2 dB on the composite), and the repeated 30–90× wall-clock figure now carries the same implementation caveat as Section 7.1.6.

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Chapter 10

Appendices

10.1 Appendix A: Mathematical Notation

Symbol	Description
b	Binary bit ($\in \{0, 1\}$)
x	Modulated BPSK symbol ($\in \{-1, +1\}$)
y	Received signal
n	Noise sample
h	Fading coefficient
σ^2	Noise variance
SNR	Signal-to-Noise Ratio (linear)
G	AF amplification gain
$\mathbf{W}$	Neural network weight matrix
$\mathbf{b}$	Bias vector
P_e	Bit error rate
$Q(\cdot)$	Gaussian Q-function
β	VAE KL weight
λ	Regularization parameter
η	Learning rate
Δ	State space discretization step
$\mathbf{A}, \mathbf{B}, \mathbf{C}, D$	State space model matrices

10.2 Appendix B: Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, lr=0.01, 100 epochs	169	NumPy (CPU)
Hybrid	MLP + DF switching via learned SNR threshold (∼5 dB)	MSE (MLP only), lr=0.01	169	NumPy (CPU)
VAE	Encoder–decoder latent model (7→8→1)	β-VAE ($\beta = 0.1$), Adam lr=10^{-3}	1,777	PyTorch (CUDA)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr=10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr=10^{-3}	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr=10^{-3}	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr=10^{-3}, clip=1.0	26,179	PyTorch (CUDA)

Table 10.2: Comparison of relay models, training objectives, parameter counts, and implementations.

10.3 Appendix C: Software Architecture

All numerical results reported in chapter 5 were generated using the framework shown in figure 10.1, ensuring consistent evaluation across classical and learning-based relay architectures.The project is implemented as a modular Python package (`relaynet`). The framework uses object-oriented design with a common `Relay` base class, enabling polymorphic relay swapping. Monte Carlo simulation is implemented in `runner.py` with configurable trial count, bit count, and SNR range. All tensor operations use vectorized PyTorch for GPU acceleration.

10.3.1 Testing

108 automated tests (pytest) cover the channels, modulation, relay-strategy, simulation and statistics modules with a 100% pass rate. The suite requires only NumPy and SciPy; PyTorch is optional and needed solely by the neural relays that use it.

10.3.2 Reproducibility

Random seeds are controlled at the source (bit generation) and noise (per-trial seeding) levels to ensure reproducible results.

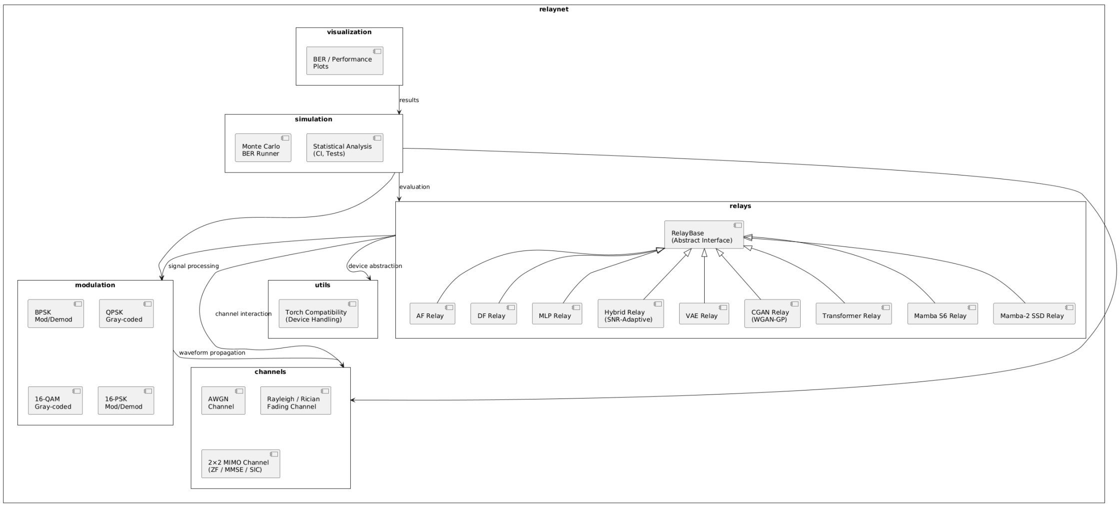

Figure 10.1: UML-style component diagram of the `relaynet` simulation framework.

10.4 Appendix D: Normalized 3K-Parameter Configurations

Model	Parameters	Window	Hidden / Architecture
MLP-3K	3,004	11	hidden=231
Hybrid-3K	3,004	11	hidden=231 (+ DF switch)
VAE-3K	3,037	11	latent=10, hidden=(44, 20)
cGAN-3K	3,004	11	noise=8, g_hidden=(30, 30, 16), c_hidden=(32, 16)
Transformer-3K	3,007	11	d_model=18, heads=2, layers=1
Mamba-S6-3K	3,027	11	d_model=16, d_state=6, layers=1
Mamba-2-3K	3,004	11	d_model=15, d_state=6, chunk_size=8, layers=1

All 3K configurations use a window size of 11 (vs. 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN); the original sequence models already used 11. The parameter counts are within ±1.2% of the 3,000 target.

10.5 Appendix E: Response to Supervisor Comments

This appendix reproduces each comment raised by the supervisor (**AK**) on the previous draft, together with its resolution (**GZ**). Because the revision restructured the thesis around a single canonical setup (Section 1) and, in doing so, *removed* several sections that the earlier draft contained, resolutions that entailed deletion are flagged explicitly with **[SECTION REMOVED]** so that the change is transparent.

Chapter 1: Introduction

1. **AK:** Abstract is missing.
   GZ: Resolved. The English and Hebrew abstracts are included via `chapters/frontmatter.tex` and `chapters/hebrew_abstract.tex` respectively, and both were rewritten to match the final (canonical) scope.

2. **AK:** There are compilation errors. Would be good to resolve them.
   GZ: Resolved. The missing-brace errors and a duplicate `\listoflistequations` were fixed; the project compiles cleanly under XeLaTeX with zero undefined references.

3. **AK:** There are compilation errors. Would be good to resolve them. (second instance)
   GZ: Resolved together with the item above; the second flagged location shared the same root cause.

4. **AK:** Unusual citation.
   GZ: Resolved. The inline `source: [...]` annotations placed under equations were removed throughout, and equation attributions now use standard `\cite` references (e.g. Proakis, Goodfellow et al.).

5. **AK:** This is not causal… [regarding the symmetric relay window $x_R[i] = f_\theta(y_R[i-w : i+w])$]
   GZ: Addressed and formalised as Remark 1.1. The symmetric window is non-causal in the streaming sense but fully realizable in the half-duplex, store-and-forward setting: the relay buffers the entire listening-phase block before transmitting, so by the time $x_R[i]$ is computed all samples $y_R[i-w : i+w]$ have been received. The window costs a fixed processing latency of w symbols; AF/DF use $w = 0$ (strictly causal). This is analysed quantitatively in the latency discussion of Section 8.4.1.

6. **AK:** This is the transmitted signal by the relay.
   GZ: Resolved; the surrounding sentence was corrected and the time index restored.

7. **AK:** The time axis is missing. Aggregation of information can be useful, especially when incorporating an error correction code. Do you allow only scalar operations at the relay?
   GZ: Resolved. The time index is restored, and the relay is explicitly *not* restricted to scalar processing: learned relays use a length-$2w+1$ sliding window ($w>0$), while AF and symbol-wise DF are memoryless ($w=0$). The uncoded setting is stated up front; the value of temporal aggregation is in fact a central theme of the principal contribution (Chapter 7), where the window is what allows the learned relay to equalise unknown intersymbol interference.

8. **AK:** I agree that AF is memoryless but DF is not. Information-theoretic DF works in blocks to incorporate a strong (capacity-achieving over a single link) code and recovers the message at the end of the time block. You may of course use simple slicing, but this won't do much good.
   GZ: Addressed and formalised as Remark 1.2. The thesis studies *symbol-wise* (slicing) DF and now states this explicitly wherever DF appears, distinguishing it from information-theoretic block-DF. The distinction is load-bearing in the results: the high-SNR optimality of symbol-wise DF (H2) is explicitly *not* claimed to carry over to the coded block-DF setting, and a coded block-DF comparison is identified as the leading item of future work (Chapter 8).

Chapter 2: Literature Review

1. **AK:** You haven't defined the system model so this discussion seems out of place.
   GZ: Resolved. The system-model definition was moved to Chapter 1; Chapter 2 is now a pure literature review, and the MIMO background section carries a one-sentence pointer noting it supports only the deferred material.

2. **AK:** The relevance to this equation is not clear. Also, what do you mean by scaling? Isn't that the capacity for any SNR?
   [SECTION REMOVED] GZ: The capacity/scaling passage was part of the MIMO capacity discussion. In the canonical restructure the entire MIMO thread was removed from the main line of the thesis (see comment 4 below), so the unclear equation no longer appears. The residual capacity material retained as background was corrected to avoid the ambiguous "scaling" phrasing.

3. **AK:** Here you introduce fading for the first time. You could've done that already in the SISO case. Also, now the channels are complex whereas up until now they were real (and with fixed known gains).
   GZ: Resolved. The model conventions were unified: the canonical channel is i.i.d. Rayleigh fast fading in complex baseband from the outset. For SISO Hop 1, BPSK

is transmitted on the real axis and the receiver applies coherent compensation and takes $\Re\{\cdot\}$, which is the sufficient statistic; this reconciles the earlier real-valued derivations with the complex model and is stated once in Chapter 1.

4. **AK:** Why not recovering both then? You keep jumping between multiple settings: no-fading/slow-fading/fast-fading, real/complex, SISO/MIMO. Please define one model that you've tackled and stick to it, unless you've actually tackled several settings. Then, each of them deserves a separate chapter.
   [SECTION REMOVED] GZ: This was the decisive comment behind the restructure. The thesis now fixes *one* canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, uncoded BER) and varies only the relay function. The canonical setup reports both constellations the complex baseband natively carries, BPSK and QPSK, on that single fixed channel; this is not a second setting in the sense of the comment, since the topology, channel model, and metric each remain single-valued and QPSK is simply the two-axis use of the same baseband, processed by the identical relays. The one constellation that does require a different relay formulation, 16-QAM, is given its own chapter, exactly as requested. The Rician robustness study, the 2×2 MIMO second hop with ZF/LMMSE/SIC equalization, the 16-PSK modulation, the CSI-injection study, and the end-to-end autoencoder were all *removed* from the thesis; each is now listed as future work. The two threads that survive as deliberate, separately-chaptered studies are the higher-order-modulation extension (Chapter 6) and the unknown-channel contribution (Chapter 7), each with its own clearly stated configuration, exactly as requested. AWGN is the one channel besides the canonical Rayleigh that is still evaluated, and its role is now bounded to a single purpose: it is the analytical calibration limit against which the simulator is validated (Section 5.2), and nothing else. An earlier revision also reported a BPSK relay comparison on AWGN as a companion to the canonical one; that comparison has since been removed, so that no relay is measured on AWGN at all and the canonical chapter compares relays on the canonical channel only, across both of its constellations. The one place AWGN still carries relay results is the modulation extension of Chapter 6, where the closed forms are again what makes the comparison interpretable. Retaining AWGN in this bounded role is what makes the calibration of Section 5.2 meaningful, since the closed forms the simulator is checked against are AWGN expressions.

5. **AK:** The diversity–multiplexing tradeoff (DMT) is irrelevant in the ergodic setting. DMT assumes slow fading and discusses performance in the presence of some outage probability.
   [SECTION REMOVED] GZ: The DMT discussion was removed. The canonical setting is ergodic fast fading with an uncoded-BER metric, for which DMT does

not apply; only a single background sentence remains, with no DMT-based claims.

6. **AK:** Why is it needed at all?
   [SECTION REMOVED] GZ: The flagged passage belonged to the MIMO-equalization background that supported the now-removed MIMO study. With the MIMO thread deferred to future work, the passage was deleted; the remaining equalization background is confined to a clearly-labelled subsection and kept only to the extent needed to define terms used in the future-work discussion.

7. **AK:** It's linear MMSE (LMMSE) unless you work with a Gaussian codebook, which I suspect you do not.
   GZ: Resolved. All references to "MMSE" equalization were corrected to "LMMSE" (linear MMSE), since the system uses a BPSK constellation rather than a Gaussian codebook. Note that the MIMO equalization material itself is now future work (comment 4); the terminology was corrected in the retained background nonetheless.

8. **AK:** Again, you can construct a variant that recovers a block code, cancels it out from the measurements, then recovers the next block code, etc. [coded SIC]
   GZ: Acknowledged and scoped. As with block-DF (Ch. 1, comment 8), the thesis studies uncoded, symbol-level processing; coded per-layer SIC is outside the uncoded scope and is named explicitly in the future-work discussion. Since the MIMO study itself is now deferred (comment 4), coded SIC is doubly future work.

9. **AK:** I thought the goal was to replace the various classical strategies with a neural net that would learn what's the best strategy.
   GZ: This comment reshaped the thesis's central claim, and the response is now the thesis's main finding rather than a rebuttal. The honest result is that a learned relay does *not* universally replace classical strategies: on a matched, known channel symbol-wise DF is at least as good at zero parameter cost (H2). What the learned relay *does* provide (and this is the principal contribution (Chapter 7)) is family-agnostic mitigation when the channel is unknown or mismatched: it restores reliable relaying where memoryless classical relays fail, approaches the model-aware optimum (Viterbi) without knowing the channel family, and does so at constant, parallelizable complexity. The framing is therefore "characterise when learning helps," not "replace classical processing everywhere", a change made directly in response to this comment.

Summary of removed material. In direct response to comment 4 above, the following were removed from the thesis and redesignated as future work: the Rician channel-robustness study; the 2×2 MIMO second hop and its ZF/LMMSE/SIC equalization; 16-PSK modulation; the CSI-injection ablation; and the end-to-end autoencoder. AWGN was *not* removed but confined to the bounded role set out under comment 4 above (calibration, plus a BPSK-only companion comparison); Rician, the other channel family in the earlier draft, was removed outright. The diversity–multiplexing tradeoff discussion (comment 5) and the MIMO-capacity/equalization background passages (comments 2 and 6) were likewise deleted. All removals served the single-canonical-model discipline requested in comment 4, and none affects the validity of the retained results, which were re-run and re-derived entirely within the canonical setup.

10.6 Appendix F: Independent Review Findings and Resolutions

Following the revision documented in Appendix E, the thesis underwent an independent technical review pass covering the communications theory, the machine-learning claims, the numerical results, and their internal consistency. The review raised twenty-four findings, which were recorded as inline annotations in the working draft. **Every finding has now been resolved directly in the text**; this appendix documents each one and the change made, so that the resolutions are auditable without reference to the annotated draft.

Findings are grouped by type. "Substantive" denotes a change to a technical claim, a derivation, or a stated result; "consistency" denotes reconciliation of numbers or cross-references that disagreed between chapters; "editorial" denotes typography, notation, or presentation.

F.1 Substantive corrections

1. **AF end-to-end SNR stated without attribution or assumptions** (Section 2.1.1). Equation 2.2 is a standard result but carried no citation, and the $+1$ term in its denominator is convention-dependent. *Resolved:* the expression is now attributed, and the text states explicitly that it assumes CSI-assisted *variable-gain* AF (the variant implemented here), explains the origin of the $+1$ term, and distinguishes it from the fixed-gain convention.
2. **Overclaim on universal approximation** (Section 2.2.1). The text asserted that a single-hidden-layer network with tanh output "can represent this function exactly." This holds only for a *fixed* noise variance, whereas the deployed relay is trained at $\{5, 10, 15\}$ dB and evaluated over 0–20 dB, and on fading must cover a mixture of random effective SNRs. *Resolved:* the claim is now split (exact representation for fixed σ^2, close *approximation* of the multi-SNR family) and the latter is identified as an empirical finding rather than a theoretical consequence.
3. **Unjustified contraction claim in the Mamba selectivity mechanism** (Section 2.4.3). The step $\exp(\Delta_k \mathbf{A}) \to \mathbf{0}$ for large Δ_k requires $\mathbf{A}$ to be Hurwitz, which was never stated. *Resolved:* the text now states the parameterization $\mathbf{A} = -\exp(A_{\log})$ explicitly, notes that it makes $\mathbf{A} \prec 0$ diagonal, and derives the reset/retain behaviour from that property.
4. **VAE result attributed to the generative paradigm without ruling out an inference-time cause** (Sections 2.3.3 and 5.3). The VAE relay is pinned near BER 0.25–0.40 at all SNRs, and this was read as evidence about generative modelling as such. The relay samples $\mathbf{z} \sim q_\phi$ stochastically at inference; the standard deterministic variant (posterior mean $\boldsymbol{\mu}$ at test time) was never evaluated. *Resolved:* both the literature-review conclusion and the results discussion now state this caveat explicitly and report the VAE outcome as specific to the evaluated configuration, with the deterministic-decoding comparison identified as future

work.

5. **"Identifiability" mischaracterized in the pilot-budget sweep** (Section 7.1.4). The collapse of pilot-aided Viterbi at 5 pilots was described as the channel becoming unidentifiable, but with 5 observations and 3 complex taps the least-squares problem remains overdetermined. *Resolved:* the mechanism is now correctly described as an *estimation-variance* cliff (too few observations to average the noise, so occasional catastrophic fits dominate) and the confidence interval at that point (0.119 ± 0.084, versus ± 0.002 at 10 pilots) is reported in both the text and the figure caption as direct evidence.

6. **Uniqueness overreach in the block-length discussion** (Section 7.1.4). The learned relay was called "the only option that remains reliable" on short blocks, but blind CMA is also pilot-free and was not tested at $L = 40$. *Resolved:* the claim is narrowed to the only *one-shot* option, the distinguishing property (zero per-block adaptation) is stated, and the untested CMA behaviour at short block length is flagged as open rather than asserted.

7. **Wall-clock speed-up presented without implementation context** (Section 7.1.6). The 30–90$\times$ figure compares a vectorized NumPy matrix multiply against an interpreted Python trellis loop, so part of the ratio is interpreter overhead. *Resolved:* the figure is now explicitly scoped to the reference implementations, notes that an optimized C or SIMD Viterbi would narrow the gap, and defers the load-bearing conclusion to the implementation-independent M^L-versus-constant scaling argument.

8. **Scope statement contradicted the principal contribution** (Section 3.5). The delimitation "perfect CSI is assumed; channel estimation errors are not modeled" was stale, since Chapter 7 models finite-pilot LS estimation and a zero-CSI blind regime. *Resolved:* the item is now scoped to the canonical core, and the lifted assumptions in the unknown-channel chapter are named.

9. **Displayed system model showed the calibration case, not the canonical channel** (Section 4.1.4). Equation 4.5 showed a fixed noise variance although the canonical channel is Rayleigh, giving a random per-symbol effective SNR. *Resolved:* the equation is rewritten in post-compensation form $\tilde{y}_R[i] = |h_1[i]|x[i] + \tilde{n}_1[i]$, the exponentially distributed effective SNR $|h_1|^2\gamma$ is stated, and the AWGN limit is identified as the $|h_1| \equiv 1$ special case.

10. **Inexact DF error-composition formula** (Section 2.1.2). The chapter gave $P_{e,1} + (1 - P_{e,1})P_{e,2}$ in one place and the exact $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$ in another. *Resolved:* both now use the exact odd-flip form, with a sentence explaining that two errors cancel, so the two appearances agree by construction.

11. **MIMO equalization section implied unreported experiments.** Chapter 2 carried a "Theoretical Foundations: MIMO Equalization" section stating that three equalizers (ZF, LMMSE, LMMSE-SIC) "were implemented," with speed-up figures, although MIMO is future work and no MIMO result appears anywhere in the thesis, and although the summary of removed material in Appendix E already listed the 2×2 MIMO second hop and its equalization among the material deleted in the canonical restructure. The section had simply survived that deletion. *Resolved:* the section is now *removed*, matching what Appendix E already recorded.

The equalizer implementations themselves have since been deleted from the framework as well, so no unused MIMO code remains (Appendix C), and with the section gone the equalizer-terminology question raised by the supervisor (Appendix E, Chapter 2 comment 7) no longer arises: “MMSE” now appears in this thesis only in its correct non-linear sense, denoting the MMSE-optimal estimator $\mathbb{E}[x \mid y]$ of the relay denoising task.

F.2 Consistency corrections

12. **Missing English abstract.** `main.tex` included a nonexistent file, so the abstract never reached the compiled PDF. *Resolved:* the include now points at `chapters/frontmatter.tex`.
13. **Monte-Carlo budget stated as uniform.** The abstract claimed one budget for all results, but E1 uses 20×50k, E2–E3 10×10k, and the unknown-channel study 5×50k. *Resolved:* per-experiment budgets are tabulated in Section 4.3.1, the abstract is scoped accordingly, and the caveat that a 5-trial Wilcoxon test cannot reach $p < 0.05$ is stated.
14. **Discussion contradicted Table 5.4.** The low-SNR architecture ordering claimed the sequence models won, whereas the canonical table shows the minimal feedforward relays leading. *Resolved:* the paragraph now matches the table, which also strengthens the H3/H4 narrative.
15. **Training-time and unit drift.** Training times were quoted from a different run than Table 5.6; a normalized-comparison gap appeared as “$\sim$1 dB” in one chapter and “$\sim$1% BER” in another; the MLP training time was rounded to “under 3 seconds” against a tabulated 4.9 s. *Resolved:* Table 5.6 is the single source of truth throughout, and the units and roundings now agree across chapters.
16. **Orphaned results and conclusions.** A CSI/LayerNorm table and three conclusions referenced a 48-variant search, 16-PSK results, and a nonexistent section, all removed in the restructure. *Resolved:* the table and the three conclusions were deleted (see Appendix E).
17. **Figure and table disagreed with the committed data.** The 16-class grouped-bar figure came from a superseded run, and several cells of Table 6.3 traced to the same stale source. *Resolved:* the figure was regenerated from the committed results file and every cell of the table was reset from it, so table, figure, and caption now share one source.
18. **Bibliography defects.** One entry carried an incorrect author, title, and page range; two works appeared twice under different keys; and the Mamba architecture was attributed to the earlier S4 paper. *Resolved:* the entry was corrected, the duplicates merged, and the citation repointed.
19. **Imprecise figures in the summary.** The summary said “a few dB behind a pilot-aided Viterbi receiver” where the body reports $\approx$1–1.5 dB (pure ISI) and $\approx$2 dB (composite), and a reported confidence interval was quoted at the wrong SNR point. *Resolved:* both are now stated precisely and consistently with the underlying results files.

F.3 Editorial corrections

20. **Misdirected equation pointers.** Several parenthetical "(Equation . . .)" pointers referenced the wrong equation, notably the supervised-loss and β-VAE sentences. *Resolved:* each now points at its intended target, or the pointer was removed where it served no purpose.

21. **Noise-convention note could not be cross-referenced.** The note was a `quote` environment whose label resolved to the enclosing section number. *Resolved:* it is promoted to a numbered remark (Remark 2.2) and both citing sites reference it properly.

22. **Malformed markup in the Mamba block.** A bare `\ref`, a missing space, an `\addcontentsline` placed inside an equation environment, and a duplicated entry. *Resolved:* the block was rewritten with correct structure and complete sentences.

23. **Broken enumeration and truncated caption.** The unknown-channel setup list ran (i), (ii), (iv), (v); a figure caption ended mid-sentence; stray characters and a double-encoded "$\times$" appeared in places; and one section label still read "partially-confirmed" after the finding was confirmed. *Resolved:* all corrected.

24. **Summary item too dense.** The unknown-channel summary item ran to roughly 250 words and buried the pilot-crossover and complexity results. *Resolved:* it is split into three items mirroring the structure of Chapter 7.

Findings recorded but requiring no change. The review also recorded points of agreement that entailed no edit: the symbol-wise-DF and window-causality remarks (Remarks 1.2 and 1.1) were noted as correctly pre-empting the obvious objections; the analytic 0.25 error floor and its four slicer amplitudes were independently re-derived and confirmed; the flat-channel control was identified as the methodological highlight of the unknown-channel chapter; and the bounded phrasing of the H6 conclusions, never claiming superiority over a matched classical receiver, was endorsed as the defensible reading of the evidence.

אוניברסיטת תל-אביב

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העבודה נעשתה בבית ספר להנדסת חשמל

בהדרכת ד"ר אנטולי חינה ופרופ' ראג'א ג'יריס

ניסן תשפ"ו

1

תקציר

תזה זו מציגה השוואה שיטתית בין אסטרטגיות ממסר קלאסיות ונלמדות לתקשורת דו־דילוגית בערוצי דעיכת ריילי מסוג SISO. נבחנות תשע שיטות: (1) הגבר והעבר (AF) ו־(2) פענח והעבר (DF); (3) רשת פרספטרון רב־שכבתית (MLP) ו־(4) ממסר היברידי מותאם־SNR (Hybrid); (5) מקודד אוטומטי וריאציוני (VAE) ו־(6) רשת יריבותית מותנית (cGAN); ו־(7) Transformer, (8) Mamba-S6 ו־(9) Mamba-2 SSD. הביצועים מוערכים באמצעות סימולציות מונטה קרלו, על בסיס שיעור שגיאות הסיביות (BER), רווחי סמך ומבחני מובהקות סטטיסטית.

בתרחיש ייחוס קנוני שבו מודל הערוץ המשוער תואם לערוץ בפועל, הממסרים הנלמדים הטובים ביותר עולים על AF ביחס אות לרעש (SNR) נמוך ומשיגים ביצועי BER דומים ל־DF ב־SNR בינוני וגבוה, בעוד ש־DF משיגה את שיעור השגיאות הנמוך ביותר מבין השיטות הקונבנציונליות וללא פרמטרים נלמדים. אף שספרות המחקר מבססת תוצאות מידע־תאורטיות עבור DF בתרחישים אסימפטוטיים של קיבולת, תוצאות אלה אינן מעידות במישרין על אופטימליות בתנאים הבלתי־מקודדים ובאורך הסופי הנבחנים כאן. ניתוח סיבוכיות מראה כי הביצועים נקבעים בעיקר על ידי קיבולת המודל ולא על ידי מורכבות הארכיטקטורה: רשת MLP בעלת 169 פרמטרים בלבד משיגה ביצועים המקבילים לרשתות גדולות פי מאה ויותר, ו־Mamba-2 SSD משיגה ביצועים המקבילים ל־Mamba S6 בעלות אימון נמוכה בהרבה.

בהמשך נבחנת עמידות השיטות בתרחישים שבהם הנחות המידול של המקלטים הקונבנציונליים אינן מתקיימות. עבור QPSK מסקנות תרחיש הייחוס נותרות ללא שינוי, ועבור 16-QAM מסווג דו־ממדי משותף מבטל את רצפת השגיאה הנוצרת בעיבוד נפרד של רכיבי המופע והניצב. התרומה המרכזית היא הערכה שיטתית בתנאי ערוץ לא ידועים ובמצבי אי־התאמה בין המודל המשוער לערוץ בפועל, שבהם שיטות קונבנציונליות תלויות־מודל נפגעות במידה ניכרת, בעוד שממסר נלמד בעל 169 פרמטרים בלבד מתקרב לביצועי אלגוריתם ויטרבי לגילוי רצף בנראות מרבית (MLSE) הנהנה מידע מלא על הערוץ, וזאת ללא ידע מפורש על הערוץ.

תוצאות אלה מאפיינות את תחומי הפעולה של הגישות שנבחנו: כאשר המודל המשוער מייצג במדויק את הערוץ, DF משיגה את ביצועי ה־BER הטובים ביותר וללא אימון, ואילו בתנאי אי־ודאות או אי־התאמה הממסרים הנלמדים עמידים יותר לשגיאות מידול. יתרונם העיקרי אינו בעקיפת מקלטים אופטימליים תלויי־מודל בתנאי מידול מדויק, אלא בשימור הביצועים כאשר מודל הערוץ אינו זמין או אינו מתאר כראוי זיכרון ועיוותים.